

Abundance after nonvanishing for compact Kähler fourfolds

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Abstract

We prove semiampleness of the actual $\mathbb{Q}$ -Cartier adjoint $K_X + \Delta$ of a normal connected compact Kähler klt fourfold whenever it is analytically nef and some positive Cartier multiple has a nonzero section. The boundary is effective and rational; the fourfold need not be projective or $\mathbb{Q}$ -factorial. In Iitaka dimension zero, a positive Cartier multiple is the trivial holomorphic line bundle.

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1 Introduction

A rational holomorphic line is *semiample* if some positive Cartier multiple is generated by global sections. Such a multiple defines a holomorphic map to projective space. The abundance question considered here is whether analytic nefness of a klt adjoint on a compact Kähler fourfold, together with one nonzero plurisection, forces this conclusion.

Let Δ be an effective rational Weil divisor on a normal complex space X . We say that $D = K_X + \Delta$ is an *actual $\mathbb{Q}$ -Cartier adjoint* when an appropriate reflexive adjoint power is an invertible holomorphic sheaf. All multiples and sections of D then refer to tensor powers of that line. This convention allows K_X and Δ to fail to be separately $\mathbb{Q}$ -Cartier. The line is *analytically nef* if, for a fixed Kähler form and every $\varepsilon > 0$, a fixed Cartier multiple has a smooth Hermitian

metric whose normalized curvature is bounded below by minus ε times that form, using local smooth potentials on X . Section 2 gives the full sheaf and metric conventions. The Iitaka condition $\kappa(X, D) \geq 0$ means that a positive Cartier multiple has a nonzero holomorphic section.

Theorem 1.1. *Let X be a normal connected compact Kähler complex space of dimension four. Let Δ be an effective rational Weil divisor such that (X, Δ) is Kawamata log terminal and the actual adjoint $D = K_X + \Delta$ is $\mathbb{Q}$ -Cartier. If D is analytically nef and $\kappa(X, D) \geq 0$, then there is an integer $m > 0$ for which mD is Cartier and*

$$H^0(X, \mathcal{O}_X(mD)) \otimes_{\mathbb{C}} \mathcal{O}_X \longrightarrow \mathcal{O}_X(mD)$$

is surjective at every point. If $\kappa(X, D) = 0$, then one may choose m so that $\mathcal{O}_X(mD) \simeq \mathcal{O}_X$.

Nonvanishing is a hypothesis. The conclusion concerns the given holomorphic adjoint line on X , without projectivity or $\mathbb{Q}$ -factoriality of X and without a numerical-dimension restriction. When the Iitaka dimension is zero, the conclusion is an actual rational-linear trivialization.

The threefold abundance theorems provide both the lower-dimensional input and the methods behind this problem. In the projective terminal case, Miyaoka treated numerical dimension one [40], and Kawamata proved abundance for minimal threefolds [35]. Passing to compact Kähler spaces requires contraction and positivity arguments which cannot be obtained by choosing a global ample divisor. Höring–Petersen developed the minimal-model and Mori-fiber-space theory for compact Kähler threefolds [33, 32]. Campana–Höring–Petersen established canonical abundance for normal ordinary $\mathbb{Q}$ -factorial compact Kähler threefolds with terminal singularities [6]; the corrected Chern-class argument is given in their appendix to Guenancia–Păun’s orbifold Bogomolov–Gieseker theorem [30, Appendix, Theorem A.2].

For lc compact Kähler threefold pairs with rational boundary and nef adjoint, Das–Ou proved semiampleness when the numerical dimension is different from two or the Iitaka dimension is positive [13, Theorem 1.1]. Their sequel treats numerical dimension two and obtains abundance for lc compact Kähler threefolds [14, Theorem 1.2 and Corollary 1.3]. These results generate the adjoint lines on the normal three-dimensional components of a fourfold boundary. They do not by themselves choose sections agreeing on its intersections or extend those sections to the fourfold. Those are the two boundary problems treated here.

For positive Iitaka dimension, Höring–Lazić–Lehn prove semiampleness for nef klt adjoints on ordinary $\mathbb{Q}$ -factorial compact Kähler spaces through dimension four [31, Theorem 4.1]. A crepant ordinary $\mathbb{Q}$ -factorial Kähler model projective over X places our pair in that scope. The sections of the pulled-back Cartier line are exactly the sections from X , so generation descends to the original line. It remains to treat Iitaka dimension zero.

From a plurisecion to a supported model

Choose $s \neq 0$ in $H^0(X, \mathcal{O}_X(mD))$, and let

$$M = \frac{1}{m} \operatorname{div}(s)$$

be its normalized effective rational divisor. If $M = 0$, the section already trivializes $\mathcal{O}_X(mD)$. Suppose that $M \neq 0$.

The birational reduction requires projective contractions with Kähler targets. We prepare them by two constructions. First, Proposition 3.2 contracts the entire null locus of a nef and big threefold class when that locus is a finite union of curves; its initial target is normal compact

analytic. Second, conormal direct-image vanishings and analytic thickenings extend a contraction of a prime floor to the ambient space in Proposition 4.3. These two constructions, combined with the specified relative MMP, rationality, descent, and positivity inputs, give the threefold contraction statement Proposition 3.1. It is used first on the floors of the fourfold program and later on lower strata in the boundary comparison argument.

The surrounding birational results are the pseudoeffective terminal canonical threefold program of Höring–Peternell, the logarithmic threefold program of Das–Hacon, and the crepant dlt models and supported fourfold construction of Das–Hacon–Păun [33, 11, 12]. The proof below states the precise imported inputs alongside the constructions just described.

On a log resolution, we raise to one the coefficients of the strict transforms of $\text{Supp } M$ and of the exceptional primes. The resulting adjoint has an effective representative supported on its entire reduced floor. The supported program then gives the model of Proposition 5.1: a normal ordinary $\mathbb{Q}$ -factorial compact Kähler dlt fourfold (V, B) with effective rational boundary and

$$A = K_V + B, \quad P \sim_{\mathbb{Q}} A, \quad S = \lfloor B \rfloor, \quad \text{Supp } P = \text{Supp } S,$$

where A is an analytically nef actual $\mathbb{Q}$ -Cartier adjoint, P is an effective nonzero rational $\mathbb{Q}$ -Cartier divisor, and $\kappa(V, A) = 0$. The equivalence is an isomorphism of actual rational holomorphic lines. Nonvanishing of P follows from exceptional negativity: if its support disappeared, the pullback of M would be a nonzero effective nef exceptional divisor.

The model also has the special projective resolution in Lemma 5.6. Its strict boundary and exceptional support have globally smooth distinct SNC components, its exceptional crepant coefficients are below one, and it is generically an isomorphism on the image of every irreducible component of an intersection of distinct strict floor components. The first boundary result, Theorem 6.1, proves that the already existing restriction $A|_S$ is semiample for this model. It uses this resolution condition and requires no ambient section.

Compatible sections on the whole floor

The normal components and lower strata carry actual adjoint lines. Lower-dimensional abundance, applied after the small models specified below, makes these lines semiample; the threefold input is Das–Ou’s lc abundance theorem [14, Corollary 1.3]. We must choose sections whose iterated residue restrictions agree through every intersection, so that they descend to sections on the reduced space S itself.

The restriction criterion sometimes extends a section without a comparison. When it requires a comparison, a perturbed program on the lower stratum reduces the condition, in a sufficiently divisible common degree, to the two coefficient-one markings on a projective-line fiber of a Mori contraction. They give a birational residue comparison, called a link. On a common projective graph, a link identifies the compatible invertible meromorphic adjoint subsheaves; this equality transports the actual residue sections and their products. The two markings can belong to distinct primes or to one degree-two branch, whose normal finite Stein space carries the exchange involution.

Fujino’s induction by pre-admissible and admissible sections and finite symmetrization is the ancestry of this compatibility construction [19, Sections 2–4]. Kollár’s sources and links describe the related algebraic configuration with two disjoint distinguished sections of a projective-line fiber [37, Definition 9, Theorem 10, and Proposition 14]; the last proposition records the compatibility of even iterated residues along these links.

For nonprojective surface strata, the links carry an additional geometric constraint. Restrictions of one Kähler form on V induce positive-square classes on their smooth minimal surface models.

A link from a three-dimensional stratum transports these classes modulo the real span of divisor classes. A returning chain begins and ends on the same surface stratum. After a common Cartier degree q is chosen, the image of all returning chains on the space of qk -plurisecions is finite for each fixed integer $k > 0$. The groupoid may have infinitely many arrows: its surface arrows are generated only by these links and their inverses, while point and curve comparisons are treated separately. Finite norm products impose the required invariance, and finite hyperplane avoidance chooses sections nonzero at prescribed points. Their residue agreement through all intersections gives generators on the whole floor.

This common-class constraint is essential. Nonprojective K3 surfaces can have automorphisms acting by a non-root-of-unity scalar on the holomorphic two-form [39, Theorems 3.5 and 4.1]. Swapnajt Das's recent preprint states abundance for compact Kähler semi-log-canonical threefolds [15, Theorem 1.3]. Its Lemmas 7.2–7.3 and Corollary 7.4 contain a closely related positive-class and ruling mechanism. We give the explicit residue construction here, including the actual meromorphic adjoint equality, the normalized degree-two branch, and the uniform fixed-degree bound for the link action. The cited surface mechanism is prior art; the stated semi-log-canonical theorem is not an input to this proof.

Lifting from the supported boundary

The second boundary result is a dimension-free statement for an arbitrary standard dlt pair. The special resolution used in the fourfold floor-generation application is not part of its data.

Theorem 1.2 (Supported lifting). *Let (V, B) be a normal irreducible compact Kähler dlt pair of positive dimension, with effective rational boundary and actual $\mathbb{Q}$ -Cartier adjoint $A = K_V + B$. Suppose that an effective nonzero rational $\mathbb{Q}$ -Cartier divisor P satisfies*

$$P \sim_{\mathbb{Q}} A, \quad \text{Supp } P = \text{Supp } \lfloor B \rfloor.$$

If the actual restriction $A|_S$ to the whole reduced subspace $S = \lfloor B \rfloor$ is semiample, then $\kappa(V, A) \geq 1$.

The support equality places the divisor of the initial section exactly on the boundary where generation is known, and makes the pair klt away from that divisor. The theorem assumes generation on the entire reduced S , including its intersections; it does not assume nefness of A .

Choose an effective Cartier multiple $G = qP$ and an actual identity $\mathcal{O}_V(G) \simeq \mathcal{O}_V(qA)$ such that its restriction to S is generated. A generating system constructs a morphism $f : S \rightarrow T = \mathbb{P}^b$ and an actual identity

$$\mathcal{O}_V(G)|_S \simeq f^* \mathcal{O}_T(1).$$

On an ambient neighborhood $W \subset V$ of each compact fiber, a root pair and a normalized cyclic cover $\pi : Z \rightarrow W$ replace the supported divisor by a reduced Cartier divisor E . Put $I = \mathcal{O}_Z(-E)$, and let $g : E \rightarrow U$ be the induced map over a parameter neighborhood $U \subset T$. The finite lifting target is that, for every integer j and every $k \geq 1$,

$$g_*(I^j / I^{j+k+1}) \longrightarrow g_*(I^j / I^{j+k})$$

is an epimorphism of sheaves of complex vector spaces on U . The quotients are viewed on the underlying space of E , and the neighborhood used to lift a particular germ may depend on j and k . Thus the assertion lifts section germs by one finite order using the boundary map g .

The obstruction to each order is a graded derivation with values in the first cohomology of a normal layer. A second root and a split residue map embed that cohomology as a direct

summand of the residue cohomology of a resolved SNC divisor, retaining classes supported at special parameters. A local differentiation identity turns any nonzero value of the derivation on a base coordinate into a map excluded by the Hodge vanishing on a smooth projective parameter cover. The same identity then kills the remaining derivation. Finally, cyclic invariants give recurring divisorial layers on V with positive twists from $\mathcal{O}_T(1)$. Their sections grow while their higher cohomology and the final loss from $H^1(V, \mathcal{O}_V)$ stay bounded. This forces $\kappa(V, A) \geq 1$.

The direct ancestors of this lifting method are the threefold arguments of Miyaoka and Kawamata. Miyaoka studies an effective pluricanonical divisor on a minimal projective terminal threefold through neighborhood covers and successive thickenings in the numerical-dimension-one case [40, Section 4]; he credits Reid with the suggestion to analyze that divisor [40, p. 220]. Kawamata’s alternative proof of that case uses neighborhood roots, residues, compatible thickenings, and a mixed Hodge complex [35, Section 4]. The proof here establishes the stated compact Kähler dlt lifting theorem.

Apply the two boundary results to the supported model. Generation of $A|_S$ and Theorem 1.2 force positive Iitaka dimension because $P \neq 0$, contradicting $\kappa(V, A) = 0$. Therefore $M = 0$, and the original section s trivializes the actual Cartier line $\mathcal{O}_X(mD)$.

Related manuscripts by OpenAI treat log abundance in the projective characteristic-zero setting [43] and supported boundary lifting [42]. The projective supplement also states rational-boundary semi-log-canonical abundance over $\mathbb{C}$ and a dlt restriction result for a nef adjoint with an effective rational representative whose support contains the floor, in every Cartier degree $m \geq 2$ [43, Corollaries 11.5–11.6]. Those projective statements are separate context; both analytic boundary arguments needed here are proved in this manuscript.

Section 2 fixes the actual-line conventions and a connectedness lemma. Sections 3 and 4 establish the contraction preparations used on fourfold floors and on lower strata. Section 5 constructs the supported model. Sections 6–9 construct compatible sections and prove generation of the existing adjoint restriction on the entire reduced floor. Section 10 constructs root neighborhoods and the split residue map; Section 11 proves the vanishing used for lifting; and Section 12 proves the finite-order surjections and section growth. Section 13 assembles the main theorem.

2 Actual adjoints and analytic conventions

We collect the conventions that keep the argument on the given holomorphic line bundles. All complex spaces are Hausdorff and countable at infinity. A normal connected complex space is irreducible: its locally irreducible components are open as well as closed. All divisors called effective are Weil divisors with nonnegative coefficients.

2.1 Adjoints, residues, and singularities

For a normal complex space X , let $j : X_{\text{reg}} \hookrightarrow X$ be the regular locus and set $\omega_X = j_*\omega_{X_{\text{reg}}}$. For an integral Weil divisor H , $\mathcal{O}_X(H)$ is the rank-one reflexive divisorial sheaf. If Δ is rational and an integer $r > 0$ clears its coefficients, the adjoint index condition is the invertibility of

$$\mathcal{L}_r = (\omega_X^{[r]} \otimes \mathcal{O}_X(r\Delta))^{**}. \quad (2.1)$$

All later degrees are chosen divisible by such an index. Tensor powers of $\mathcal{L}_r$, rather than a separately chosen global canonical divisor, define $\mathcal{O}_X(m(K_X + \Delta))$. Equality in $\text{Pic}(X)_{\mathbb{Q}}$ means an isomorphism of holomorphic line bundles after a common positive multiple. The notation

$P \sim_{\mathbb{Q}} K_X + \Delta$ for a rational $\mathbb{Q}$ -Cartier divisor means precisely $\mathcal{O}_X(mP) \simeq \mathcal{O}_X(m(K_X + \Delta))$ in such a degree.

A local frame of $\mathcal{L}_r$ restricts on the regular locus to a meromorphic r -pluricanonical form with the allowed boundary poles. Pulling that form back differentially along a proper bimeromorphic model $p : W \rightarrow X$ with W smooth defines a rational crepant subboundary G_W by

$$K_W + G_W = p^*(K_X + \Delta). \quad (2.2)$$

This equality records both the rational divisor of the meromorphic map and the actual isomorphism of the pulled-back line. It is independent of the local frame: a change of frame is a holomorphic unit, and both pullbacks change by its pullback. At a prime divisor E on W ,

$$a(E; X, \Delta) = 1 - \text{coeff}_E(G_W) = 1 + \text{coeff}_E(K_W - p^*(K_X + \Delta))$$

is the log discrepancy. We use the same definition on higher smooth models. The pair is lc if all these numbers are nonnegative and klt if they are positive. We use the standard Kollár–Mori dlt notion, as imported for analytic pairs in [12, Definition 2.7(4)]; in particular, a dlt pair is klt away from its coefficient-one floor. Theorem 6.1 uses in addition the globally simple projective resolution supplied by Lemma 5.6 in the fourfold application. Theorem 1.2 uses the standard dlt notion without this additional resolution hypothesis.

On an SNC pair, adjunction to an intersection of distinct coefficient-one components is the iterated meromorphic residue. In a degree divisible by the adjoint index and by two, interchanging two residue operations does not change the resulting pluriform. We always use such even degrees when comparing paths through boundary strata. Equality of the pulled-back invertible subsheaves of meromorphic pluriforms is stronger than an abstract isomorphism of lines: the former fixes the residue transport of sections. This stronger equality is what we mean by a *crepant residue comparison*. It will be proved for every comparison used below.

Reflexive extension is used in the following precise form. An isomorphism between rank-one reflexive sheaves on a normal space, defined away from an analytic subset of codimension at least two, extends uniquely if it is given there by the same meromorphic identification. In particular, a meromorphic pluriadjoint identity transported through a bimeromorphic map that extracts no prime divisor can be tested in codimension one and then extended. An isomorphism merely in numerical cohomology would not support this operation.

We use ordinary global $\mathbb{Q}$ -factoriality: every global Weil divisor has a Cartier multiple, and a reflexive power of the canonical sheaf is invertible. This is the convention of [12, Definition 2.7]. It does not assert factoriality of all analytic germs. The input of Theorem 1.1 has no such factoriality assumption.

2.2 Kähler positivity and section spaces

A Kähler form on a complex space is given by smooth strictly plurisubharmonic local potentials in local embeddings into complex Euclidean spaces. Smooth metrics and their curvature are interpreted in the same way. For a rational line represented by L_r , we use the metric criterion for analytic nefness stated in the introduction. It is independent of the index and of the fixed Kähler form. Restriction to a closed analytic subspace preserves the inequalities and hence nefness. For a holomorphic map from a compact Kähler space, pullback also preserves nefness: after pulling back the metric inequality, bound the pulled-back fixed form by a constant multiple of a fixed Kähler form on the source. This argument will be used for resolutions and for restrictions to possibly

singular strata. Curve intersection tests occur only in the relative projective MMP calculations where their use is justified.

For an actual rational line D , its Iitaka dimension is $-\infty$ if all positive Cartier powers have no sections; otherwise it is the maximum dimension of the meromorphic images of their complete systems. On an irreducible compact space, two linearly independent sections of one line have a nonconstant meromorphic quotient. Consequently, if $\kappa(D) = 0$, every Cartier degree has at most one independent section. Conversely, unbounded dimensions of the section spaces of positive powers force $\kappa(D) \geq 1$: one of those spaces has two independent sections.

We will repeatedly use two elementary consequences of normality. If $p : Y \rightarrow X$ is a proper bimeromorphic morphism between normal spaces, then $p_*\mathcal{O}_Y = \mathcal{O}_X$. Hence, for a line L on X ,

$$H^0(Y, p^*L) = H^0(X, L).$$

If p^*L is generated, then L is generated: at any $x \in X$ choose $y \in p^{-1}(x)$ and a pulled-back section nonzero at y . The corresponding section of L is nonzero at x . If a positive multiple of D has a nowhere-vanishing section, that section is an actual isomorphism $\mathcal{O}_X \simeq \mathcal{O}_X(mD)$.

The second consequence concerns exceptional divisors. If $E \geq 0$ is p -exceptional and integral, then

$$p_*\mathcal{O}_Y(E) = \mathcal{O}_X. \quad (2.3)$$

Indeed, a section is a meromorphic function on X that is holomorphic away from the codimension-at-least-two image of E ; normal Hartogs extension makes it holomorphic everywhere. The same statement applies to rational exceptional divisors after clearing their denominators. Combined with projection formula, it identifies the section spaces in the exceptional comparisons below.

2.3 A connectedness and rationality lemma

The following sheaf calculation is used both for the conormal layers of a prime floor and for descent of compatible boundary sections. Its klt case also supplies rationality and the Cohen–Macaulay property for the strata used later.

Lemma 2.1. *Let (T, B_T) be a normal irreducible effective lc analytic pair with actual $\mathbb{Q}$ -Cartier adjoint, and let $p : U \rightarrow T$ be a projective resolution with smooth source whose crepant subboundary G has SNC support. Put*

$$R = G^{\leq 1}, \quad H = \lceil -G^{<1} \rceil.$$

Then H is effective and exceptional, has no component in common with R , and

$$p_*\mathcal{O}_R = \mathcal{O}_{p(R), \text{red}}. \quad (2.4)$$

In particular $R \rightarrow p(R)$ has connected fibers. If the pair is klt, then T has rational singularities and is Cohen–Macaulay.

Proof. The strict boundary coefficients belong to $[0, 1]$, so positive coefficients of H occur only on exceptional primes. Coefficient by coefficient,

$$H - R = -\lfloor G \rfloor, \quad (H - R) - (K_U + \{G\}) = -p^*(K_T + B_T).$$

The fractional boundary $\{G\} = G - \lfloor G \rfloor$ is effective SNC with coefficients below one. The displayed adjoint difference is relatively nef and big for the bimeromorphic p : it is relatively trivial, and the generic relative dimension is zero. Analytic relative Kawamata–Viehweg vanishing

[12, Theorem 2.41] gives $R^i p_* \mathcal{O}_U(H - R) = 0$ for $i > 0$. Exceptional Hartogs extension gives $p_* \mathcal{O}_U(H) = \mathcal{O}_T$. The sequence for the Cartier divisor R therefore yields a surjection

$$\mathcal{O}_T \longrightarrow p_* \mathcal{O}_R(H).$$

It factors through $p_* \mathcal{O}_R$. Multiplication by the canonical section of H embeds $\mathcal{O}_R$ into $\mathcal{O}_R(H)$, because R is reduced SNC and shares no component with H . The surjection consequently makes that embedding an isomorphism after pushforward and makes $\mathcal{O}_T \rightarrow p_* \mathcal{O}_R$ surjective. Its kernel is exactly the reduced ideal of the proper image $p(R)$: a holomorphic function vanishes after restriction to R precisely when it vanishes on that image. This proves (2.4); Stein factorization gives connected fibers.

For the last assertion, take a klt resolution, so $R = 0$. The same vanishing and Hartogs calculation identify $Rp_* \mathcal{O}_U(H) \simeq \mathcal{O}_T$. The factorization

$$\mathcal{O}_T \longrightarrow Rp_* \mathcal{O}_U \longrightarrow Rp_* \mathcal{O}_U(H) \simeq \mathcal{O}_T$$

is the identity, as can be checked on degree zero. It splits the first map. Apply proper coherent duality. Canonical higher direct images of the resolution vanish: the canonical torsion-freeness theorem [21, Theorem 2.9 and Proposition 2.11] applies locally on the base, and these higher images vanish generically for a bimeromorphic map. The required local Kählerness follows by restricting to a small Stein base neighborhood and combining a relative ample metric with a pulled-back strictly plurisubharmonic potential. Thus the dual split surjection is the trace

$$p_* \omega_U[\dim U] \longrightarrow \omega_T^\bullet,$$

where $\omega_T^\bullet$ is the dualizing complex. It follows that this complex is concentrated in degree $-\dim U$. On that cohomology sheaf the trace is also injective: its source is torsion-free and the map is generically an isomorphism. Hence it is an isomorphism. Biduality gives $Rp_* \mathcal{O}_U = \mathcal{O}_T$. This is rational singularity, and the concentration of the dualizing complex is the Cohen–Macaulay property. $\square$

3 Threefold contractions for the boundary argument

The boundary argument uses the following specialization of Das–Hacon–Păun’s threefold contraction theorem [12, Theorem 5.5]. It supplies contractions both on the floors of the supported fourfold program and on lower strata in the restriction argument. Throughout these preparations, a real $(1, 1)$ -class on a singular space is a Bott–Chern class defined by smooth local potentials.

Proposition 3.1 (Threefold contraction of an adjoint plus a Kähler class). *Let (Z, Γ) be a normal connected compact Kähler klt threefold with effective rational boundary and actual $\mathbb{Q}$ -Cartier adjoint $A = K_Z + \Gamma$. Suppose*

$$\alpha = c_1(A) + [\omega] \quad \text{is nef}, \quad \omega \quad \text{is Kähler.} \tag{3.1}$$

Then there is a projective surjection $f : Z \rightarrow Y$ with $f_ \mathcal{O}_Z = \mathcal{O}_Y$, where Y is normal compact Kähler with rational singularities, and a Kähler class $[\omega_Y]$ satisfying $\alpha = f^*[\omega_Y]$. The actual rational line $-A$ is f -ample. A compact curve is contracted exactly when its α -degree is zero.*

The proposition assumes no bigness of α , pseudo-effectivity of A , or factoriality of Z . Its proof is given in Subsection 4.4, after the required constructions. The separate exposure input [12,

Corollary 5.3] supplies a class of the form (3.1) exposing any negative extremal ray in the cited threefold cone theorem. In that case the proposition contracts exactly the curves in that ray.

The preparation has three parts. First we contract the entire finite curve null locus of a nef and big class, obtaining a normal compact analytic target. Next, conormal direct-image vanishings and analytic thickenings extend a contraction of a prime floor to its ambient space. Finally, these two constructions combine with the named projective MMP, descent, positivity, and nonbig inputs to prove the proposition and its floor restriction. The Kähler target in the proposition is part of this combined argument, beyond the finite-null construction alone.

3.1 Contraction of the full finite null locus

We first prove the finite-null contraction assertion of [11, Proposition 6.2] used in the nef and big cases. For an irreducible positive-dimensional analytic subspace V , its top intersection with such a class is computed on a resolution of V . Write

$$\text{Null}(\alpha) = \bigcup_{\substack{V \subset X \text{ irreducible} \\ \dim V > 0, \alpha^{\dim V} \cdot V = 0}} V.$$

Proposition 3.2 (A finite null locus). *Let X be a normal connected ordinary $\mathbb{Q}$ -factorial compact Kähler threefold with klt singularities. Let α be a nef and big real $(1,1)$ -class. Suppose that $C = \text{Null}(\alpha)$ is a finite union of curves. There is a proper bimeromorphic morphism $\phi : X \rightarrow X'$ to a normal compact analytic space such that ϕ is an isomorphism on $X \setminus C$, and the underlying sets of its nontrivial fibers are exactly the connected components of C .*

Proof. The assertion is the identity map if C is empty. Otherwise give C its reduced structure and write $C_1, \dots, C_s$ for its irreducible components.

We will construct a positive line on a resolution and use it to give the conormal of the full analytic inverse image a positive presentation. Grauert's contraction criterion will then apply to that conormal.

An exceptional divisor and one actual positive line

Choose a projective resolution $\pi : \widehat{X} \rightarrow X$ with smooth source which is an isomorphism over X_{reg} and has pure divisorial exceptional locus [12, Theorem 2.13]. Compactness makes the resolution a finite composition of blowups. Its source is compact Kähler: metrics of a relatively ample line can be patched using a partition of unity from the base, preserving positivity on the vertical tangent spaces, and a sufficiently large pulled-back Kähler form makes the curvature positive in every direction. Compactness supplies one constant for this last step.

Put $\beta = \pi^*\alpha$. Pullback of the metric nef approximations shows that β is nef. We check the positive top intersection needed on the smooth source. By bigness choose a Kähler current $T_X \in \alpha$ with $T_X \geq \omega_X$, where ω_X is a Kähler form on X . Pulling back its local plurisubharmonic potentials gives

$$T = \pi^*T_X \in \beta, \quad T \geq \eta := \pi^*\omega_X.$$

The pulled potentials are not identically minus infinity, since $\widehat{X}$ dominates X . The form η is smooth and semipositive. Fix a Kähler form $\kappa_{\widehat{X}}$, and for $\delta > 0$ choose a positive form $b_\delta \in \beta + \delta[\kappa_{\widehat{X}}]$. Wedge the positive current $T - \eta$, successively, with b_δ^2 , $\eta \wedge b_\delta$, and η^2 . Integration and $\delta \downarrow 0$ give

$$\beta^3 \geq \beta^2[\eta] \geq \beta[\eta]^2 \geq [\eta]^3 > 0. \quad (3.2)$$

The last inequality holds because η is positive on a nonempty open set. All these products are on the smooth compact $\widehat{X}$; only a current wedged with smooth semipositive forms was used.

For an irreducible d -dimensional positive-dimensional subspace $V \subset \widehat{X}$, projection of its fundamental cycle gives

$$\beta^d \cdot V = \begin{cases} 0, & \dim \pi(V) < d, \\ \deg(V/\pi(V)) \alpha^d \cdot \pi(V), & \dim \pi(V) = d. \end{cases} \quad (3.3)$$

The second case maps a null subspace into C . In the first case its image is contained in the image of the exceptional locus, hence in $\text{Sing } X$. Normality of the threefold makes that singular locus at most one-dimensional. Thus every null subspace of β maps into the set $C \cup \text{Sing } X$ of dimension at most one.

The smooth nef positive-volume criterion [12, Lemma 2.35] makes β big by (3.2). Collins–Tosatti [8, Theorem 1.1] identifies its non-Kähler locus on $\widehat{X}$ with $\text{Null}(\beta)$. Choose the Kähler current $S \in \beta$ supplied by [3, Section 2.5.1 and Theorem 3.17(ii)]. In that convention its analytic singularities are defined by one coherent ideal $\mathfrak{a}$ on $\widehat{X}$ and one coefficient $c > 0$, with $E_+(S) = V(\mathfrak{a}) = \text{Null}(\beta)$ as sets. Principalize this global ideal by a finite sequence $g : Y \rightarrow \widehat{X}$ of blowups with smooth centers, using [12, Remark 2.14]. The source Y is smooth and compact Kähler. If $\mathfrak{a}\mathcal{O}_Y = \mathcal{O}_Y(-D_{\mathfrak{a}})$, the logarithmic presentation and Poincaré–Lelong, in the cited normalization, give

$$g^*S = \theta + [G], \quad G = cD_{\mathfrak{a}} \geq 0,$$

where G is a finite real divisor supported over the singular set and θ is a global smooth closed form. If $S \geq \epsilon \kappa_{\widehat{X}}$, the equality off G and continuity show that $\theta \geq \epsilon g^* \kappa_{\widehat{X}}$ everywhere.

Discarding blowups along Cartier centers, which are isomorphisms, choose for this finite composition an effective g -exceptional integral divisor F with $\mathcal{O}_Y(-F)$ g -ample. Such a divisor is obtained from the exceptional tautological divisors, giving earlier pullbacks sufficiently large positive weights. A smooth representative v of $c_1(F)$ can be chosen so that $g^* \kappa_{\widehat{X}} - \delta v$ is Kähler for some $\delta > 0$: the curvature of $-F$ is positive on the vertical kernels, and compactness bounds the horizontal and mixed terms. If g is an isomorphism, take $F = 0$. Consequently $\theta - \epsilon \delta v$ is Kähler. With $p = \pi g$, we obtain

$$p^* \alpha = \kappa_0 + [E], \quad \kappa_0 = [\theta - \epsilon \delta v] \text{ Kähler}, \quad E = G + \epsilon \delta F \geq 0. \quad (3.4)$$

Every prime of G maps into $C \cup \text{Sing } X$, and every prime of F maps to a set of codimension at least two in $\widehat{X}$. Hence every prime of E is p -exceptional. The morphism p is projective: projective morphisms compose over the compact base X by [12, Remark 2.11].

Openness of the Kähler cone on the smooth Y permits a nonnegative rational divisor E' with the same exceptional prime support and coefficients sufficiently close to those of E so that

$$\kappa' = p^* \alpha - [E'] \text{ is Kähler.}$$

Choose $r > 0$ clearing its denominators, and put

$$H = rE' \geq 0, \quad L = \mathcal{O}_Y(-H). \quad (3.5)$$

Thus H is an integral exceptional Cartier divisor and L is an actual holomorphic invertible sheaf.

We verify relative ampleness on every full fiber of p . Let a be a smooth representative of α with local potentials on X , let Θ_L be the curvature of a smooth metric on L , and choose a Kähler representative for κ' . Equality of Bott–Chern classes on Y gives a global smooth function φ with

$$r\kappa' = rp^*a + \Theta_L + dd^c \varphi.$$

Adjust the metric of L by φ . On a base chart $a = dd^c \rho$, adding $r\rho \circ p$ to its local weights makes their curvature $r\kappa'$. On a fiber, that added function is constant. The restricted weights therefore have strictly plurisubharmonic ambient extensions. The same weights define positivity on the full possibly nonreduced fiber: nilpotents do not change their values or the absolute values of transition units. The compact analytic positivity criterion and the proper fiberwise criterion [22, Definition 2.8, Corollary 1.12, and Remark 3.2] now show that L is p -ample.

Numerical ampleness on the projective inverse image

The compact reduced curve C is projective, even if it is reducible or disconnected. Indeed, choose one point on each irreducible component which is smooth on all of C and is outside the other components. Their sum is an effective Cartier divisor. Its canonical section is nonzero and has a zero on each positive-dimensional irreducible subspace of C . The positivity lemma in Section 3.4 of [26], followed by Section 3.2, Satz 2, makes its line positive and ample. Fix a very ample line A on C .

Put

$$D = (p^{-1}C)_{\text{red}}, \quad f_D : D \rightarrow C, \quad L_D = L|_D.$$

The reduced base change of p is projective over C . Since C is compact and projective and the final base is a point, the compact-base projective composition assertion makes D projective [12, Remark 2.11]. Chow and GAGA allow D and its holomorphic invertible sheaves to be treated as a projective scheme over $\mathbb{C}$ [50]. The space D can be reducible, nonnormal, and of mixed dimension. The restriction L_D is the restriction of the actual line L ; a component of D can lie inside H .

Choose an ample line A_D on D with a smooth curvature form Θ_{A_D} . For one integer $q > 0$ sufficiently large,

$$qr\kappa'|_D - \Theta_{A_D}$$

is positive. To see this on the singular space, extend the local weights of A_D to finitely many relatively compact ambient charts covering D , and bound their Hessians by the ambient Kähler form. This is the local-potential positivity convention. If $\Gamma \subset D$ is an integral curve, its image is a point or one of the C_i ; projection on the normalizations gives $(p^*\alpha) \cdot \Gamma = 0$. Therefore

$$\deg_\Gamma(L_D^q \otimes A_D^{-1}) = qr\kappa' \cdot \Gamma - \Theta_{A_D} \cdot \Gamma > 0. \quad (3.6)$$

The class of $L_D^q \otimes A_D^{-1}$ is consequently in the dual of the closed cone of curves of the projective scheme D . The numerical space is finite-dimensional. Kleiman's line-bundle criterion [20, Section 4.8 and Theorem 4.10] says that the ample A_D is positive on every nonzero class of the closed curve cone. Its minimum on a compact unit slice is positive, so its class is in the interior of the dual cone. The sum of an element of the dual cone and an element of its interior is again in the interior. Thus L_D^q , and hence L_D , is ample. The ample subtraction in (3.6) supplies the uniform margin; strict positivity on individual curves alone would not give this conclusion.

The ample line L_D is the positivity needed for the contraction. The remaining ideal calculations transfer it to a positive presentation of the conormal of the full analytic inverse image, including its nonreduced structure.

The full scheme inverse image

Let $\mathfrak{c} = \mathcal{I}_C$, and define

$$J = \mathfrak{c}\mathcal{O}_Y = \text{im}(p^*\mathfrak{c} \rightarrow \mathcal{O}_Y), \quad Z = V(J) = Y \times_X C, \quad f : Z \rightarrow C. \quad (3.7)$$

The space Z has its full analytic subspace structure, including possible multiplicities and embedded components, and $Z_{\text{red}} = D$. Put $K_m = p_* L^m$ for $m \geq 1$. It is a coherent ideal of $\mathcal{O}_X$: effectivity of H gives $L^m \subset \mathcal{O}_Y$, and $p_* \mathcal{O}_Y = \mathcal{O}_X$ by normality and proper bimeromorphicity.

We use relative Serre vanishing in the following exact form. For a proper holomorphic map, a relatively ample invertible sheaf, a fixed coherent twist, and a compact part of the base, its positive higher direct images vanish there for every sufficiently large tensor power. This is [2, Definitions 2.1–2.2 and Proposition 2.5]; it permits nonreduced complex spaces and assumes they are separated and countable at infinity. Those conditions hold for the present spaces and the relatively compact opens used next.

On a base open U choose generators $g_1, \dots, g_t$ of $\mathfrak{c}$. Let Q_U be the coherent kernel in

$$0 \longrightarrow Q_U \longrightarrow \mathcal{O}_{p^{-1}U}^{\oplus t} \xrightarrow{(p^*g_1, \dots, p^*g_t)} J|_{p^{-1}U} \longrightarrow 0.$$

Choose finitely many such opens with compact subsets whose interiors cover C . Relative Serre vanishing for $Q_U \otimes L^m$ on these compact subsets and for $J \otimes L^m$ over C gives one integer M for which the needed $R^1 p_*$'s vanish for every $m \geq M$. Tensor the displayed presentation by L^m and push forward. Inside K_m , the image of $K_m^{\oplus t}$ is exactly multiplication by the g_i . Hence, at every stalk over C ,

$$p_*(J \otimes L^m) = \mathfrak{c}K_m \quad (m \geq M). \quad (3.8)$$

This is an image computation; it uses no projection formula for the possibly nonflat ideal $\mathfrak{c}$.

Push forward the exact sequence

$$0 \longrightarrow J \otimes L^m \longrightarrow L^m \longrightarrow L^m|_Z \longrightarrow 0.$$

The other R^1 vanishing and (3.8) give the canonical isomorphism of coherent $\mathcal{O}_C$ -modules

$$P_m := K_m / \mathfrak{c}K_m = K_m \otimes_{\mathcal{O}_X} \mathcal{O}_C \xrightarrow{\sim} f_*(L^m|_Z), \quad m \geq M. \quad (3.9)$$

In particular the full inverse Z , rather than only its reduction, occurs in this equality. No flatness or Cartier assumption on J has been used.

A positive presentation of the conormal

Let $\mathfrak{n} = \ker(\mathcal{O}_Z \rightarrow \mathcal{O}_D)$. This coherent nilradical satisfies $\mathfrak{n}^e = 0$ for one e , by local Noetherianity and compactness of Z . For

$$T_m = L^m|_Z \otimes f^* A^{-2},$$

the filtration $\mathfrak{n}^j T_m$ has quotients

$$(\mathfrak{n}^j / \mathfrak{n}^{j+1}) \otimes L_D^m \otimes f_D^* A^{-2}, \quad 0 \leq j < e.$$

Their coefficient sheaves are fixed and coherent on the projective D . Serre vanishing for the ample L_D kills their H^1 for every sufficiently large m . Induction on this finite filtration then gives

$$H^1(Z, T_m) = 0 \quad (m \gg 0).$$

Only H^1 of the subobject and quotient is needed at each step, so the embedded nilpotent layers are included.

Write $F_m = f_*(L^m|_Z)$. Projection formula with the invertible A^{-2} and the low-degree Leray sequence give

$$H^1(C, F_m \otimes A^{-2}) = H^1(C, f_* T_m) \hookrightarrow H^1(Z, T_m) = 0. \quad (3.10)$$

This does not assert vanishing of $R^1 f_* T_m$. Embed $i : C \hookrightarrow \mathbb{P}^n$ by A . By GAGA the coherent sheaf F_m is algebraic. The sheaf $i_*(F_m \otimes A^{-1})$ is 0-regular: its degree-one condition is (3.10), and all higher conditions vanish since its support has dimension at most one. Castelnuovo–Mumford regularity [54, Tag 08A8, Lemma 33.35.12] gives a surjection

$$A^{\oplus n_m} \longrightarrow F_m = P_m \longrightarrow 0 \quad (m \gg 0). \quad (3.11)$$

Coherent torsion is allowed in this argument.

Fix one such m . We first check that $C \subset V(K_m)$. For each C_l , the projective D contains an integral curve Γ dominating C_l : take a component dominating C_l and cut by sufficiently general ample hyperplanes, taking the component itself when it is a curve. From $(p^* \alpha) \cdot \Gamma = 0$ and $\kappa' \cdot \Gamma > 0$ we obtain $E' \cdot \Gamma < 0$. If Γ were not contained in the support of the effective Cartier divisor H , its canonical section on the normalization of Γ would give nonnegative degree, a contradiction. Thus an irreducible component of H contains Γ . Its image contains C_l and has dimension at most one by exceptionality, so that image equals C_l . At every point of C_l a unit germ on X pulls back to a unit and cannot vanish along that component. Hence K_m is a proper ideal at every point of C .

Outside $p(H)$, the ideal K_m equals $\mathcal{O}_X$. The image of each exceptional prime has codimension at least two in the normal threefold. It follows that $V(K_m)_{\text{red}}$ is a finite union of curves and points. Since it contains every C_l , each C_l is one of its irreducible curve components. Let B be the union of its irreducible components other than the selected curves C_l , and put $\mathfrak{b} = \mathcal{I}_B$. Define

$$I = (K_m : \mathfrak{b}^\infty).$$

This is a coherent ideal with one global finite saturation exponent. Indeed, for each integer $j \geq 0$, the ideal $(K_m : \mathfrak{b}^j)$ is the kernel of the coherent morphism $\mathcal{O}_X \rightarrow \mathcal{H}om(\mathfrak{b}^j, \mathcal{O}_X/K_m)$. The ascending chain stabilizes at each stalk by Noetherianity. Equality of two successive coherent ideals then holds on a neighborhood, and the colon recursion gives all later equalities there. Compactness supplies one exponent for all of X .

On $X \setminus B$ we have $I = K_m$. On $X \setminus C$, the analytic Nullstellensatz puts a local power of $\mathfrak{b}$ inside K_m , so $I = \mathcal{O}_X$. Closure of the dense parts $C_l \setminus B$ now gives

$$V(I)_{\text{red}} = C, \quad \sqrt{I} = \mathfrak{c}.$$

The set $B \cap C$ is finite. The induced map

$$P_m = K_m / \mathfrak{c} K_m \longrightarrow Q := I / \mathfrak{c} I$$

is therefore an isomorphism away from finitely many points of C . After tensoring by A^{-2} , its kernel and cokernel are still point-supported. Splitting through its image, the two long exact sequences, (3.10), and the vanishing of positive cohomology of point-supported sheaves give $H^1(C, Q \otimes A^{-2}) = 0$. Here a coherent sheaf on the projective curve has no H^2 . The same regularity argument as above gives

$$A^{\oplus N_Q} \longrightarrow Q \longrightarrow 0. \quad (3.12)$$

Since $I \subset \mathfrak{c}$, right exactness of restriction gives the actual conormal identity

$$Q = I / \mathfrak{c} I = (I / I^2) \otimes_{\mathcal{O}_X} \mathcal{O}_C. \quad (3.13)$$

Its support is all of C . At each point there, I is a nonzero proper finitely generated ideal and $\mathfrak{c}$ lies in the maximal ideal, so Nakayama gives $I / \mathfrak{c} I \neq 0$. Nonzeroness of the ideal also follows from its one-dimensional zero set in the normal threefold.

The coherent conormal contraction criterion

For a coherent ideal, Grauert's associated normal linear space is defined by its local linear relations. Restricting a finite presentation of I to C presents the module in (3.13). Thus the reduced associated space in Sections 3.6–3.7 of [26] is

$$N_I = (\text{Specan}_C \text{Sym } Q)_{\text{red}}.$$

This definition does not require Q to be locally free. The surjection (3.12) embeds N_I as a closed reduced linear subspace of $\text{Tot}((A^{-1})^{\oplus N_Q})$.

Put $V_A = H^0(C, A)^*$. The very ample embedding $C \hookrightarrow \mathbb{P}(V_A)$, where $\mathbb{P}(V_A)$ parametrizes lines, identifies A^{-1} with the restricted tautological line. Consequently $\text{Tot}((A^{-1})^{\oplus N_Q})$ is closed in $C \times V_A^{N_Q}$. Its projection to $V_A^{N_Q}$ is proper because C is compact, and it is an analytic embedding off the zero section: on the open set where its j -th vector is nonzero, projectivization to that vector's line, followed by the inverse of the embedding of C on its image, recovers the base point holomorphically. The squared Euclidean norm of this projection is strictly plurisubharmonic off the zero section. Its positive sublevels are relatively compact neighborhoods of that section with strongly pseudoconvex boundaries; scalar multiplication gives a nonzero radial derivative at every positive level. Restricting them to N_I proves exactly the weak negativity of this reduced linear space, including when C is singular or disconnected and Q has torsion.

Grauert's Section 3.7, Satz 8, applies to any coherent ideal with the given compact zero set and with weakly negative associated normal linear space [26]. It does not require the ideal to be radical or locally principal. The ambient X is reduced and normal, and its compact zero set C is a union of curves with no isolated points. The criterion therefore makes C exceptional. Section 2, Definition 3 and Satz 5, of that source give a global proper surjective holomorphic map $\phi : X \rightarrow X'$ with $\phi(C)$ discrete, $\phi^{-1}(\phi(C)) = C$ as underlying sets, and

$$X \setminus C \simeq X' \setminus \phi(C).$$

The Remmert reduction used there has connected fibers. Hence each connected component of C has one image point and two components cannot have the same image. Its normality statement for a normal source (p. 337) makes the Remmert-reduced target neighborhood normal; gluing it to the unchanged normal complement makes X' normal. Compactness of X makes X' compact, and the complement isomorphism makes ϕ bimeromorphic. This proves the proposition. $\square$

The proposition supplies the normal compact analytic contraction and its fiber sets. The later assertions that a target is Kähler and that a class descends to a Kähler class are separate inputs.

Lemma 3.3 (Finiteness when there is no null surface). *Let X be a normal connected compact Kähler threefold, and let α be nef and big. If no irreducible surface has zero top intersection with α , then $\text{Null}(\alpha)$ is a finite union of curves, possibly empty.*

Proof. Use a projective resolution $\pi : \hat{X} \rightarrow X$ with smooth compact Kähler source as above. The class $\beta = \pi^* \alpha$ is nef with positive cube by (3.2). Collins–Tosatti makes $\text{Null}(\beta)$ a proper analytic set.

Each irreducible component Z of this analytic set is itself null. It has positive dimension, since each of its points lies on a positive-dimensional null subspace. Suppose instead that $\beta^{\dim Z} \cdot Z > 0$. Resolve Z projectively. The restricted pulled class on the smooth compact Kähler resolution is nef with positive top intersection, so Collins–Tosatti gives it a proper null set. Choose a point of Z outside the image of that set, the target nonisomorphism locus of the resolution, the singular

locus of Z , and the other components of $\text{Null}(\beta)$. Such a point exists because these are proper analytic subsets of Z . By the definition of $\text{Null}(\beta)$, a null irreducible subspace passes through it. That subspace lies in Z , since the point is on no other component. Its strict transform is null for the restricted class, contradicting the choice of the point.

There are finitely many components of the compact analytic null set. By (3.3), each of its surface components is exceptional, since a nonexceptional one would map to a null surface on X . Thus their images, as well as those of its curve components, have dimension at most one. For any downstairs null curve C_0 , choose an irreducible component V of $\pi^{-1}C_0$ dominating C_0 . Proper surjectivity supplies such a component. If $\dim V = 1$, cycle projection gives $\beta \cdot V = \deg(V/C_0) \alpha \cdot C_0 = 0$; if $\dim V > 1$, it gives zero by dimension drop. Thus V lies in an upstairs null component whose image contains C_0 . Every downstairs null curve is therefore contained in the finite union of images. An irreducible curve in a finite union of curves and points is one of the curve components. There are therefore only finitely many downstairs null curves, as asserted. $\square$

4 Conormal layers and the contraction interfaces

We next prove the ordinary conormal statement used to extend a contraction of one prime floor to an ambient contraction. It concerns divisorial ideals and their actual reflexive powers. We then construct analytic base thickenings from the resulting direct images and apply the analytic blowdown criterion to obtain the ambient contraction.

Proposition 4.1 (Conormal layers of a prime floor). *Let $(X, S + B)$ be an ordinary plt pair, where X is a normal connected compact Kähler space, S is an irreducible effective $\mathbb{Q}$ -Cartier prime, $B \geq 0$ is a rational $\mathbb{Q}$ -Cartier divisor not containing S , and $K_X + S + B$ is an actual $\mathbb{Q}$ -Cartier adjoint. Suppose $\dim S \leq 3$. Write the actual residue adjunctions as*

$$(K_X + S)|_S = K_S + \Theta, \quad (K_X + S + B)|_S = K_S + B_S.$$

Let $\phi : S \rightarrow W$ be a projective surjection to a normal compact analytic space with $\phi_\mathcal{O}_S = \mathcal{O}_W$. Suppose that the actual rational lines $-(K_S + B_S)$ and $-S|_S$ are ϕ -ample. For $k \geq 0$, put*

$$\mathcal{I}_k = \mathcal{O}_X(-kS), \quad \mathcal{E}_k = \mathcal{I}_k / \mathcal{I}_{k+1}.$$

Then S is normal, (S, B_S) is klt, and each $\mathcal{E}_k$ is a rank-one reflexive sheaf on S . For any positive global Cartier index q of S , there is a finite effective rational Weil divisor Ξ_k on S , with $q\Xi_k$ integral, such that

$$\mathcal{E}_k^{[q]} \simeq (\mathcal{O}_X(-kqS)|_S) \otimes \mathcal{O}_S(-q\Xi_k), \quad 0 \leq \Xi_k \leq \Theta \leq B_S. \quad (4.1)$$

*Here $\mathcal{E}_k^{[q]} = (\mathcal{E}_k^{\otimes q})^{**}$ is formed on S , and $\mathcal{O}_S(-q\Xi_k)$ is a divisorial reflexive sheaf, which need not be invertible. Moreover*

$$R^i \phi_* \mathcal{E}_k = 0 \quad (i > 0, k \geq 1). \quad (4.2)$$

Proof. Choose an integer $N > 0$ clearing the actual adjoint index and the Cartier indices of $S + B$. On the common codimension-one locally free open, and hence everywhere by reflexive extension,

$$\omega_X^{[N]} \simeq \mathcal{O}_X(N(K_X + S + B)) \otimes \mathcal{O}_X(-N(S + B)).$$

Thus this reflexive canonical power is an actual line. In particular $K_X + S$ is an actual rational adjoint; no global canonical Weil generator is being chosen. Subtracting the effective $\mathbb{Q}$ -Cartier

boundary from discrepancies shows that (X, S) is plt and X is klt. Ordinary plt adjunction [9, Lemma 5.3] gives normal S and klt (S, B_S) . With compatible residue embeddings, restriction of the canonical section of a Cartier multiple of B gives

$$B_S = \Theta + B|_S, \quad B|_S \geq 0.$$

The coefficient computation below also proves $\Theta \geq 0$. The notation $S|_S$ always denotes the restriction of the actual rational normal line.

The analytic quotient in codimension two

Fix a prime P on S . Near a general point choose a holomorphic equation v for a Cartier multiple mS , and normalize the full finite root cover

$$t^m = v.$$

The root algebra is reduced before normalization. It is free with basis $1, t, \dots, t^{m-1}$ over the normal local domain. Its generic Kummer algebra is reduced in characteristic zero because $v \neq 0$, and freeness makes the map to that generic algebra injective. A nilpotent element must therefore be zero. Finite analytic normalization [34, Part B, Section 4, Corollaries 2–3] gives a finite normal cover $\tau : Y \rightarrow U$ with its complete μ_m -action. It can be disconnected, and we retain all components. The invariant subalgebra of its normalization is $\mathcal{O}_U$: in the total meromorphic algebra the invariants are the meromorphic field of U , and normality of U identifies its integral elements. Thus the quotient is the original normal space.

This cover is étale at every codimension-one point. Off S , the equation is a root of a unit. At a general smooth point of S , write $v = ux^m$ for a unit u ; after choosing a holomorphic root of u , normalization is a disjoint union of the branches $t = \zeta u^{1/m}x$. The lifted floor

$$S_Y = \operatorname{div}(t)$$

is reduced and Cartier. Its prime coefficients are one, and on a normal space the principal divisorial ideal with these coefficients is the radical ideal of their union.

The finite discrepancy formula is valid here in the analytic category. Normalize a base change of a model carrying a tested divisorial valuation. The Hurwitz formula for the corresponding discrete valuation rings and the actual identity $K_Y + S_Y = \tau^*(K_X + S)$ give

$$a(F; Y, S_Y) = e(F/E) a(E; X, S). \quad (4.3)$$

Extensions of the valuation of S are exactly the height-one valuations of the lifted floor. All exceptional log discrepancies remain positive, so the lifted pair is plt. Applying the same formula without the floor shows that the components of Y are klt.

The lifted floor is normal. One can use the ordinary plt adjunction just cited, or the following independent connectedness argument. On a projective SNC resolution of the lifted pair, its coefficient-one locus is the union of the strict floor components, with no exceptional component. Two such components cannot meet: blowing up their intersection would give an exceptional log discrepancy zero. They are therefore disjoint and smooth. Lemma 2.1 gives the equality of their direct-image structure sheaf with that of the reduced floor. Factoring the map through the finite normalization of the floor, and applying normal bimeromorphic Hartogs extension on that normalization, identifies this direct image with the normalization sheaf. The equality forces the

normalization to be an isomorphism. This use of the connectedness lemma is independent of any contraction existence assertion.

Choose a component P' above P and a general point on it. The normal S_Y is smooth there, since P' has codimension one in it, and P' is smooth after a further generic choice. The ambient Y is smooth there as well. Indeed, in its local ring R' the nonzerodivisor t has regular quotient $R'/(t)$. Lifting its minimal generators and adjoining t bounds the embedding dimension of R' by $\dim R'$, so R' is regular.

Remove the finitely many proper fixed loci on P' . At a remaining point its stabilizer $G \subset \mu_m$ fixes P' pointwise. Holomorphic linearization can preserve the smooth flag $P' \subset S_Y \subset Y$: average lifts of an eigenbasis of the cotangent space inside the respective invariant ideals, and use their independent differentials as local coordinates. The coordinates tangent to P' are fixed. The stabilizer has no ineffective element: its action sends t to ζt , and the nonzerodivisor t forces $\zeta = 1$ for an element acting trivially on the local germ. A nonidentity stabilizer element cannot fix a transverse hyperplane, because τ is étale in codimension one. Both transverse characters are consequently faithful. If $r = |G|$, rescaling the generator gives the analytic germ

$$(X, S, P) \simeq ((\mathbb{C}^2, \{x = 0\}), 0) / \mu_r(1, a) \times \mathbb{C}^{\dim X - 2}, \quad \gcd(a, r) = 1. \quad (4.4)$$

The smooth case is $r = 1$. The local Cartier index of S is r : x^r descends, whereas invariance of a unit times x^d , evaluated at the fixed point, forces $r \mid d$.

On the quotient floor let $z = y^r$. Residue of the r -th log-canonical power changes $(dy)^{\otimes r}$ into a nonzero constant times $z^{-(r-1)}(dz)^{\otimes r}$; the tangential factors are unchanged. The actual meromorphic residue embedding therefore gives

$$\text{coeff}_P \Theta = \frac{r-1}{r}. \quad (4.5)$$

This derivation is on the analytic cover and does not infer a higher-dimensional analytic quotient theorem from an algebraic slice.

Depth and the canonical multiplication

The sheaves $\mathcal{S}_k$ are maximal Cohen–Macaulay on X . Here is the index-cover argument at every point of X . On a small open about an arbitrary point, repeat the full normalized root construction using a local equation for a Cartier multiple mS . Codimension-one étaleness and the discrepancy calculation above do not require the generic choice on P . The klt components of this full cover Y are Cohen–Macaulay by the last assertion of Lemma 2.1; its proof uses relative vanishing, canonical torsion-freeness, and coherent duality. The total meromorphic Kummer algebra has basis $1, t, \dots, t^{m-1}$, even when it splits. Its character- j holomorphic summand on this open is

$$\mathcal{O}_X(jS)t^j \quad (0 \leq j < m).$$

Indeed regularity in each height-one discrete valuation ring requires the coefficient to have order at least $-j$ along S and at least zero elsewhere. Normality supplies the equality of these sheaves. Up to tensoring by the actual Cartier line of a multiple of mS , these are all the $\mathcal{S}_k$. Averaging over μ_m splits them as modules.

For the depth assertion, let $R = \mathcal{O}_{X,x}$ have dimension d and choose a system of parameters. Every normalization component dominates the base open: the generic Kummer algebra is a separable product of fields, and the full normalization retains all of those components. Thus each local ring of Y above x has dimension d , and finiteness makes the extended parameter ideal

primary for its maximal ideal. Cohen–Macaulayness upstairs makes the same parameter sequence regular. Testing at all upstairs maximal ideals makes it regular on the finite semilocal R -module $(\tau_* \mathcal{O}_Y)_x$. It remains regular on each split summand, and hence on $\mathcal{I}_k$. No flatness of the finite cover is used.

Divisor orders give $\mathcal{I}_j \mathcal{I}_\ell \subset \mathcal{I}_{j+\ell}$; also $\mathcal{I}_1$ is the reduced ideal of S . Thus $\mathcal{E}_k$ is a coherent sheaf on S . The depth lemma in

$$0 \longrightarrow \mathcal{I}_{k+1} \longrightarrow \mathcal{I}_k \longrightarrow \mathcal{E}_k \longrightarrow 0$$

gives depth at least $\dim X - 1$ along S . The quotient has generic rank one on the irreducible S , so its support is all of S and has local dimension $\dim X - 1$. Depth cannot exceed this dimension, and equality follows. Thus it is Cohen–Macaulay on S , has no embedded associated component, and is torsion-free and S_2 . Normality of S makes it rank-one reflexive.

In (4.4), exactness of finite-group invariants identifies

$$\mathcal{E}_k = (x^k \mathbb{C}\{y, \mathbf{u}\})^{\mu_r},$$

where $\mathbf{u}$ denotes the fixed tangential coordinates. Let $j_{k,P}$ be the unique integer with

$$0 \leq j_{k,P} < r, \quad k + a j_{k,P} \equiv 0 \pmod{r}.$$

Each invariant series factors as $x^k y^{j_{k,P}} h(y^r, \mathbf{u})$. Hence this is a free rank-one module at the general point of P , and its r -th multiplication into the degree kr layer is

$$(x^k y^{j_{k,P}})^r = (x^r)^k z^{j_{k,P}}.$$

Relative to the local frame $(x^r)^k$ of the actual line $\mathcal{O}_X(-krS)|_S$, its vanishing order is $j_{k,P}$. By (4.5),

$$\xi_{k,P} := \frac{j_{k,P}}{r} \quad \text{satisfies} \quad 0 \leq \xi_{k,P} \leq \text{coeff}_P \Theta. \quad (4.6)$$

Now choose the global Cartier index q in the statement. Multiplication of the divisorial filtration defines a canonical global map

$$\mu_k : \mathcal{E}_k^{\otimes q} \longrightarrow \mathcal{E}_{kq} = \mathcal{O}_X(-kqS)|_S =: \mathcal{A}_k. \quad (4.7)$$

Replacing one factor by $\mathcal{I}_{k+1}$ raises its product into $\mathcal{I}_{kq+1}$, so the map is well defined. The equality on the right follows from $\mathcal{I}_{kq+1} = \mathcal{I}_1 \otimes \mathcal{O}_X(-kqS)$. The image $\mathcal{J}_k$ is a coherent rank-one subsheaf of the actual line $\mathcal{A}_k$. Its reflexive hull is that line tensored with the divisorial ideal of an effective integral Weil divisor. At every codimension-one point the source of μ_k is free and the map is injective. Thus its reflexive hull is canonically the same $\mathcal{J}_k^{**}$.

The local index r divides q , so the order of μ_k at P is $q j_{k,P}/r$. Define $\Xi_k = \sum_P \xi_{k,P} P$. It is finite, either from the coherent image on the compact S or from the bound by $\Theta \leq B_S$. Taking the reflexive image in (4.7) gives exactly (4.1). The canonical map into the specified target line supplies the global identity and excludes an undetermined line-bundle factor.

Vanishing on a small floor model

We use the projective relative MMP in dimension at most three to obtain a small ordinary $\mathbb{Q}$ -factorial model $p : S' \rightarrow S$. For precision, take a projective log resolution of (S, B_S) , principalizing the coherent ideal of its finite boundary support; no individual $\mathbb{Q}$ -Cartier assumption on B_S is

required. Give each exceptional prime an effective rational coefficient below one and strictly above its crepant coefficient. For the resulting effective SNC klt boundary $C_{\widehat{S}}$,

$$K_{\widehat{S}} + C_{\widehat{S}} = \rho^*(K_S + B_S) + F, \quad F \geq 0$$

with F exceptional and positive on every exceptional prime. This is the actual meromorphic adjoint identity. The adjoint is relatively pseudo-effective. The morphism is a projective surjection between normal compact analytic spaces, the source is smooth and ordinary $\mathbb{Q}$ -factorial, and its dimension is at most three. Thus [11, Proposition 2.26] applies. On its relative minimal model the transform of F is exceptional and relatively nef, so negativity [55, Lemma 1.3] makes it zero. No step extracts a prime, hence the resulting $p : S' \rightarrow S$ is small and projective. The source is ordinary globally $\mathbb{Q}$ -factorial and compact Kähler, and codimension-one comparison gives the actual crepant identity

$$K_{S'} + B'_S = p^*(K_S + B_S) \quad (4.8)$$

with the strict boundary and a klt pair. This uses the relative projective MMP as an external input; it is separate from the global contraction construction considered here.

The finite strict transform Ξ'_k is $\mathbb{Q}$ -Cartier by ordinary global $\mathbb{Q}$ -factoriality, as is B'_S . Put $B'_k = B'_S - \Xi'_k$. It is effective by (4.1). Pullback of the effective $\mathbb{Q}$ -Cartier Ξ'_k lowers the crepant coefficients of the klt pair, so (S', B'_k) is klt.

Work over a connected relatively compact Stein open $U \subset W$ whose closure lies in a chosen larger open. Properness and connected fibers make S_U connected, and normality then makes it irreducible. Relative generation for a large twist by a ϕ -ample line [12, Theorem 2.12], followed by Cartan generation on the Stein base, gives a meromorphic section of $\mathcal{E}_k|_{S_U}$: take the quotient of a nonzero section of the twist and a nonzero section of the twisting line. The same argument applies to the actual line $\mathcal{A}_k$. Write

$$\mathcal{E}_k|_{S_U} = \mathcal{O}_{S_U}(G_k), \quad \mathcal{A}_k|_{S_U} = \mathcal{O}_{S_U}(H_k)$$

for the resulting integral Weil divisor G_k and Cartier divisor H_k . Applying the actual identity (4.1) to these meromorphic sections gives a genuine principal relation

$$qG_k \sim H_k - q\Xi_k.$$

Let G'_k be the strict transform on S'_U . A small map has no local exceptional prime either. Strict transform therefore takes H_k to its Cartier pullback and a principal divisor to that of the pulled-back meromorphic function. It follows that

$$qG'_k \sim p^*H_k - q\Xi'_k. \quad (4.9)$$

After clearing the global indices of Ξ'_k , the right side is Cartier. Hence the locally chosen integral G'_k is $\mathbb{Q}$ -Cartier. This local assertion follows from the displayed identity; ordinary global $\mathbb{Q}$ -factoriality supplied the index of the globally defined Ξ'_k . The actual adjoint in (4.8), together with the globally $\mathbb{Q}$ -Cartier boundary B'_S , also makes a reflexive canonical power on S' an actual line. The same local generation argument supplies a meromorphic canonical generator if a local Weil representative is needed for the vanishing statement.

Equations (4.8) and (4.9) give

$$G'_k - (K_{S'} + B'_k) \sim_{\mathbb{Q}} p^*H, \quad H = -(K_S + B_S) - kS|_S. \quad (4.10)$$

The right side pulls back the defined full adjoint and normal rational lines, and the locally chosen G'_k is $\mathbb{Q}$ -Cartier by (4.9). The line H is ϕ -ample. Projective analytic morphisms compose over a neighborhood of a compact subset of the final base [12, Remark 2.11]. Here W is compact, so ϕp is projective over the whole base; equivalently one may use the compact closures in the local argument. Both p and ϕp are proper surjections.

The difference in (4.10) is nef and big for both maps in the convention of [12, Definition 2.40]. Nefness follows by projecting vertical curves: a curve for ϕp maps to a ϕ -vertical curve or a point, and the degree for p is zero. For the relative Iitaka condition, clear the index and write $\mathcal{H} = \mathcal{O}_S(hH)$ for a ϕ -ample line. Normality and proper bimeromorphicity give $p_*\mathcal{O}_{S'} = \mathcal{O}_S$, so projection formula gives

$$(\phi p)_* p^* \mathcal{H}^m = \phi_* \mathcal{H}^m, \quad p_* p^* \mathcal{H}^m = \mathcal{H}^m.$$

For large m , the first relative complete system is p followed by the relative embedding from $\mathcal{H}^m$. Its image has relative dimension $\dim S' - \dim W$. The second has image the base S , of relative dimension zero, equal to $\dim S' - \dim S$. These are exactly the two maximal relative Iitaka dimensions. For p , relative bigness is this zero-relative-dimension condition; the pulled line has degree zero on its exceptional curves.

Analytic Kawamata–Viehweg vanishing [12, Theorem 2.41] now applies to the effective rational klt boundary B'_k , the integral $\mathbb{Q}$ -Cartier divisor G'_k , and each of the two proper surjections. It yields, locally over U ,

$$R^i p_* \mathcal{O}_{S'}(G'_k) = 0, \quad R^i (\phi p)_* \mathcal{O}_{S'}(G'_k) = 0 \quad (i > 0).$$

The bijection of prime valuations for the small map, together with normal divisorial extension, identifies

$$p_* \mathcal{O}_{S'}(G'_k) = \mathcal{O}_S(G_k) = \mathcal{E}_k.$$

This is an equality of meromorphic sections satisfying the same codimension-one order conditions. Leray gives (4.2) on U , hence everywhere. This proves the proposition. $\square$

For $k \geq 1$, put $S_k = V(\mathcal{I}_k)$. The inclusions $\mathcal{I}_1^k \subset \mathcal{I}_k \subset \mathcal{I}_1$ show that $\sqrt{\mathcal{I}_k} = \mathcal{I}_1$, so these thickenings have the common underlying floor. Their conormal sequence is

$$0 \longrightarrow \mathcal{E}_k \longrightarrow \mathcal{O}_{S_{k+1}} \longrightarrow \mathcal{O}_{S_k} \longrightarrow 0.$$

Use the underlying continuous map $|\phi| : |S| \rightarrow |W|$ to put $\mathcal{B}_k = |\phi|_* \mathcal{O}_{S_k}$. Then $\mathcal{B}_1 = \mathcal{O}_W$, and (4.2) gives epimorphisms of sheaves of rings on $|W|$

$$\mathcal{B}_{k+1} \twoheadrightarrow \mathcal{B}_k \quad (k \geq 1). \tag{4.11}$$

This is the immediate cohomological assertion used in [9, Claim 5.9]. Exactness is stalkwise, and no splitting of the augmentation $\mathcal{B}_k \rightarrow \mathcal{O}_W$ is assumed. The next construction realizes these ringed spaces analytically and constructs the corresponding maps from S_k .

4.1 Analytic realization of the floor thickenings

Retain the divisorial ideals $\mathcal{I}_k$, the thickenings S_k , the map $\phi : S \rightarrow W$ from Proposition 4.1, and the sheaves $\mathcal{B}_k$ in (4.11). We construct their analytic structure from the already analytic source thickenings.

Lemma 4.2 (Analytic base thickenings). *For every $k \geq 1$, put*

$$W_k = (|W|, \mathcal{B}_k).$$

Then W_k is a complex space with canonical reduction W , and the canonical sheaf evaluation defines a proper surjective holomorphic map $\phi_k : S_k \rightarrow W_k$ with

$$(\phi_k)_* \mathcal{O}_{S_k} = \mathcal{O}_{W_k}.$$

Its reduction is ϕ . The quotients $\mathcal{B}_{k+1} \twoheadrightarrow \mathcal{B}_k$ define closed analytic embeddings $W_k \hookrightarrow W_{k+1}$. They commute with the embeddings $S_k \hookrightarrow S_{k+1}$ and the maps ϕ_k .

Proof. Canonical rings and their successive kernels.

All direct images defining $\mathcal{B}_k$ first use the continuous map $|\phi| : |S| \rightarrow |W|$. The equality $\mathcal{B}_1 = \mathcal{O}_W$ is the canonical equality for the holomorphic map ϕ . Apply derived direct image for abelian sheaves to the conormal sequence. Its derived image on an $\mathcal{O}_S$ -module is the underlying sheaf of the analytic higher direct image. To see this formally, an injective $\mathcal{O}_S$ -module is flasque: for open sets $V \subset U$, the extension-by-zero modules satisfy $j_{V!} \mathcal{O}_V \hookrightarrow j_{U!} \mathcal{O}_U$. Adjunction identifies $\mathrm{Hom}_{\mathcal{O}_S}(j_{U!} \mathcal{O}_U, J)$ with $\Gamma(U, J)$, so injectivity of J makes the restriction of sections surjective. Flasque abelian sheaves are acyclic for continuous direct image. An injective module resolution therefore computes both the analytic module higher images and their underlying abelian-sheaf higher images. Thus (4.2) gives exact sequences

$$0 \longrightarrow \mathcal{F}_k := \phi_* \mathcal{E}_k \longrightarrow \mathcal{B}_{k+1} \longrightarrow \mathcal{B}_k \longrightarrow 0 \quad (k \geq 1). \quad (4.12)$$

The displayed quotients are ring maps; exactness here concerns their underlying abelian sheaves. Proper direct-image coherence for the already holomorphic ϕ [29, Theorem 1.1, p. 15-01] makes $\mathcal{F}_k$ a coherent $\mathcal{O}_W$ -module.

In $\mathcal{O}_{S_{k+1}}$, the ideal $\mathcal{E}_k$ is square-zero and is annihilated by $\mathcal{I}_1$: use $\mathcal{I}_k^2 \subset \mathcal{I}_{2k} \subset \mathcal{I}_{k+1}$ and $\mathcal{I}_1 \mathcal{I}_k \subset \mathcal{I}_{k+1}$. Consequently the action of $\mathcal{B}_{k+1}$ on the ideal $\mathcal{F}_k$ factors through $\mathcal{B}_1 = \mathcal{O}_W$. This is the $\mathcal{O}_W$ -module structure on the square-zero ideal used in the following construction.

The composites of (4.12) are epimorphisms $\mathcal{B}_k \rightarrow \mathcal{O}_W$. Their kernels are

$$\mathcal{N}_k = |\phi|_*(\mathcal{I}_1 / \mathcal{I}_k), \quad \mathcal{N}_k^k = 0,$$

by left exactness and $\mathcal{I}_1^k \subset \mathcal{I}_k$. Since W is reduced, $\mathcal{N}_k$ is exactly the nilradical. Each stalk of $\mathcal{B}_k$ is local: an element lifting a unit in $\mathcal{O}_{W,w}$ has an inverse by a finite geometric series in the nilpotent kernel, whereas an element mapping to the maximal ideal is not a unit. Its residue field is $\mathbb{C}$. Thus W_k is already a locally ringed space, and it remains to construct local analytic presentations for it.

A local analytic presentation from source functions.

Fix k and $w \in W$. Choose a local closed analytic embedding of a neighborhood of w into an open polydisc. Lift the finitely many germs of its coordinate restrictions through $\mathcal{B}_k \rightarrow \mathcal{O}_W$, represent the lifts on a common neighborhood, and shrink the polydisc. This gives a closed embedding $i : U \hookrightarrow D \subset \mathbb{C}^d$ and sections $b_1, \dots, b_d \in \mathcal{B}_k(U)$ reducing to the coordinate restrictions. Put $Z = S_{k,U}$, the open subspace of S_k with underlying set $\phi^{-1}(U)$. By definition, $\mathcal{B}_k(U) = H^0(Z, \mathcal{O}_Z)$.

For a possibly nonreduced complex space, a tuple of holomorphic functions defines a holomorphic map to affine space, functorially on structure sheaves [28, Theorem 1.1, p. 10-02]. Apply this theorem to the b_j . Their point values lie in D , so the map factors as

$$g : Z \longrightarrow D.$$

The same tuple theorem identifies its actual analytic restriction to the reduction with $i\phi$, because the reduced coordinate functions are those of $i\phi$. In particular $|g| = i|\phi|$. The restriction $\phi^{-1}(U) \rightarrow U$ is proper and i is closed, so g is proper; nilpotents do not change properness of the underlying continuous map.

The holomorphic map g makes $\mathcal{A} = g_*\mathcal{O}_Z$ a genuine $\mathcal{O}_D$ -algebra. Its underlying module is coherent by the proper direct-image theorem [29, Theorem 1.1]. Directly from the underlying maps, as sheaves of $\mathbb{C}$ -algebras,

$$\mathcal{A} = i_*(\mathcal{B}_k|_U).$$

Reduction gives an epimorphism $\mathcal{A} \rightarrow i_*\mathcal{O}_U$. It is $\mathcal{O}_D$ -linear because the actual analytic restriction $g|_S = i\phi$ identifies the reduced action of every ambient holomorphic function with its restriction to U . The target is the coherent quotient of $\mathcal{O}_D$ by the ideal of U . Hence its kernel $\mathcal{N}$ is a coherent $\mathcal{O}_D$ -module. It is also a nilpotent ideal. Shrink D once more and choose sections $n_1, \dots, n_r$ generating $\mathcal{N}$ as a module. They are actual sections of $\mathcal{O}_Z$ whose reductions vanish. Since $\mathcal{O}_D \rightarrow i_*\mathcal{O}_U$ is an epimorphism, subtraction of an ambient lift of a reduction gives the stalkwise module equality

$$\mathcal{A} = \mathcal{O}_D \cdot 1 + \sum_{\nu=1}^r \mathcal{O}_D \cdot n_\nu. \quad (4.13)$$

No splitting of the reduction map is used here.

Apply the tuple theorem again to the b_j and n_ν . It gives

$$H = (g, n_1, \dots, n_r) : Z \rightarrow P = D \times \mathbb{C}^r.$$

If $j : U \hookrightarrow P$ is $u \mapsto (i(u), 0)$, then $|H| = j|\phi|$, since the n_ν are nilpotent. Thus H is proper. The sheaf $\mathcal{A}' = H_*\mathcal{O}_Z$ is a coherent $\mathcal{O}_P$ -module and the unit is an algebra map

$$\theta : \mathcal{O}_P \rightarrow \mathcal{A}'.$$

As sheaves of rings $\mathcal{A}' = j_*(\mathcal{B}_k|_U)$, so its stalk at $j(u)$ is $\mathcal{B}_{k,u} = \mathcal{A}_{i(u)}$, and its stalk outside $j(U)$ is zero. The stalk identification uses the cofinality of restrictions of ambient neighborhoods among neighborhoods in the closed subspace U .

The map θ is surjective on every stalk. At $j(u)$, any element of the target has, by (4.13), the form $a_0 + \sum a_\nu n_\nu$ with $a_\nu \in \mathcal{O}_{D,i(u)}$. The projection of H to D is g , and the extra coordinate t_ν pulls back to n_ν . The element is therefore the image of the convergent germ

$$a_0(z) + \sum_{\nu=1}^r a_\nu(z)t_\nu \in \mathcal{O}_{P,j(u)}.$$

At all other stalks the target is zero. The kernel $\mathcal{Q} = \ker \theta$ is a coherent analytic ideal [27, pp. 9-07–9-08]. It defines a closed complex subspace $Y = V(\mathcal{Q}) \subset P$ by [27, Definition 2.2 and adjacent construction, p. 9-10]. Its support is exactly $j(U)$, since the target stalk there surjects onto the nonzero ring $\mathcal{O}_{U,u}$. Under the homeomorphism with U , its structure sheaf is $\mathcal{B}_k|_U$, and its reduction is U .

Every germ of $\mathcal{Q}$ pulls back to zero by the definition of θ . The closed-subspace factorization criterion [27, pp. 9-04–9-05] therefore factors H holomorphically through Y . Its sheaf map, under the identified structure sheaf, is the canonical evaluation

$$|\phi|^{-1}(\mathcal{B}_k|_U) \rightarrow \mathcal{O}_Z.$$

Indeed every germ of $\mathcal{B}_k|_U$ lifts through the surjective θ to an ambient germ, whose pullback is precisely its value as a section on the inverse image. This proves the assertion for every germ, not only the chosen coordinates.

Gluing and closed transition maps.

The ringed space $(|W|, \mathcal{B}_k)$ was fixed before any of the local choices. The presentations just constructed make each of its restrictions an analytic space by the local definition and adjacent construction in [27, Definitions 2.1–2.2 and adjacent prose]. On overlaps the identifications are the identity on the same sheaf of $\mathbb{C}$ -algebras, and hence are analytic isomorphisms satisfying the cocycle condition [27, Definition 2.4 and Proposition 2.5]. This proves that W_k is a complex space. It retains the Hausdorff topology of W , and its reduction is canonically W . Write $\varepsilon_k : |\phi|^{-1}\mathcal{B}_k \rightarrow \mathcal{O}_{S_k}$ for the global sheaf evaluation. It is locally the holomorphic factorization above, so it defines ϕ_k . Its reduction is ϕ , and its underlying map is $|\phi|$; thus it is proper and surjective. The direct-image equality is the definition of its target structure sheaf, with the canonical unit as the identification.

For the closed transition, use the construction with H at level $k+1$ and restrict its coordinate functions to the closed subspace $S_{k,U}$. They give another proper holomorphic map to the same P , with underlying map $j|\phi|$. Its unit is the composite

$$\mathcal{O}_P \rightarrow j_*(\mathcal{B}_{k+1}|_U) \rightarrow j_*(\mathcal{B}_k|_U).$$

The second quotient stays surjective under the closed embedding j , as seen on its stalks. The last sheaf with this ambient action is coherent by proper direct image. The two units therefore have nested coherent ideal kernels, so their analytic subspaces give the required closed inclusion. These local inclusions glue because their ring maps are the fixed quotients. Naturality of the evaluation maps gives

$$\varepsilon_k \circ |\phi|^{-1}(\mathcal{B}_{k+1} \rightarrow \mathcal{B}_k) = (\mathcal{O}_{S_{k+1}} \rightarrow \mathcal{O}_{S_k}) \circ \varepsilon_{k+1},$$

which proves commutativity with the source inclusions. No retraction $W_k \rightarrow W$, Cartesian square, or realization of the infinite tower as one formal completion is asserted or needed. $\square$

4.2 Extension of a prime-floor contraction

Proposition 4.3 (Extension from a prime floor). *Under the hypotheses of Proposition 4.1, there are a normal compact analytic space Z , a closed embedding $W \hookrightarrow Z$, and a proper bimeromorphic morphism $F : X \rightarrow Z$ such that $F|_S$ is the given map ϕ followed by that embedding and*

$$X \setminus S \simeq Z \setminus W, \quad F^{-1}(|W|) = |S|, \quad |F^{-1}(w)| = |\phi^{-1}(w)| \quad (w \in W). \quad (4.14)$$

The natural map $\mathcal{O}_Z \rightarrow F_\mathcal{O}_X$ is an isomorphism, and the fibers are connected. For a sufficiently divisible global index ℓ of S , the actual line $\mathcal{O}_X(-\ell S)$ is F -ample; in particular F is projective. More generally an actual rational line on X whose restriction to S is ϕ -ample is F -ample. The target is in Fujiki's class $\mathcal{C}$.*

Proof. Choose a positive global index ℓ such that ℓS is Cartier and $\mathcal{O}_S(-\ell S)$ is ϕ -ample. Put

$$A = S_\ell = \ell S, \quad A' = W_\ell, \quad f = \phi_\ell : A \longrightarrow A'.$$

Here A is the full effective Cartier divisor, including its nilpotents; its local equation on normal X is a nonzerodivisor. Lemma 4.2 makes f a proper surjective holomorphic map of complex spaces and gives the natural equality $f_*\mathcal{O}_A = \mathcal{O}_{A'}$.

Cartier multiplication gives $\mathcal{I}_\ell \mathcal{I}_j = \mathcal{I}_{j+\ell}$. Hence, for every integer $n > 0$, on the full space A one has

$$\mathcal{O}_A(-nA) = \mathcal{I}_{n\ell} \otimes \mathcal{O}_A = \mathcal{I}_{n\ell} / \mathcal{I}_{(n+1)\ell}. \quad (4.15)$$

Its finite filtration by $\mathcal{I}_j / \mathcal{I}_{(n+1)\ell}$, for $n\ell \leq j \leq (n+1)\ell$, has successive quotients $\mathcal{E}_j$ for $n\ell \leq j < (n+1)\ell$. All indices are at least one. These filtration terms are $\mathcal{O}_A$ -modules, because $\mathcal{I}_\ell \mathcal{I}_j \subset \mathcal{I}_{(n+1)\ell}$.

Let $a : S \hookrightarrow A$ and $i : W \hookrightarrow A'$ be the closed reduction embeddings. The compatibility in Lemma 4.2 gives $fa = i\phi$. Pushforward by a closed embedding is exact, so the composition identity for derived direct images of abelian sheaves gives

$$R^q f_*(a_* \mathcal{E}_j) = i_* R^q \phi_* \mathcal{E}_j = 0 \quad (q > 0).$$

The long exact sequences of the finite filtration in (4.15) therefore give

$$R^q f_* \mathcal{O}_A(-nA) = 0 \quad (q > 0, n > 0). \quad (4.16)$$

This uses no vanishing for $\mathcal{E}_0$ and keeps the full Cartier thickening throughout.

The actual line $\mathcal{O}_A(-A)$ restricts to $\mathcal{O}_S(-\ell S)$. The underlying map of f is ϕ , so the reductions of their fibers, as closed subspaces of X , coincide: a reduced closed analytic subspace is determined by its support. Thus $\mathcal{O}_A(-A)$ is ample on the reduction of every f -fiber. For a line on a compact nonreduced complex space, ampleness on the reduction implies ampleness on the whole space. Indeed the positive metric from [22, Corollary 1.12] has local strictly plurisubharmonic weights in common ambient embeddings, by its Lemma 2.4. The same weights define a metric on the original line, since the pointwise absolute values of transition units are unchanged by nilpotents. The converse positive-metric criterion gives ampleness on the full space. The proper fiberwise criterion [22, Definition 3.1 and Remark 3.2] now makes $\mathcal{O}_A(-A)$ f -ample.

We apply the analytic blowdown criterion stated in [9, Theorem 4.2, p. 14], attributed there to [18, Theorem 2, p. 495]. For a reduced complex space, an effective Cartier divisor A with its full possibly nonreduced structure, and a proper surjective holomorphic map $f : A \rightarrow A'$, this criterion assumes f -ampleness of $\mathcal{O}_A(-A)$ and

$$R^1 f_* \mathcal{O}_A(-nA) = 0 \quad \text{for every } n > 0.$$

It supplies a blowing down: a complex target Z containing A' as an embedded subspace, a proper surjective holomorphic map $F : X \rightarrow Z$ whose restriction to A is f , and an analytic isomorphism of the complements. In its universal form it also supplies the natural equality of sheaves of rings

$$F_* \mathcal{S}_{X,A,f} = \mathcal{O}_Z. \quad (4.17)$$

Here $\mathcal{S}_{X,A,f}$ consists along A of the germs in $\mathcal{O}_X$ whose restrictions lie locally in $\text{im}(f^{-1}\mathcal{O}_{A'} \rightarrow \mathcal{O}_A)$, and is $\mathcal{O}_X$ away from A . This is a subsheaf of rings; no module coherence of this particular sheaf is used. The hypotheses of the stated criterion follow from (4.16) and the preceding ampleness argument. No flatness or reducedness of A or A' is required.

The embedded A' is closed because it is compact. Its reduction embeds the original W as a closed subspace of Z . The compatible reductions give $F|_S = \phi$ followed by this embedding. Removing a subspace depends only on its support, so the complement isomorphism is $X \setminus S \simeq Z \setminus W$. It follows that $F^{-1}(|W|) = |S|$, and the prescribed restriction to A gives the fiber-set equalities in (4.14). Outside W the fibers are single reduced points. These assertions concern underlying fiber

sets over W ; they require no Cartesian square of the closed thickenings. The fibers are connected because the floor fibers are connected. The target is compact as the image of compact X .

We next prove normality; it is not an extra conclusion assumed from the blowdown criterion. For every open $V \subset Z$, a section $s \in \mathcal{O}_X(F^{-1}V)$ restricts to a section on the open space

$$A \cap F^{-1}V = f^{-1}(V \cap A').$$

The equality $f_*\mathcal{O}_A = \mathcal{O}_{A'}$ makes this restriction the pullback of a unique section on $V \cap A'$. Thus every germ of s along A satisfies the defining condition of $\mathcal{S}_{X,A,f}$. The reverse inclusion follows from $\mathcal{S}_{X,A,f} \subset \mathcal{O}_X$. Consequently

$$F_*\mathcal{S}_{X,A,f} = F_*\mathcal{O}_X, \quad F_*\mathcal{O}_X = \mathcal{O}_Z$$

as sheaves of rings, the second equality by (4.17). This reasoning uses equality of the displayed open subspaces, not of any closed fiber products.

The equality first makes Z reduced. The complement is dense: the inverse image of a nonempty open of Z is a nonempty open of X , and therefore meets $X \setminus S$. The complement isomorphism thus makes F bimeromorphic. Let $\nu : Z' \rightarrow Z$ be the finite analytic normalization. The finite normalization exists by [34, Part B, Section 4, Corollaries 2–3]. The dominant bimeromorphic map from normal X lifts to Z' . Pullback along this lift gives inclusions, injective on the dense common complement,

$$\mathcal{O}_Z \subset \nu_*\mathcal{O}_{Z'} \subset F_*\mathcal{O}_X = \mathcal{O}_Z.$$

The normalization is consequently an isomorphism and Z is normal. The natural direct-image equality also gives connected fibers by analytic Stein factorization.

Finally consider the actual global line $\mathcal{O}_X(-\ell S)$. Its restriction is ample on every reduced F -fiber: this is clear off W , and over W the reduction is the same embedded reduced space as that of the corresponding ϕ -fiber. The nonreduced upgrade and the proper fiberwise criterion used above therefore make this one line F -ample. The same proof works for any actual ambient rational line with ϕ -ample restriction after clearing its global index. In particular it works for the negative full adjoint in Proposition 4.1. Thus F is projective. A projective resolution with smooth source of the compact Kähler X is again compact Kähler, and its composite with F is bimeromorphic. Hence Z lies in Fujiki's class $\mathcal{C}$. $\square$

The proposition supplies the ambient step for floor data satisfying its stated hypotheses, including the two actual ample-line conditions. In the threefold argument below it extends divisor-to-curve contractions. In Section 5 it extends a selected floor contraction of the supported fourfold program. When that floor contraction cuts out the selected ray, the exact fiber description keeps the same contracted curves, and the actual relatively ample lines make the ambient step projective. Its target Kähler class is then established in the supported-program argument.

4.3 Descent of a dominating current

Lemma 4.4 (Descent of a dominating current). *Let $p : Y \rightarrow Z$ be a proper bimeromorphic morphism between normal compact complex spaces, with Y Kähler. Let η be a smooth real closed $(1,1)$ -form with local smooth potentials on Z . Suppose that a positive closed current T with local potentials represents the local-potential Bott–Chern class $p^*[\eta]$, and that $T \geq \kappa$ for a Kähler form κ on Y . For every smooth positive Hermitian form g on Z , understood in local ambient embeddings, there are a constant $c > 0$ and a global quasi-plurisubharmonic function u_Z such that*

$$\eta + i\partial\bar{\partial}u_Z \geq cg. \tag{4.18}$$

This is a current in $[\eta]$ with local potentials, and it has the bigness property preceding [12, Theorem 2.29].

Proof. We make the potential in the class equality explicit. Use the normal-space potential description of Boucksom–Guedj [4, Section 4.6.1], specifically Lemma 4.6.1 and the paragraph following Definition 4.6.2. The soft-sheaf descriptions of Bott–Chern cohomology by smooth functions and distributions first give a global real distribution u_Y for which

$$T = p^*\eta + i\partial\bar{\partial}u_Y.$$

Locally write $T = i\partial\bar{\partial}v$ with v plurisubharmonic and $p^*\eta = i\partial\bar{\partial}h$ with h smooth. Then $u_Y + h - v$ is pluriharmonic as a distribution. By the cited lemma it is a smooth real part of a holomorphic germ. Thus u_Y is locally represented by a quasi-plurisubharmonic function. These representatives agree as distributions on overlaps and hence, after taking their upper-semicontinuous representatives, agree pointwise. They give one global quasi-plurisubharmonic u_Y , which is locally integrable and locally bounded above.

In local ambient embeddings, p^*g is represented by a smooth semipositive form and κ by a smooth positive definite form. A finite relatively compact cover of Y therefore gives a constant $C > 0$ with $p^*g \leq C\kappa$. Consequently

$$p^*\eta + i\partial\bar{\partial}u_Y \geq C^{-1}p^*g.$$

The map p is a modification between normal spaces, and u_Y has the local upper bound needed in the current descent argument. [12, Corollary 2.32 and its proof, p. 18] gives a global quasi-plurisubharmonic u_Z with (4.18) for $c = C^{-1}$. It is an L^1 potential, and the formula keeps the descended current in the chosen Bott–Chern class. The local potentials of η give local potentials for that current. No Kähler form on Z has been used. $\square$

4.4 Threefold contractions used in the boundary argument

We prove Proposition 3.1 by applying the cited threefold MMP with two analytic constructions supplied here: Proposition 3.2 contracts the finite null loci, and Proposition 4.3 extends the prime-floor contractions. The projective relative MMP, the nonbig argument through a surface contraction, and the terminal canonical threefold program remain external inputs. We verify the hypotheses at each use of the two constructions, then supply the required target positivity; neither construction alone asserts that its target is Kähler. We first record the cone observation used in the big and floor cases.

For a normal compact space Z in Fujiki’s class $\mathcal{C}$, put $N^1(Z) = H_{BC}^{1,1}(Z)$ in the local-potential convention. Let $N_1(Z)$ be the vector space of real closed currents of bidimension $(1, 1)$, modulo $T_1 \equiv T_2$ when $T_1(\eta) = T_2(\eta)$ for every real closed $(1, 1)$ -form η with local potentials; the pairing is evaluation. Following [33, Definitions 3.6 and 3.8, pp. 8–9], let $\text{NA}(Z) = \overline{\text{NA}}(Z) \subset N_1(Z)$ denote the closed cone generated by the numerical classes of positive closed currents. Thus both notations used below refer to this same closed cone.

Lemma 4.5 (Positivity on the current cone). *Let Z be a normal compact Kähler analytic variety with rational singularities, and let κ be a Kähler class. In the finite-dimensional local-potential numerical spaces put*

$$C = \overline{\text{NA}}(Z), \quad \Sigma = \{z \in C : \kappa \cdot z = 1\}.$$

Then Σ is compact. If a real local-potential $(1, 1)$ -class ℓ is strictly positive on Σ , then ℓ is Kähler.

Proof. By [12, Lemma 2.3 and Proposition 2.4], the natural pairing is perfect, $\text{Nef}(Z) = C^*$, and the Kähler cone is open with closure the nef cone. In particular κ lies in the interior of C^* . A sequence in Σ with unbounded norm would, after division by its norm and passage to a subsequence, give a nonzero class $z \in C$ with $\kappa \cdot z = 0$, contradicting that interior property. The slice is closed and hence compact.

If it is nonempty, let $\mu = \min_{\Sigma} \ell > 0$. Homogeneity gives $\ell - \frac{\mu}{2}\kappa \in C^* = \text{Nef}(Z)$. A smooth local-potential nef approximation for this last class, bounded below by $-\mu\kappa/4$, becomes a positive representative after adding $\mu\kappa/2$. Thus ℓ is Kähler. If Σ is empty, then $C = \{0\}$ by the interior property; the same conclusion follows because every class is nef and a positive small multiple of κ can be subtracted first. All cones here are the local-potential cones in the cited perfect pairing. $\square$

The spaces used below have rational singularities: this is Lemma 2.31 of [11] for dlt pairs and its Remark 2.14 for klt varieties.

Proof of Proposition 3.1. The proof of [12, Theorem 5.5] first uses a projective relative small $\mathbb{Q}$ -factorialization and then [11, Theorem 1.7]. The latter proof passes to a small strongly $\mathbb{Q}$ -factorial model and treats a nef nonbig class by its Theorem 5.5 and a nef big class by its Theorem 6.4. Here strong $\mathbb{Q}$ -factoriality means that every global coherent reflexive rank-one sheaf has an invertible reflexive power. It implies ordinary global $\mathbb{Q}$ -factoriality; it does not assert the analogous property for all analytic open subsets. The small models use the projective relative MMP of [11, Proposition 2.26 and Lemma 2.27], whose maps are already projective. Compactness of the final bases is retained when projective maps are composed, as required by [12, Remark 2.11].

The imported programs and target descent. The nef nonbig branch follows the chain Theorem 5.5, Corollary 5.4, and Theorem 5.2 of [11]. It uses Step 4 in the proof of [32, Theorem 1.4, p. 242], which contracts a negative-definite collection of curves on a smooth compact Kähler surface and descends through a graph. We use that nonbig argument as an external input. The pseudo-effective cone lineage also uses the terminal canonical threefold program: Assumption 10.1 of [13] invokes its nonvanishing Theorem 9.1, whose proof at p. 50 invokes [32, Theorem 1.1] for a terminal K -MMP and Mori fiber space. We retain that terminal program as an external input. Together with the projective relative MMP just specified, these are the birational inputs used below.

For an already constructed negative-ray contraction with ordinary $\mathbb{Q}$ -factorial compact Kähler dlt source and relatively ample negative adjoint, [6, Proposition 3.1] gives rationality of the target and descent of the supporting class. We use the proof of its Corollary 3.1 for target positivity. The corollary assumes non-uniruledness to obtain a nef support; our ray already has an exposed nef support. Its remaining argument descends that class, proves positivity on the target current cone, and lifts target curves through the projective morphism. These steps apply to the data just stated. The proposition and corollary numbers here are those of the journal edition. We now verify the finite-null and prime-floor constructions to which this argument will be applied.

Nef and big classes with a finite null locus. Three geometric situations use Proposition 3.2: a small negative ray, the entire null locus at a stopping model, and a small ray with pseudo-effective adjoint. The big threefold program uses strongly $\mathbb{Q}$ -factorial klt sources. The last situation can also occur for ordinary dlt data, whose underlying space is ordinary $\mathbb{Q}$ -factorial and klt, as checked below. In every case the relevant locus is the full top-intersection null locus of the proposition.

For a small negative ray, the small branch of [11, proof of Theorem 4.16, p. 40] has, after its boundary perturbation, a strongly $\mathbb{Q}$ -factorial klt source and a nef big supporting class

$a = (K + \Gamma') + \kappa$ with κ Kähler. At this input [11, Lemma 4.5] says that an a -null surface is covered by a -null curves. All those curves lie in the exposed ray. Such a covering of a surface contradicts the smallness of that ray. Lemma 3.3 therefore makes $\text{Null}(a)$ a finite union of curves, and Proposition 3.2 constructs its contraction. This argument permits a non-pseudo-effective underlying adjoint.

For the entire null locus at the final model X_n of the program in [11, Claim 6.5 and proof of Theorem 6.4, pp. 50–51], the pair is strongly $\mathbb{Q}$ -factorial klt, its class a_n is nef and big, and every a_n -null curve has nonnegative degree for the running adjoint A_n . These are the stopping data from that program. Its null-surface argument, using Lemma 4.5 of the same source, would cover a null surface by a_n -null curves on which the remainder $a_n - c_1(A_n)$ has positive degree. Their A_n -degrees would then be negative, contrary to the stopping data. There is therefore no null surface. Lemma 3.3 again gives finitely many null curves. Apply Proposition 3.2 to their entire union. This use has no single-ray hypothesis and needs only the proper bimeromorphic map to a normal compact analytic target supplied by that proposition.

For a small ray with pseudo-effective adjoint, the branch in the proof of Theorem 6.4 at p. 50 invokes the small case of Theorem 2.23(1) of the same source; this case can also occur in its Theorem 3.1, Claim 3.2. Let A be the negative adjoint and b an exposed nef support for the ray. It may be scaled so that $b - c_1(A)$ is Kähler. To check this usual support step, normalize the closed cone of positive bidimension- $(1, 1)$ classes by a Kähler class. Its normalized slice is compact. The continuous function $-c_1(A)$ is strictly positive on its intersection with the exposed ray, hence on a neighborhood of that intersection. On the compact complement, the support has a positive minimum. A large multiple of the support minus $c_1(A)$ is consequently strictly positive on the full slice. Lemma 4.5 makes this difference Kähler. Since A is pseudo-effective, the scaled b is big. The same Lemma 4.5 and the smallness of the ray exclude null surfaces, so Lemma 3.3 and Proposition 3.2 apply. At the outer Theorem 6.4 use the pair is klt. If this step is made for the ordinary dlt data in Claim 3.2, the underlying space is still klt, which is the singularity condition of Proposition 3.2; a small decrease of the boundary preserves negativity when a klt adjoint is needed in a subsequent result.

In each of the two single-ray situations just described, the constructed map contracts exactly that ray. A positive-dimensional reduced fiber is a finite connected union of compact curves, hence is projective. For the corresponding negative adjoint A , a positive global Cartier multiple of the actual line $-A$ has positive degree on every irreducible component of that fiber, and is therefore ample on the fiber. It is ample also on every zero-dimensional fiber. Ampleness passes from a reduction to its nilpotent thickening by the positive-metric criterion [22, Lemma 2.4 and Corollary 1.12]. The proper fiberwise criterion [22, Definition 3.1 and Remark 3.2] now makes this one actual line relatively ample, so the contraction is projective. Normality and proper bimeromorphicity give connected fibers. The source is ordinary $\mathbb{Q}$ -factorial klt, or dlt in the indicated variant, and the exposed nef support is unchanged. Thus [6, Proposition 3.1 and proof of Corollary 3.1], applied as specified above, gives the Kähler target and the descended supporting class. Thus Proposition 3.2 supplies the normal compact analytic contraction, and this postprocessing supplies projectivity and positivity in the single-ray cases. The entire-null cleanup uses the different argument that follows.

In that cleanup, write $\mu : X_n \rightarrow Z$ for the entire-null contraction just constructed. The graph argument in [11, proof of Theorem 6.4, pp. 51–52] descends its composite with the program to a morphism $\psi : X_0 \rightarrow Z$, where X_0 is the original strongly $\mathbb{Q}$ -factorial klt source of that theorem. All steps of the program are trivial for the transported supporting class. Before applying its Lemma 2.44 to ψ , restore the original boundary Γ^{orig} and the original nef big difference ω^{orig} on

X_0 , before the internal boundary replacement. The original supporting class is numerically trivial on its contracted curves, and hence

$$-(K + \Gamma^{\text{orig}}) \equiv_{\psi} \omega^{\text{orig}}. \quad (4.19)$$

The left side is ψ -nef and is ψ -big for the bimeromorphic map. The rationality result of Lemma 2.44 thus applies to this original Q-Gorenstein klt pair and makes the target Z rational. This step uses the original nef big difference; the transported remainder $a_n - c_1(A_n)$ on the stopping model is not substituted into (4.19).

For the entire-null contraction μ , the source X_n is a normal compact Kähler klt threefold, so it has rational singularities. The target is normal and compact, and is in Fujiki's class $\mathcal{C}$: a projective resolution of X_n with smooth source is compact Kähler, and its composite with μ is bimeromorphic. The exceptional image $E_Z = \mu(\text{Null}(a_n))$ is finite. Every curve contracted by μ is a_n -null. Apply [11, Lemma 2.11] to this map between normal compact rational spaces in class $\mathcal{C}$. It gives a smooth real closed $(1, 1)$ -form η_Z with local smooth potentials such that

$$a_n = \mu^*[\eta_Z] \quad \text{in the local-potential Bott–Chern group.} \quad (4.20)$$

The normalized graph constructed ψ earlier; this application of Lemma 2.11 uses μ .

We check the three analytic positivity hypotheses on Z . Fix a smooth positive Hermitian form g_Z in local ambient embeddings. The bimeromorphic nef descent theorem [33, Lemma 3.13] applies to the normal compact threefold Z in class $\mathcal{C}$ and the normal compact threefold X_n . It makes $[\eta_Z]$ analytically nef from (4.20); it does not assume that Z is Kähler. If its defining approximation uses a different positive reference form, compact comparison with g_Z and rescaling the approximation parameter give, for every $\delta > 0$, a smooth function f_δ with

$$\eta_Z + i\partial\bar{\partial}f_\delta \geq -\delta g_Z. \quad (4.21)$$

Changing a smooth representative of the same local-potential class only changes f_δ by a global smooth potential.

The big class a_n on the compact Kähler source contains a positive closed current T_n with local potentials and $T_n \geq \kappa_n$ for a Kähler form κ_n . In view of (4.20), Lemma 4.4 applied to μ gives a global quasi-plurisubharmonic, hence L^1 , function u_Z and a constant $c > 0$ satisfying

$$\eta_Z + i\partial\bar{\partial}u_Z \geq cg_Z. \quad (4.22)$$

This establishes the required current bigness on the not yet Kähler target in the selected class $[\eta_Z]$.

Finally let $V \subset Z$ be an irreducible reduced analytic subspace of dimension $d > 0$. Its strict transform V' is the closure of the inverse image of $V \setminus E_Z$. It has dimension d , maps bimeromorphically to V , and is not contained in $\text{Null}(a_n)$. Nef approximations on X_n , their restriction to the integration cycle of V' , and Stokes' theorem give $a_n^d \cdot V' \geq 0$. Equality would put V' into the null locus by its definition, so the inequality is strict. The proper cycle identity $\mu_*[V'] = [V]$ and projection formula therefore give

$$\int_{V_{\text{reg}}} \eta_Z^d = a_n^d \cdot V' > 0. \quad (4.23)$$

This includes $V = Z$; it requires neither V nor V' to be normal. Additivity over the top-dimensional irreducible components gives the same positivity for every positive-dimensional compact reduced analytic subspace.

Equations (4.21), (4.22), and (4.23) are the three hypotheses of [12, Theorem 2.29]. Apply that theorem directly to η_Z . It gives a smooth potential making η_Z positive definite, so $[\eta_Z]$ is Kähler and Z is compact Kähler. This verifies the positivity conclusion at the cleanup contraction.

Divisorial contractions. In the point case, let P be the selected prime surface and $\nu : P^\nu \rightarrow P$ its normalization. The nef-dimension-zero condition for the supporting class is taken on P^ν ; the whole normalized fiber has zero restricted class by [32, Theorem 3.19(a) and its proof, p. 234]. Use [9, Corollary 4.4], whose proof invokes its Lemma 4.3, at the stated data: a $\mathbb{Q}$ -factorial compact Kähler source, a nef big exposed support, ray curves covering P , and zero restriction on P^ν . The corollary constructs the proper analytic point contraction and its proof supplies an actual ample line $\mathcal{O}_P(-mP)$ for some $m > 0$. Thus P is projective, and ampleness persists on the full possibly nonreduced fiber by the positive-metric criterion used above. The full-fiber criterion makes $-mP$ relatively ample.

Every curve of P lifts finitely to its normalization. The zero restricted support therefore gives zero degree on every such curve, so its ambient class lies in the exposed ray. The negative adjoint restricted to P is consequently numerically a fixed positive multiple of the negative normal line. Both have actual rational Cartier multiples. Numerical invariance of ampleness on the projective P and then the full-fiber criterion make the negative adjoint relatively ample. The same [6, Proposition 3.1 and proof of Corollary 3.1] then supplies rationality, class descent, and target positivity for this negative-ray contraction. The point-contraction corollary was used to construct the underlying proper analytic map.

For a divisor-to-curve step, Proposition 4.3 supplies the ordinary extension used both in [11, Theorem 3.1, Claim 3.2] and in its replay in Theorem 4.16 of that source. We verify the two ample-line hypotheses for this use. Write (M, Δ_M) for the strongly $\mathbb{Q}$ -factorial compact Kähler dlt threefold there, $A = K_M + \Delta_M$, P for the selected prime floor, and R for its exposed ray. The selected ray satisfies

$$A \cdot R < 0, \quad P \cdot R < 0.$$

Adjunction makes P a normal compact Kähler dlt surface and makes $A|_P$ its actual full adjoint. For its inclusion $i : P \rightarrow M$, put

$$F = \{z \in \overline{\text{NA}}(P) : i_* z \in R\}.$$

Fix a Kähler class κ on M and a nonzero $r \in R$. For $z \in F$ with $\kappa|_P \cdot z = 1$, positivity of $\kappa|_P$ gives $i_* z = r/(\kappa \cdot r)$. Thus on the full compact normalized vertical face

$$-(A|_P) \cdot z = \frac{-A \cdot r}{\kappa \cdot r} > 0, \quad -(P|_P) \cdot z = \frac{-P \cdot r}{\kappa \cdot r} > 0. \quad (4.24)$$

This uses the cone of positive current classes and includes its limits.

Let ζ be the nef support exposing R . Its restriction has null cone F . The compact-slice argument just used makes $\lambda\zeta|_P - c_1(A|_P)$ Kähler for sufficiently large λ : near F the negative adjoint has the uniform positive bound in (4.24), and away from F the support has a positive minimum. Lemma 4.5 on the rational compact Kähler surface turns this bound into a Kähler class. The ordinary dlt case of the surface theorem [10, Theorem 2.32] therefore gives a projective surjection $\gamma : P \rightarrow W_P$ with connected fibers onto a normal compact Kähler space, contracting exactly F , with the support pulled back from a Kähler class on W_P . This use is of the surface theorem.

Applying the same compact-slice estimate after adding a sufficiently large pullback of that target Kähler class makes each of $-A|_P$ and $-P|_P$ relatively Kähler over W_P . They have actual global rational Cartier multiples by the factoriality of M . The proper fiberwise positivity criterion makes both lines γ -ample, including when a fiber is the whole surface. Put $H = \lfloor \Delta_M \rfloor - P$. For small rational $\varepsilon > 0$, the pair $(M, \Delta_M - \varepsilon H)$ is plt with sole floor P . Indeed on a dlt resolution the

pullback of the effective globally $\mathbb{Q}$ -Cartier divisor H is effective. Subtracting it preserves the strict inequality for every exceptional crepant coefficient and lowers all the other strict floor coefficients below one. The negative degree of $A - \varepsilon H$ on R persists for small ε , so the first relative ample line persists by (4.24); the second is unchanged. These are precisely the hypotheses of Proposition 4.3.

That proposition constructs the extension, whose fiber sets show that the ambient map contracts exactly the original ray and is an isomorphism off P . The negative original adjoint is relatively ample by the same full-fiber criterion. Return at once to Δ_M and A for the running program. For this projective negative-ray contraction, [6, Proposition 3.1 and proof of Corollary 3.1] applies with the already exposed support, exactly as specified at the start of the proof. It gives rationality, class descent, and Kähler target positivity. Strong factoriality is preserved by [11, Lemma 2.5], and an ordinary projective flip, when needed, is supplied by its Theorem 2.24. Thus the original support and boundary are the ones transported to the next step.

Return to the original threefold. The finite-null and divisorial constructions now supply the indicated steps of the cited big-case program, with their required positivity. Together with the external inputs stated at the start, they give the asserted contraction for the data in (3.1) by the proof of [11, Theorem 1.7]. The proof of [12, Theorem 5.5] descends the contraction from the small model. A pulled-back Kähler class has degree zero on a constant curve and positive degree on a nonconstant curve, as seen on its normalization. This proves the asserted curve criterion and completes Proposition 3.1. $\square$

For the separate exposure input, the proof of Theorem 5.2 and Corollary 5.3 of [12] uses the same Theorem 1.7 for the supporting contractions, followed by the projective relative cone theorem and convex separation. Its contraction inputs are therefore the ones just established.

4.5 The actual floor split

We verify the use of that specialization for the dlt floor in the fourfold supported program. Let $(X, S + B)$ be one of its compact ordinary $\mathbb{Q}$ -factorial Kähler dlt pairs, with reduced floor S , coefficients of B below one, and actual adjoint $A = K_X + S + B$. Fix a prime T of S , and write

$$H = S - T, \quad A_0 = A - H = K_X + T + B.$$

The finite global divisor H is $\mathbb{Q}$ -Cartier by ordinary global factoriality, and A_0 is an actual rational line. On a dlt resolution, subtracting the effective pullback of H lowers the exceptional crepant coefficients and removes the strict transforms of the other floor components. Only the strict transform of T has coefficient one. Hence $(X, T + B)$ is plt. Ordinary plt adjunction [9, Lemma 5.3] gives a normal T , an effective rational boundary B_0 with (T, B_0) klt, and the actual restriction $A_0|_T = K_T + B_0$.

Choose one positive integer m clearing the indices of A, A_0, H . The canonical section of $\mathcal{O}_X(mH)$ restricts nontrivially to T because T is not a component of H . Set

$$B' = \frac{1}{m} \operatorname{div}(s_{mH}|_T) \geq 0.$$

It is an effective rational $\mathbb{Q}$ -Cartier divisor. Use compatible local meromorphic residue embeddings for the two adjunctions. Their ratio in codimension one on T is multiplication by $s_{mH}|_T$. Reflexive

extension on normal T then gives the divisor split and actual line identities

$$\begin{aligned} B_T &= B_0 + B', \\ \mathcal{O}_T(m(K_T + B_0)) &\simeq \mathcal{O}_X(mA_0)|_T, \\ \mathcal{O}_T(mB') &\simeq \mathcal{O}_X(mH)|_T. \end{aligned} \tag{4.25}$$

where $A|_T = K_T + B_T$ is the full dlt adjunction. This proves the split without assuming T to be $\mathbb{Q}$ -factorial.

Suppose, as in the supported program, that $\alpha_T = c_1(A|_T) + [\omega_T]$ is nef and ω_T is Kähler. These properties hold by restricting the ambient smooth metric nef approximations and local strictly plurisubharmonic potentials. For a sufficiently small rational $\varepsilon > 0$, put

$$B_\varepsilon = B_0 + (1 - \varepsilon)B', \quad \omega_\varepsilon = \omega_T + \varepsilon\theta_{B'},$$

where $\theta_{B'}$ is the normalized curvature of a smooth metric on a Cartier multiple of B' . The full adjunction is lc and the lower adjunction is klt. Affineness of discrepancies in this convex combination makes (T, B_ε) klt. On finitely many compact local embedding charts, the Hessian of $\theta_{B'}$ is bounded and that of ω_T is uniformly positive. For the chosen small ε , ω_ε is therefore Kähler and

$$\alpha_T = c_1(K_T + B_\varepsilon) + [\omega_\varepsilon].$$

If $\dim T = 3$, Proposition 3.1 applies and gives a projective connected-fiber contraction $T \rightarrow W$ to a normal compact Kähler target with α_T pulled back from a Kähler class. This is the use of [12, Corollary 5.6] in its Theorem 7.2. The restricted class α_T is not assumed big, and its null face may have several rays. The contraction follows from the assembled argument in the preceding subsection, using Proposition 4.3 and the specified external results.

5 Reduction to a supported nef boundary

The positive-Iitaka-dimensional case is an application of a known Kähler abundance theorem. The purpose of this section is to prepare the remaining case for the two boundary arguments: from a nonzero divisor of a section in Iitaka dimension zero, we construct a nef dlt fourfold whose adjoint has a nonzero effective representative supported on exactly its reduced floor. We also construct the particular log resolution used to index and compare all strata of that floor.

Proposition 5.1 (Supported nef model). *Under the hypotheses of Theorem 1.1, suppose $\kappa(X, D) = 0$ and the normalized rational divisor M of a nonzero plurisection of D is nonzero. Then there are a normal ordinary $\mathbb{Q}$ -factorial compact Kähler dlt fourfold (V, B) , with effective rational boundary, and a nonzero effective rational $\mathbb{Q}$ -Cartier divisor P such that*

$$A = K_V + B \text{ is analytically nef,} \quad A \sim_{\mathbb{Q}} P, \quad \text{Supp } P = \text{Supp } [B], \quad \kappa(V, A) = 0.$$

The pair has the projective resolution described in Lemma 5.6.

The resolution has globally smooth distinct SNC strict boundary and exceptional components, all exceptional crepant coefficients are below one, and it is generically an isomorphism on the image of every intersection component of distinct strict floor primes. These properties let the floor argument index its strata by actual boundary intersections. The proposition will follow from the construction below; the final step uses the original nef adjoint to show that P cannot disappear.

We use the following analytic negativity theorem at several points. If $h : W \rightarrow Z$ is a proper bimeromorphic morphism of normal irreducible analytic spaces and E is a rational $\mathbb{Q}$ -Cartier divisor with $-E$ relatively nef, then

$$E \geq 0 \iff h_*E \geq 0.$$

This is [55, Lemma 1.3], after clearing an index. In particular, an exceptional relatively nef divisor is nonpositive. When h is projective, nonnegative degrees on its contracted curves give the relative nefness used in this theorem; see [55, Appendix B]. This relative curve test will not be used as a definition of nefness on a nonprojective compact Kähler space.

5.1 Positive Iitaka dimension

Proposition 5.2. *Under the hypotheses of Theorem 1.1, if $\kappa(X, D) \geq 1$, then D is semiample on X .*

Proof. Das–Hacon–Păun’s dlt modification theorem [12, Theorem 6.1] applies to a compact Kähler lc fourfold with effective rational boundary and $\mathbb{Q}$ -Cartier adjoint. It gives a projective bimeromorphic morphism

$$g : (X', \Delta') \longrightarrow (X, \Delta)$$

with X' compact Kähler and ordinary $\mathbb{Q}$ -factorial, the pair (X', Δ') dlt, and the adjoint crepant. The input is klt, so equality of discrepancies makes the output klt as well. The crepant equality is an equality of actual rational lines: pull a local pluriadjoint frame to a common resolution with smooth source as in (2.2), compare the crepant coefficients, and extend the identical meromorphic map across codimension two on the normal X' . Thus for a common index

$$\mathcal{O}_{X'}(m(K_{X'} + \Delta')) \simeq g^*\mathcal{O}_X(mD).$$

There is no possible undetected flat-line difference in this comparison.

The pulled-back line is analytically nef by the metric criterion. Also, normality and projection formula identify all its sections with those of $\mathcal{O}_X(mD)$. The meromorphic maps agree on the common dense open, so their Iitaka dimensions agree. Theorem 4.1 of Höring–Lazić–Lehn [31] states that a nef adjoint of a compact ordinary $\mathbb{Q}$ -factorial Kähler klt pair of positive Iitaka dimension is semiample, unconditionally in dimension at most four. Its analytic positivity and canonical-sheaf conventions are the ones in use here. Apply it to (X', Δ') , then enlarge the generated degree to a multiple of the original index. All sections are pullbacks. A section nonzero at a chosen point above $x \in X$ descends to a section nonzero at x ; Nakayama’s lemma gives the evaluation surjectivity on X . $\square$

5.2 A log-smooth supported representative

We next begin with a divisor of a section. No nefness is needed for the preparation in this subsection. The later proof that the support survives the minimal model program will use the nefness of the original adjoint.

Lemma 5.3. *Let (X, Δ) be a normal connected compact Kähler klt pair with effective rational boundary and actual $\mathbb{Q}$ -Cartier adjoint D . Assume $\kappa(X, D) = 0$, and let $M \geq 0$ be a rational $\mathbb{Q}$ -Cartier divisor with $M \sim_{\mathbb{Q}} D$. There are a projective modification $p : Y \rightarrow X$ with Y smooth*

compact Kähler, an effective rational SNC boundary B_Y , and an effective rational divisor P_Y such that

$$K_Y + B_Y \sim_{\mathbb{Q}} P_Y, \quad \text{Supp } P_Y = \text{Supp } \lfloor B_Y \rfloor, \quad \kappa(Y, K_Y + B_Y) = 0.$$

The SNC support has globally smooth distinct components, and B_Y has coefficient one on every p -exceptional prime and on every strict transform of a prime in M .

Proof. Principalize the reduced coherent ideal of $\text{Supp}(\Delta + M)$, using [12, Theorem 2.13 and Remark 2.14]. This use of an ideal is important: the original boundary need not be $\mathbb{Q}$ -Cartier. The resulting modification has smooth source, is projective, and has locally normal crossing strict and exceptional support. Mark each of its finitely many global prime components separately as an ordered Cartier boundary and apply the ordered-boundary resolution of [53, Theorems 1.1.3 and 1.1.13]. Its final indexed components and their intersections are smooth, and its complete support consists of strict transforms and new exceptional components [53, Lemmas 2.1.10 and 2.2.9(iv)]. Denote the composite by $p : Y \rightarrow X$. It is projective over the compact base. A relative ample metric plus a sufficiently large pullback of a Kähler form makes Y compact Kähler.

Let G_Y be the crepant subboundary for D , defined by the actual meromorphic pluriadjoint pullback. Then $p_*G_Y = \Delta$, and klt gives $\text{coeff}_E(G_Y) < 1$ at every prime. Define B_Y only after the preceding resolution: assign coefficient one to the actual p -exceptional primes and the strict transforms of primes in M , and retain the coefficients of Δ on the other strict boundary primes. An exceptional label in the resolution bookkeeping that is not an actual exceptional divisor does not change this rule. This is an effective rational SNC boundary, hence dlt.

Set

$$P_Y = p^*M + (B_Y - G_Y). \tag{5.1}$$

Pullback of the holomorphic equation of a Cartier multiple of M makes p^*M effective. On a strict prime of M , the second summand has coefficient $1 - \text{coeff}(\Delta) > 0$; on an exceptional prime it has coefficient $1 - \text{coeff}(G_Y) > 0$; on any other strict boundary prime it is zero. Thus P_Y is effective and its reduced support is exactly the floor of B_Y . The actual identity $p^*D \sim_{\mathbb{Q}} p^*M$, together with (2.2), gives $K_Y + B_Y \sim_{\mathbb{Q}} P_Y$.

Choose an integer $t > 0$ at least the ratios of the coefficients of P_Y to those of p^*M on every nonexceptional prime of M . The positive part of $P_Y - tp^*M$ is then an effective rational exceptional divisor E , and

$$P_Y \leq tp^*M + E.$$

For every sufficiently divisible n , exceptional Hartogs extension and projection formula give

$$H^0(Y, \mathcal{O}_Y(n(tp^*M + E))) = H^0(X, \mathcal{O}_X(ntM)).$$

The space on the right has dimension one: it is nonzero and $\kappa(X, D) = 0$. The displayed divisor inequality bounds $h^0(Y, \mathcal{O}_Y(nP_Y))$ by one, while effectivity makes it nonzero. This controls every sufficiently divisible degree. If two independent sections existed in another Cartier degree, their powers s_0^v and $s_0^{v-1}s_1$ would remain independent in a degree divisible by the fixed common index, a contradiction. Therefore $\kappa(Y, P_Y) = 0$. $\square$

5.3 Projective steps with Kähler targets

Apply the preceding lemma in dimension four. We follow the supported construction of Das–Hacon–Păun [12, Theorem 7.2] from the compact ordinary $\mathbb{Q}$ -factorial Kähler dlt pair (Y, B_Y) . We isolate its three-dimensional floor-contraction input through Subsection 4.5 and use Proposition 4.3

for the ambient contraction. The construction below gives projective steps with Kähler targets and supplies the graphs needed for the discrepancy comparison.

Write (X_i, B_i) for a nonnef running pair and $A_i = K_{X_i} + B_i$. At the start of this step, induction from P_Y gives an effective rational divisor P_i with

$$P_i \sim_{\mathbb{Q}} A_i, \quad \text{Supp } P_i = \text{Supp } \lfloor B_i \rfloor. \quad (5.2)$$

The equivalence is an isomorphism of actual rational holomorphic lines. It holds initially by the supported-model construction, and the one-step transform identity proved below supplies it at the next step only after the current step has been completed. Thus the representative used now is already available before constructing the current target.

The floor-contraction input in the proof of Theorem 7.2, using its Corollary 5.6, is the following assertion. There are a global irreducible floor prime T , a Kähler form ω_i on X_i , a projective contraction $\varphi : T \rightarrow W$ to a normal compact Kähler space with $\varphi_* \mathcal{O}_T = \mathcal{O}_W$, and a Kähler form ω_W . The class $\alpha_i = c_1(A_i) + [\omega_i]$ is nef and big but not Kähler, and

$$c_1(A_i|_T) + [\omega_i|_T] = \varphi^*[\omega_W]. \quad (5.3)$$

Here the equality is in the local-potential Bott–Chern group. The contracted compact curves are exactly those whose classes on T lie in

$$F_T = (c_1(A_i|_T) + [\omega_i|_T])^\perp \cap \text{NA}(T).$$

This face is generated by finitely many curve classes whose images in X_i lie in one A_i -negative ray $R_i \subset \alpha_i^\perp$, with $T \cdot R_i < 0$. For the global form of this selection, apply Lemma 7.1 in the setup of Theorem 7.2 together with Claim 7.3 and its proof. We choose one representative per proportionality ray among the countably many compact curve classes to meet the lemma’s nonproportionality hypothesis. A Kähler degree is positive on each curve, so proportional effective representatives differ by a positive scalar. Lemma 7.1 permits at most one representative on which the selected class vanishes, and Claim 7.3 and its proof supply it. Consequently the selection gives the global implication

$$\alpha_i \cdot C = 0 \implies [C] \in R_i \quad \text{for every compact irreducible curve } C \subset X_i. \quad (5.4)$$

For a positive global Cartier multiple mT , the line $\mathcal{O}_T(-mT)$ is φ -ample. The use of Corollary 5.6 here is the actual-line specialization proved in Subsection 4.5, with the inputs stated in Subsection 4.4. In particular, the restriction to T need not be big and T need not be $\mathbb{Q}$ -factorial. The numerical selection of T , the ray, and the negative normal line is the one in the supported construction. Its threefold contraction is supplied by the bridge, using Proposition 4.3 and the external results stated there.

Write $B_i = T + C$, where $C \geq 0$ has no component T . Ordinary $\mathbb{Q}$ -factoriality makes C a global rational $\mathbb{Q}$ -Cartier divisor. For a positive rational $\varepsilon < 1$, put

$$B_i(\varepsilon) = T + (1 - \varepsilon)C, \quad A_i(\varepsilon) = A_i - \varepsilon C = K_{X_i} + B_i(\varepsilon).$$

The second identity is an identity of actual rational adjoint lines. The perturbed pair is plt. Indeed, choose a log resolution $f : U \rightarrow X_i$ with smooth source that witnesses the standard dlt criterion for (X_i, B_i) . Its strict boundary and exceptional primes have simple normal crossings, and all exceptional crepant coefficients are below one. If G is its crepant boundary, the new boundary on this resolution is $G - \varepsilon f^*C$. The divisor f^*C is effective: pull back a local holomorphic equation

for an effective Cartier multiple of C . It follows that all exceptional coefficients remain below one. The strict transform of T is the only coefficient-one prime, and all other strict coefficients are below one. The plt criterion on this resolution now applies. This is the ordinary Kollár–Mori convention used in Section 2 of [9]. If $C = 0$, the same argument applies to the unchanged pair (X_i, T) .

Choose a smooth Hermitian metric on an actual Cartier multiple of C , and let θ_C be its normalized curvature form. On a finite relatively compact local embedding cover of the compact X_i , choose smooth ambient extensions of the local potentials and metric weights. After shrinking the charts, the Levi forms for ω_i have a uniform positive lower bound on their compact closures, while those for θ_C are bounded. Thus $\omega_i + \varepsilon\theta_C$ is Kähler for one sufficiently small positive rational ε , chosen for this running step. No positivity of $C|_T$ is asserted. Equation (5.3) gives

$$-c_1(A_i(\varepsilon)|_T) + \varphi^*[\omega_W] = [\omega_i|_T + \varepsilon\theta_C|_T]. \quad (5.5)$$

The right side is Kähler. Locally on W , a potential for ω_W can be pulled back, so this equality says precisely that $-A_i(\varepsilon)|_T$ is relatively Kähler over W . After clearing its index, the same local metric weights restrict to positive weights on every full φ -fiber. The proper fiberwise criterion [22, Corollary 1.12, Definition 3.1 and Remark 3.2] therefore makes this actual rational line φ -ample. The other required positivity, that of $-T|_T$, is unchanged. With the boundary $(1 - \varepsilon)C$, these are the hypotheses of Proposition 4.1; its conormal direct-image vanishings hold for the present floor contraction.

Proposition 4.3 now constructs a proper bimeromorphic morphism $c : X_i \rightarrow Z_i$ to a normal compact analytic space. It restricts to φ on T , embeds W as the reduced image of T , has exactly the φ -fiber sets over W , and is an isomorphism away from T . Its fibers are connected. A sufficiently divisible global line $\mathcal{O}_{X_i}(-\ell T)$ is c -ample, and the actual rational line $-A_i(\varepsilon)$ is c -ample as well. The target is in Fujiki’s class $\mathcal{C}$. The proposition proves these assertions through the analytic thickenings and the full-fiber criterion. For this constructed contraction, we use the rationality assertion of [9, Remark 5.11]. The target Kähler class is proved below.

Every curve contracted by c is a φ -contracted curve in T , and therefore has class in R_i . Conversely a curve in R_i lies in T , since $T \cdot R_i < 0$ whereas the effective Cartier section of a multiple of T has nonnegative degree on the normalization of any curve not contained in T . The floor assertion then contracts it. The constructed morphism thus contracts precisely the compact curves whose classes lie in the original negative ray. The boundary transported in the supported program remains B_i ; the plt boundary is used only to construct c .

Claim 5.4 (Kähler positivity on the current target). *Retain the current pair (X_i, B_i) , its effective rational divisor P_i with the actual line identity in (5.2), the nef and big class $\alpha_i = c_1(A_i) + [\omega_i]$ with ω_i Kähler, and the selected ray and floor in (5.3)–(5.4). Let $c : X_i \rightarrow Z_i$ be the projective contraction with connected fibers just constructed. It is an isomorphism away from T , has the embedded floor image W and the exact φ -fiber sets, and contracts precisely the curves in R_i . Its target is a normal compact analytic space in class $\mathcal{C}$ with rational singularities. For every other floor prime, retain the projective contraction with connected fibers supplied by Subsection 4.5, whose target is normal compact Kähler and whose pulled-back Kähler class is the restricted α_i . Then there is a Kähler form ω_{Z_i} on Z_i such that*

$$\alpha_i = c^*[\omega_{Z_i}] \quad \text{in } H_{BC}^{1,1}(X_i), \quad (5.6)$$

where the group is defined by local smooth potentials.

Proof. Bott–Chern descent first supplies a target class whose pullback is α_i . We prove that this class is nef, contains a current dominating a positive Hermitian form, and has positive top intersection on every positive-dimensional compact reduced subspace. These are the three conditions in [12, Theorem 2.29] that make the target class Kähler.

The descended class and its floor restriction. In this paragraph and the next three, write $X = X_i$, $Z = Z_i$, and $\alpha = \alpha_i$. We use $dd^c = \frac{i}{2\pi}\partial\bar{\partial}$, the normalization for which $dd^c\phi$ is the normalized Chern curvature of a metric locally written as $e^{-\phi}$. A fixed positive rescaling of potentials gives the $i\partial\bar{\partial}$ convention in the cited analytic inequalities.

Both ends of c are normal compact spaces in class $\mathcal{C}$ with rational singularities, and α is zero on every contracted curve. The Bott–Chern descent [11, Lemma 2.11] therefore gives a class β on Z and a smooth real closed representative β_0 , with local smooth potentials, such that

$$c^*\beta = \alpha \quad \text{in } H_{BC}^{1,1}(X). \quad (5.7)$$

Here and below the group is the local-potential Bott–Chern group. Its smooth representative description, including adjustment by a global smooth dd^c -potential, is [33, Definition 3.1 and Remark 3.2].

Let $i_W : W \hookrightarrow Z$ be the embedded floor image and put $\delta = i_W^*\beta$. The actual identity $c|_T = i_W \circ \varphi$, (5.3), and (5.7) give

$$\varphi^*\delta = \alpha|_T = \varphi^*[\omega_W] \quad \text{in } H_{BC}^{1,1}(T). \quad (5.8)$$

Thus the pullback of $\delta - [\omega_W]$ is zero and hence nef. Apply [12, Lemma 2.38] only to $\varphi : T \rightarrow W$: it is proper and surjective, and both spaces are normal compact Kähler. The lemma makes $\delta - [\omega_W]$ nef. For a smooth representative e of that difference, choose a nef approximation $e + dd^cu \geq -\omega_W/2$. Then $\omega_W + e + dd^cu$ is a Kähler representative of δ . In particular its restriction gives a positive current in the class $\beta|_V$ and a positive top integral for every positive-dimensional irreducible $V \subset W$. Only the nefness of $\delta - [\omega_W]$ enters this deduction.

Nefness on the target. For every positive-dimensional irreducible reduced compact subspace $V \subset Z$, we first produce a current

$$T_V = \beta_0|_V + dd^cq_V \geq 0 \quad (5.9)$$

with a global real distribution q_V on V . Currents and their positivity on the pure-dimensional reduced V are defined by duality with smooth test forms from local ambient embeddings; see [16, Section 1, Definitions 1.1–1.2, pp. 14–15]. We will use exactly this positive-current meaning of pseudoeffectivity on a possibly nonnormal subspace. The preceding Kähler representative of δ gives (5.9) for $V \subset W$, with a smooth potential.

Suppose $V \not\subset W$, and let $V' \subset X$ be the closure of the inverse image of $V \setminus W$. The isomorphism $X \setminus T \simeq Z \setminus W$ makes V' irreducible and its map to V proper and bimeromorphic. Choose a projective resolution $r : U \rightarrow V'$ with smooth source. The restricted Kähler form makes V' a compact Kähler space. A relatively positive metric for the projective resolution, plus a sufficiently large pullback of this form, makes U a compact Kähler manifold. This metric construction applies to the possibly nonnormal V' .

Let $\nu : V^\nu \rightarrow V$ be normalization. The dominant map from the normal U factors through ν , giving a proper bimeromorphic map $p : U \rightarrow V^\nu$, hence a modification between normal compact spaces. For the other map $j : U \rightarrow X$, functoriality of the actual maps and (5.7) gives

$$p^*\nu^*(\beta|_V) = j^*\alpha.$$

The class on the right is nef. Indeed, pull back the smooth nef approximations from X ; on compact U , the pullback of their reference form is bounded above by a fixed multiple of a Kähler form η on U , so rescaling the error gives the nef inequalities.

Set $\gamma = p^*\nu^*(\beta_0|_V)$. Choose smooth approximations $\gamma + dd^c f_n \geq -n^{-1}\eta$. The closed positive currents $\gamma + dd^c f_n + n^{-1}\eta$ have bounded η -mass by Stokes' theorem. Weak compactness gives a positive closed limit in $[\gamma]$. The $\partial\bar{\partial}$ -lemma on the compact Kähler manifold U , and the global potential description of a positive current, give a global quasi-plurisubharmonic q_U with

$$\gamma + dd^c q_U \geq 0.$$

These are also the closed-cone statement following Definition 1.6 and the potential description in (3.1) and its following paragraph in [17, pp. 1253 and 1260]. Apply [12, Corollary 2.32 and its proof, p. 18] to p , with target form $\nu^*(\beta_0|_V)$ and constant zero in its current inequality. The locally integrable, locally bounded above q_U supplies the required potential. We obtain a global quasi-plurisubharmonic q_ν satisfying

$$T_\nu = \nu^*(\beta_0|_V) + dd^c q_\nu \geq 0. \quad (5.10)$$

This applies the normal-modification descent to the displayed local-potential inequality on the still general normal compact target.

It remains to push through the finite normalization. Locally write $\beta_0|_V = dd^c h$ with h smooth. The function $\nu^*h + q_\nu$ is locally integrable and locally bounded above, and its Hessian is $T_\nu \geq 0$. On normal V^ν , its upper regularization is plurisubharmonic. The finite trace is weakly plurisubharmonic and its Hessian is the positive current ν_*T_ν by [16, Corollary 1.11 and Proposition 1.13(a), p. 22]. Since ν has generic degree one, change of variables off the proper analytic exceptional sets gives $\nu_*\nu^*(\beta_0|_V) = \beta_0|_V$ as currents; those sets have measure zero for the smooth top-degree integrands. Proper push of distributions commutes with dd^c . Hence

$$T_V = \nu_*T_\nu = \beta_0|_V + dd^c(\nu_*q_\nu) \geq 0. \quad (5.11)$$

The traced distributions are locally integrable and glue because q_ν is global. On a locally reducible V , these local potentials may be only weakly plurisubharmonic; the argument below uses their associated positive currents.

We now apply the sufficient direction of the restriction criterion [12, Theorem 2.36 and Remark 2.37], spelling out its singular-current input. Proposition 3.3(iv), p. 1262, and the following paragraph, pp. 1262–1263, of [17] applies to a compact complex space and a smooth class containing a closed positive current. It gives nefness if the restrictions to the irreducible components of every positive Lelong level set are nef. Induct on the dimension of each irreducible $V \subset Z$; points are automatic. For a positive-dimensional V , use (5.9). The Siu analyticity assertion accompanying that proposition makes each positive Lelong level set analytic, and it is proper. Indeed, if a fixed positive level filled V , pack disjoint radius- r balls in a coordinate ball of V_{reg} . The Lelong mass lower bound on each ball would force the locally finite mass of the $(1,1)$ -current to grow at least as a positive multiple of r^{-2} when r tends to zero. This is impossible. The level set components therefore have smaller dimension, and their restricted classes are nef by induction. The proposition makes $\beta|_V$ nef. Taking $V = Z$ proves analytic nefness of β . This uses the positive-current formulation above also on nonnormal subspaces, where the local potentials may be only weakly plurisubharmonic.

Bigness from the supported section. Use the effective actual representative already present in (5.2). Choose $m > 0$ clearing the indices and an isomorphism of holomorphic lines

$$L = \mathcal{O}_X(mA_i) \simeq \mathcal{O}_X(mP_i).$$

The canonical section of the right side gives a nonzero holomorphic section s of L with divisor mP_i . Choose a smooth metric h on L . In a local frame write $h = e^{-\phi}$, $s = f$, and put

$$\vartheta = \frac{1}{m} dd^c \phi, \quad \ell = \frac{1}{m} \log |s|_h^2.$$

The first formula defines the normalized global curvature form. The second is a global quasi-plurisubharmonic function, locally $m^{-1}(\log |f|^2 - \phi)$. The holomorphic f extends in a local ambient embedding and is not identically zero on any nonempty open subset of that chart in X . Local integrability is Proposition 1.8 of [16, p. 19], and the unnumbered prose after its proof gives the current positivity

$$\vartheta + dd^c \ell = \frac{1}{m} dd^c \log |f|^2 \geq 0.$$

The smooth form $\vartheta + \omega_i$ represents $\alpha = c^* \beta$, so the smooth local-potential representative description supplies a global smooth v with $\vartheta + \omega_i = c^* \beta_0 + dd^c v$. Thus

$$c^* \beta_0 + dd^c(v + \ell) \geq \omega_i. \quad (5.12)$$

This is a global quasi-plurisubharmonic potential obtained from the supported section at this running step.

Fix any smooth positive Hermitian form g_Z on Z in local ambient embeddings. Local holomorphic coordinate lifts of c make $c^* g_Z$ semipositive in those embeddings. A finite relatively compact cover of X and comparison with the positive definite ambient representatives of ω_i give one $C > 0$ with $c^* g_Z \leq C \omega_i$. Corollary 2.32 of [12] applies to the normal modification c and (5.12). It gives a global quasi-plurisubharmonic, hence L^1 , function v_Z with

$$\beta_0 + dd^c v_Z \geq C^{-1} g_Z. \quad (5.13)$$

This is the required dominating current in the selected target class.

Top intersections and the other floor. Let $V \subset Z$ be irreducible of dimension $k > 0$. The case $V \subset W$ was proved using the Kähler class δ . Otherwise use its strict transform V' above. The proper integration cycle identity $c_*[V'] = [V]$, projection formula, and (5.7) give

$$\int_{V_{\text{reg}}} \beta_0^k = \int_{V'_{\text{reg}}} \alpha^k. \quad (5.14)$$

The right side denotes the integral of any smooth representative of α ; Stokes' theorem makes it independent of that choice. Neither subspace needs to be normal.

If $V' \not\subset \text{Supp } P_i$, take the projective resolution $r : U \rightarrow V'$ with smooth Kähler source used above. The pullback of s is nonzero and defines the effective rational divisor $E = m^{-1} \text{div}(r^* s)$ on U . Put $a = r^*(\alpha|_{V'})$ and $w = r^*[\omega_i|_{V'}]$. The class a is nef, w has a semipositive smooth representative, and the actual section gives $a - w = c_1(E)$. Therefore

$$\int_U a^k - \int_U w^k = \sum_{j=0}^{k-1} c_1(E) \cdot a^{k-1-j} w^j \geq 0. \quad (5.15)$$

For each summand, choose a smooth representative of $a + \varepsilon[\eta]$ which is semipositive, where η is a fixed Kähler form on U . Restrict it and the semipositive representative of w to every effective divisor component, integrate, and let ε tend to zero. This proves its nonnegativity, including the degree statement when $k = 1$. Projection formula gives $\int_U w^k = \int_{V'_{\text{reg}}} \omega_i^k > 0$. Equations (5.14)–(5.15) give the desired strict positivity.

It remains that V' lies in a component T' of $\text{Supp } P_i$. It meets $X \setminus T$, so $T' \neq T$. The actual floor specialization in Subsection 4.5 gives

$$g : T' \rightarrow \overline{W}, \quad \alpha|_{T'} = g^* \kappa, \quad (5.16)$$

where g is projective with connected fibers, its source and target are normal compact Kähler, and κ is a Kähler class. The equality is in the local-potential Bott–Chern group. Every irreducible curve in a g -fiber has α -degree zero, so (5.4) puts its class in R_i , and c contracts it.

It follows that $h = c|_{T'}$ is pointwise constant on every full g -fiber. To see this also for nonreduced fibers, their reductions are connected projective varieties, possibly reducible. If the image of such a reduction were not a point, one irreducible component Q would have nonconstant image; otherwise its image would be a connected finite set. At a smooth point of Q where a local coordinate of h has nonzero differential, general hyperplanes through that point cut an irreducible projective curve component whose tangent is not in the differential's kernel. That curve has nonconstant image, a contradiction. If $\dim Q = 1$, take Q itself.

There is consequently a holomorphic factorization

$$h = b \circ g, \quad b : \overline{W} \rightarrow Z. \quad (5.17)$$

For clarity, the specialization gives $g_* \mathcal{O}_{T'} = \mathcal{O}_{\overline{W}}$; this also follows from analytic Stein factorization, connected fibers, and the normality of the two spaces. Proper surjectivity makes g a closed quotient map, so the pointwise constancy first defines a continuous b . For an open $O \subset Z$ embedded in a polydisc, restrict to the open $b^{-1}(O)$. The coordinate functions of h on its full inverse image descend through $g_* \mathcal{O}_{T'} = \mathcal{O}_{\overline{W}}$ to a holomorphic tuple. It lies pointwise in the embedded O ; its defining ideal therefore vanishes on the reduced open $b^{-1}(O)$. The resulting local holomorphic maps glue to (5.17). This factorization uses pointwise constancy on the underlying full fibers.

Put $H = g(V')$, a closed irreducible analytic subspace of $\overline{W}$. Since $c|_{V'}$ is bimeromorphic onto V , (5.17) yields

$$k = \dim c(V') = \dim b(H) \leq \dim H \leq \dim V' = k.$$

Thus $g|_{V'}$ is generically finite onto H , of some integer degree $d > 0$. The projection formula using (5.16) now gives

$$\int_{V_{\text{reg}}} \beta_0^k = \int_{V'_{\text{reg}}} (g^* \kappa)^k = d \int_{H_{\text{reg}}} \kappa^k > 0. \quad (5.18)$$

The last integral is a positive Kähler volume. The one-way factorization $h = b \circ g$ and this degree calculation give the needed implication in [12, proof of Theorem 7.2, p. 48].

These cases cover every irreducible positive-dimensional V . Additivity over top-dimensional irreducible components gives strict positivity for every positive-dimensional compact reduced subspace. We have proved analytic nefness of β , the dominating current (5.13), and all positive top intersections. The normal compact Z and the smooth real closed β_0 therefore satisfy the three hypotheses of [12, Theorem 2.29]. That theorem gives a positive smooth representative of β . Its local smooth potentials make this a Kähler form ω_{Z_i} . Equation (5.7) gives (5.6), and Z_i is compact Kähler. $\square$

For a small step, the construction of its flipped morphism also has to respect the hypotheses of the relative canonical-model theorem. Use the current representative in (5.2), and write $P_i = \sum p_j S_j$ with every $p_j > 0$ and S_j the floor primes. At a nontrivial step this list is nonempty, since otherwise $A_i \sim_{\mathbb{Q}} 0$ is nef. Independently choose a rational $0 < \varepsilon < 1$ small enough that

$$B_i^\varepsilon = B_i - \varepsilon P_i$$

is effective. Its floor coefficients are below one. On a dlt resolution, subtracting the pullback of the effective $\mathbb{Q}$ -Cartier divisor εP_i can only lower the exceptional crepant coefficients, which were already below one. Hence (X_i, B_i^ε) is klt. Its actual adjoint satisfies

$$K_{X_i} + B_i^\varepsilon = A_i - \varepsilon P_i \sim_{\mathbb{Q}} (1 - \varepsilon) A_i. \quad (5.19)$$

The projective contraction is bimeromorphic, so this adjoint is relatively big. Corollary 3.7 of [12] therefore applies at its stated klt scope. The actual isomorphism in (5.19) identifies common Cartier Veroneses of its relative algebra and of the relative A_i -algebra. On the compact base the latter is thus finitely generated. A further Veronese is generated in degree one, and its coherent degree-one piece embeds its relative Proj in the associated relative projective space. The tautological line is globally relatively ample. The ordinary small flip supplied by Theorem 7.2 is this relative canonical model; in a common degree the tautological line agrees with the flipped adjoint line on the common codimension-one open and hence everywhere by reflexive extension. The flipped morphism is projective and the flipped adjoint is positive on its contracted curves. Thus Corollary 3.7 is applied to the klt perturbation, whose common Veronese is identified with that of the original adjoint.

The transform identities follow by induction from P_Y , one completed step at a time. Divisorial contractions remove their contracted primes and flips change no prime valuation. On the open containing all codimension-one points of the destination, the fixed meromorphic pluriadjoint section identifies its line with the strict transform P_{i+1} . After clearing the new indices, reflexive extension gives

$$P_{i+1} \sim_{\mathbb{Q}} A_{i+1}, \quad \text{Supp } P_{i+1} = \text{Supp } [B_{i+1}].$$

This establishes the identity needed for the perturbation at the next step. It uses ordinary $\mathbb{Q}$ -factoriality only for the finitely many global prime transforms.

We have now constructed a projective step, proved that its target is Kähler, and supplied the effective actual representative needed to repeat the construction. The factorial/dlt preservation and the special-termination argument in [12, proof of Theorem 7.2] apply to the unchanged original boundary. That construction, with Proposition 4.3 at its ambient extension steps and the target positivity proved above, therefore gives a finite supported sequence to an analytically nef adjoint.

For a flip, take the main component of $X_i \times_{Z_i} X_{i+1}$ as its graph; its projections are projective. For a divisorial contraction the source is its graph. The main component of the iterated fiber product of these finitely many graphs, followed by normalization, is a normal common graph Γ . Normalization is finite and projective, and projective morphisms compose over a compact base [12, Remark 2.11]. Thus Γ is projective over every running model. Projective resolutions of it supply all common resolutions with smooth source used below. Their sources are compact Kähler by the relative metric construction.

On a common projective resolution of one step with smooth source W_i , let $r : W_i \rightarrow X_i$ and $s : W_i \rightarrow X_{i+1}$ be the maps, and put

$$E_i = r^* P_i - s^* P_{i+1}. \quad (5.20)$$

It is s -exceptional by the codimension-one comparison. For an s -contracted curve, its image under r is a point or a curve in the negative source contraction fiber, so E_i has nonpositive degree. Negativity gives $E_i \geq 0$. In any common Cartier degree,

$$\begin{aligned} H^0(X_i, \mathcal{O}_{X_i}(mP_i)) &= H^0(W_i, \mathcal{O}_{W_i}(mr^*P_i)) \\ &= H^0(W_i, \mathcal{O}_{W_i}(ms^*P_{i+1} + mE_i)) \\ &= H^0(X_{i+1}, \mathcal{O}_{X_{i+1}}(mP_{i+1})). \end{aligned}$$

The last equality is exceptional Hartogs extension for s . The identifications agree with the fixed section on the common open and therefore respect multiplication. A common Veronese section ring is unchanged through the finite chain. The power argument in Lemma 5.3 handles other degrees, so the final pair (V, B) , with $A = K_V + B$ and P the final transform, satisfies

$$A \sim_{\mathbb{Q}} P \geq 0, \quad \text{Supp } P = \text{Supp } [B], \quad \kappa(V, A) = 0. \quad (5.21)$$

5.4 Discrepancy increase and a resolution of the floor

We now show that the final pair has a resolution which is generically unchanged along every floor intersection. The point is that a negative step strictly increases discrepancies over all its nontrivial fibers, including those on the positive side of a flip.

Lemma 5.5 (Support on a projective fiber). *Let $h : W \rightarrow Z$ be a projective morphism between normal analytic spaces with connected fibers, let F be a scheme fiber, and let $E \geq 0$ be a rational $\mathbb{Q}$ -Cartier divisor on W . Suppose that $E \cdot C \leq 0$ for every irreducible curve C in F . Then either $\text{Supp } E \cap F = \emptyset$ or $F \subseteq \text{Supp } E$.*

Proof. Clear the index and pass to the reduction of F . If an irreducible component F_α is not contained in $\text{Supp } E$ but meets it, the restricted canonical section cuts a nonempty effective Cartier divisor on the integral projective F_α . This component has positive dimension: a point that is an irreducible component cannot meet another component, and a connected zero-dimensional fiber is a point. Cutting by sufficiently general hyperplanes of a very ample line on F_α gives a curve on which E has strictly positive degree, a contradiction. If some components are contained in the support and others are not, connectedness gives a noncontained component meeting a contained one and the same contradiction. These alternatives prove the claim, with no reducedness assumption on the original fiber. $\square$

Apply the lemma to a common resolution over the contraction base of one step and to E_i in (5.20). The maps to the contraction base have connected fibers: they are proper bimeromorphic over a normal space. On a base-fiber curve C , a divisorial step has $E_i \cdot C = (r^*A_i) \cdot C \leq 0$. For a flip,

$$E_i \cdot C = (r^*A_i) \cdot C - (s^*A_{i+1}) \cdot C \leq 0,$$

because the source adjoint is negative on its contracted curves and the flipped adjoint is positive on its contracted curves. Above any point where either side has a nontrivial fiber, a curve in the common fiber can be chosen to dominate a curve of that side: take a component over the curve and cut by relative ample hyperplanes. The corresponding term in the preceding degree is then strictly negative. Thus E_i meets the common fiber, and Lemma 5.5 puts the entire fiber in $\text{Supp } E_i$.

Use the fixed meromorphic pluriadjoint section to calculate discrepancies on the resolution. Its divisor as a section of the old pulled-back line is r^*P_i , and as a section of the new line is s^*P_{i+1} .

The two meromorphic pluriforms agree. Their difference is consequently the difference of the two crepant boundary divisors, so for every prime F

$$\text{coeff}_F(E_i) = a(F; X_{i+1}, B_{i+1}) - a(F; X_i, B_i). \quad (5.22)$$

The same formula applies on higher projective resolutions after pullback.

For each contraction base Z_i , let $C_i \subset Z_i$ consist of the points whose fiber on the source or, for a flip, on the destination has positive dimension. This is a closed analytic subset of the base by properness and the fiber-dimension theorem. A proper bimeromorphic morphism to a normal space is an isomorphism near any zero-dimensional fiber, since it is finite there. Write $v : \Gamma \rightarrow V$ and $q_i : \Gamma \rightarrow Z_i$ for the projections from the normal common graph to the final model and the contraction bases. Set

$$D_\Gamma = \bigcup_i q_i^{-1}(C_i), \quad B_V = v(D_\Gamma), \quad U_V = V \setminus B_V. \quad (5.23)$$

The sets D_Γ and B_V are closed analytic, the latter by properness. We claim that

$$v^{-1}(B_V) = D_\Gamma. \quad (5.24)$$

Indeed, at a point $\gamma \notin D_\Gamma$, both sides of every step are isomorphisms near the corresponding point of Z_i . Restricting to these neighborhoods makes their iterated graph a common normal open, so normalization does not change it and v is an isomorphism near γ . Thus $\{\gamma\}$ is open in its v -fiber, and it is closed because the spaces are Hausdorff. The fibers of v are connected: a proper bimeromorphic morphism between normal spaces has $v_*\mathcal{O}_\Gamma = \mathcal{O}_V$, and its Stein factorization has connected fibers. That fiber is therefore $\{\gamma\}$, so it cannot meet D_Γ . This proves (5.24).

The corresponding open $U_Y \subset Y$ is isomorphic to U_V through every step, because every successive lift there is unique. No step extracts a divisor, so the generic point of any prime on V follows a surviving prime through the chain and avoids the contraction centers. Therefore

$$\text{codim}_V(V \setminus U_V) \geq 2. \quad (5.25)$$

Every final log-canonical center tested by a prime on a projective resolution meets U_V . To see this, suppose the final discrepancy of that prime is zero. Initial log canonicity, $E_i \geq 0$, and (5.22) force all intermediate discrepancies and all coefficients in the E_i to be zero. If its center lay over some C_i , the fiber-support conclusion would place that center in $\text{Supp } E_i$. A local equation of a positive Cartier multiple of E_i would then have positive order in the valuation, a contradiction. If its center on V were contained in B_V , its center on the proper graph Γ would be contained in $v^{-1}(B_V) = D_\Gamma$. Irreducibility would then put that entire center in one $q_i^{-1}(C_i)$. On a higher projective resolution carrying the prime, the center would lie over C_i , giving the contradiction just proved. Thus it meets U_V , as claimed. We only need this statement for primes on the projective resolutions constructed here and for the blowups of their floor strata.

Lemma 5.6 (A globally simple dlt resolution). *The final pair (V, B) has a projective log resolution $\pi : \hat{V} \rightarrow V$ with the following properties:*

1. *the strict boundary and exceptional support have globally smooth distinct SNC components;*
2. *every π -exceptional crepant coefficient is strictly below one;*
3. *if $\hat{Z}$ is an irreducible component of the nonempty intersection of distinct strict transforms of floor primes, then π is generically an isomorphism on $\pi(\hat{Z})$.*

Moreover π is an isomorphism over U_V .

Proof. Write $b : \Gamma \rightarrow Y$ for the normal common graph. Choose a relatively very ample and generated line $\mathcal{L}$ for b ; compactness permits one global power. Its evaluation embeds Γ in $\mathbb{P}_Y(\mathcal{E})$, where $\mathcal{E} = b_*\mathcal{L}$. This coherent sheaf is torsion-free of rank one: a section killed by a nonzero function vanishes on the dense isomorphism locus and hence everywhere. Since Y is smooth, $\mathcal{E}^{**}$ is a line, and

$$\mathcal{J} = \mathcal{E} \otimes (\mathcal{E}^{**})^{-1} \subset \mathcal{O}_Y$$

is a coherent ideal equal to $\mathcal{O}_Y$ on U_Y .

Apply the ideal principalization functor of [53, Theorems 1.1.11 and 1.1.13] to $\mathcal{J}$. For smooth input it is supported on the cosupport of the ideal, so it avoids U_Y ; its total transform is invertible on a smooth space Y_1 . The resulting invertible quotient of the pulled-back $\mathcal{E}$ defines a map $Y_1 \rightarrow \mathbb{P}_Y(\mathcal{E})$. It lands in the closed subspace Γ on a dense open, and hence everywhere because Y_1 is reduced. The map $Y_1 \rightarrow \Gamma$ is projective: its graph is a closed immersion into the base change of the projective map $Y_1 \rightarrow Y$. Its composite to V is projective and is an isomorphism over U_V .

This principalization is not required to have transverse intermediate centers. Instead, after it is complete, mark as separate ordered Cartier boundary components all strict initial boundary primes, all primes in the principalized support, and all divisorial exceptional primes. On U_Y this is the original globally simple SNC list. Apply the ordered-boundary functor of [53, Theorems 1.1.3 and 1.1.13]. Its centers lie over the original labeled bad locus and therefore avoid U_Y . The final indexed components and intersections are smooth, and the total support consists of the strict support and new exceptional support. The already invertible ideal remains invertible, so the map to Γ persists.

Finally apply the same two stages to the pullback of the reduced coherent ideal of $V \setminus U_V$. In the second stage mark, each distinct prime once, every component of the prior resolved support, every component of the newly principalized support, and every divisorial exceptional prime. Its total transform has divisorial support equal to the entire inverse image of $V \setminus U_V$; every such divisor maps to a subset of codimension at least two by (5.25). Outside this support the resulting map $\pi : \widehat{V} \rightarrow V$ is an isomorphism. Conversely a point on this support cannot be a local isomorphism point, because the divisor germ through it would map to a hypersurface contained in $V \setminus U_V$. Thus this is exactly the nonisomorphism locus. The complete support is globally simple SNC and includes every strict boundary component of V .

Each π -exceptional prime has center in $V \setminus U_V$. The preceding discrepancy argument shows that its log discrepancy is positive, proving property (2). Let $\widehat{Z}$ be an irreducible intersection of k distinct transformed floor components. At its general point exactly those components occur, by SNC. If $k = 1$, its prime valuation has log discrepancy zero. If $k \geq 2$, the blowup of the smooth stratum has log discrepancy $\text{codim}(\widehat{Z}) - k = k - k = 0$. This is a prime on a projective resolution, so its image meets U_V . Since π is an isomorphism there, it is generically an isomorphism on that image. This proves property (3). $\square$

5.5 The support survives

Proof of Proposition 5.1. All statements except $P \neq 0$ have been established. Suppose that $P = 0$. Every component of P_Y , and hence every component of p^*M , has disappeared. On a common projective resolution with smooth source $r : W \rightarrow Y$, $s : W \rightarrow V$, set $Q = r_*p^*M$. It is effective and nonzero: the pullback of a nonzero effective Cartier multiple has a nonzero strict transform. It is s -exceptional. Indeed, a prime of W mapping onto a prime of V corresponds, by no extraction,

to a surviving prime of Y ; its coefficient in Q is zero under the supposition that every component of p^*M disappeared.

The actual rational line of Q is $(pr)^*D$. It is analytically nef on the compact Kähler W , by pullback of the metric inequalities for the original D . In particular it is relatively nef for the projective s . Exceptional negativity gives $Q \leq 0$, contradicting its effectivity and nonvanishing. Thus $P \neq 0$. $\square$

The construction uses the original section but supplies no new ambient section. Its floor line is already defined by restricting a Cartier multiple of A to the reduced subspace $S = \lfloor B \rfloor$. The next sections will generate that line on all of S , after which supported lifting contradicts the last equality in Proposition 5.1.

6 Adjunction on the entire dlt floor

We now prove the boundary-generation statement applied to the model supplied by Proposition 5.1. The adjoint line already exists on the reduced floor as a restriction from the ambient space. Our task is to construct enough sections of that line which agree through all of its intersections. This section sets up the strata and the actual residue identities; the next three sections construct the compatible sections.

Theorem 6.1 (Generation on the whole dlt floor). *Let (V, B) be a normal ordinary $\mathbb{Q}$ -factorial compact Kähler dlt fourfold with effective rational boundary and analytically nef actual adjoint $A = K_V + B$. Suppose that it has a projective log resolution $\pi : \hat{V} \rightarrow V$ for which the strict boundary and exceptional support have globally smooth distinct SNC components, every exceptional crepant coefficient is below one, and π is generically an isomorphism on the image of every irreducible component of an intersection of distinct strict transforms of coefficient-one boundary primes. Then the actual restriction $A|_S$ to the entire reduced floor $S = \lfloor B \rfloor$ is semiample.*

If S is empty the conclusion is vacuous. Otherwise its components are three-dimensional. We may work on each connected component of V and take a common multiple over the finitely many components, so in the proof we assume V connected and hence irreducible. The theorem needs no section of the ambient adjoint. Its resolution hypothesis is precisely the output of Lemma 5.6, so it does not add a hypothesis to Theorem 1.1.

6.1 Local adjunction calculations

Lemma 2.1 supplies the connectedness and rationality properties used for successive strata. We next compare an adjunction coefficient on a normal surface slice.

Lemma 6.2 (A normal surface slice for an adjunction coefficient). *Let Z be a normal Cohen–Macaulay irreducible analytic space of dimension $n \geq 2$, let D be a normal prime divisor, and let $B = D + B'$ be an effective rational boundary with actual $\mathbb{Q}$ -Cartier adjoint. Fix a divisible degree m , a local frame of $\mathcal{O}_Z(m(K_Z + B))$, and its meromorphic m -residue on D . For a prime $\Gamma \subset D$, a general point of Γ admits a local slice by $n - 2$ holomorphic parameters with the following properties. The slice T is a normal surface germ, the cut $C = D \cap T$ is a smooth curve germ transverse to Γ , and the restricted line is the actual m -adjoint line of the effective sliced boundary on T . After division by the base parameter volume, the order along Γ of the residue on D equals the order at $C \cap \Gamma$ of the surface residue on C . For $n = 2$, the slice is the surface germ itself.*

Proof. Work in a relatively compact local embedding in $\mathbb{C}^N$, and refine the regular loci of Z , its singular locus, D , Γ , the boundary primes, and their intersections into a locally finite collection of smooth strata. After a neighborhood shrink only finitely many relevant strata meet the chosen compact closure. Choose a linear map $\ell : \mathbb{C}^N \rightarrow \mathbb{C}^{n-2}$ whose differential is an isomorphism on the tangent space of Γ at a general smooth point where D is smooth. Such points exist because D is normal. For the restriction of ℓ to each smooth stratum of dimension at least $n-2$, Sard's theorem makes its critical values a measure-zero subset of the parameter space; strata of smaller dimension have measure-zero image [48]. The image of Γ contains a neighborhood of the chosen value. We may therefore choose a nearby value regular for all these restrictions and still meeting Γ at a general point. The corresponding level is transverse to every stratum it meets.

At regular points of Z the level is a smooth surface. The singular locus of a normal n -fold has dimension at most $n-2$; its strata therefore meet this level only discretely. Every irreducible component of the level has dimension at least two by the principal ideal theorem. It cannot be contained in that discrete singular intersection, and hence has dimension exactly two by its smooth dense part. The $n-2$ level equations have height $n-2$ at every point. In the Cohen–Macaulay local rings of Z they are a regular sequence. The quotient is consequently Cohen–Macaulay of dimension two. It is generically reduced and regular in codimension one, since its possible singular points are discrete; Serre's criterion makes it normal. Transversality on D gives a smooth curve at the cut point, and the fact that $d\ell$ is an isomorphism on $T\Gamma$ makes this curve transverse to Γ .

Each boundary prime cuts an effective curve with its original rational coefficient, counted with its positive intersection multiplicity. At a generic codimension-one point of the normal surface, the ambient space and the slice are smooth and transverse to these primes. Dividing the ambient canonical form by the parameter volume identifies the restriction of the frame with the pluriadjoint frame of this sliced effective boundary. Both are rank-one reflexive sheaves on the normal surface, so the identification extends from codimension one and is an actual line identity.

To compare the orders, on the smooth D near a general point of Γ complete the parameters to coordinates $(z_1, \dots, z_{n-2}, u)$ with $\Gamma = (u = 0)$. Write the meromorphic residue as

$$u^c h(z, u) (du \wedge dz_1 \wedge \dots \wedge dz_{n-2})^{\otimes m},$$

where h is a unit at a general point of Γ . Choose the regular level above also outside the proper zero set of its leading coefficient. Division by the base volume and restriction to that level then has order exactly c on the cut curve. On the ambient smooth locus the two operations commute: in coordinates with $D = (x = 0)$, both take the residue of $h(x, u, z)(dx/x \wedge du \wedge dz)^{\otimes m}$ to $h(0, u, z)(du)^{\otimes m}$ on the level, up to the ordering sign. The preceding reflexive identification makes the surface term the residue of the same actual frame. The restricted original residue is already a meromorphic form on the smooth curve C , so equality on its dense smooth-ambient part extends across the cut point. This proves the order assertion. In even degree the ordering sign is one. $\square$

6.2 The indexed strata and their actual adjoints

Fix the resolution in Theorem 6.1, and let $\widehat{S}_i$, for i in a finite set I , be the strict transforms of the components of S . For a nonempty subset $J \subset I$, each irreducible component $\widehat{Z}$ of $\bigcap_{i \in J} \widehat{S}_i$ is smooth of dimension $4 - |J|$. Call its reduced image $Z = \pi(\widehat{Z})$ a stratum and write $\pi_Z : \widehat{Z} \rightarrow Z$. We also include V , with the empty index set and resolution $\widehat{V}$. The special resolution makes each π_Z bimeromorphic and generically an isomorphism. The image determines its index set and its resolving component: on the generic isomorphism locus, two different choices would force an SNC

intersection to have two different codimensions or local branches. There are finitely many strata, since the intersections are compact analytic spaces. Imposing one additional index $j \notin J$ cuts $\widehat{Z}$ in a smooth, possibly disconnected divisor. Its components give the incidences from Z to the next strata.

To perform adjunction inductively, allow each unused floor coefficient either to remain one or to become $1/2$. More precisely, for a stratum indexed by J , choose $e_i = 1$ for $i \in J$ and $e_i \in \{1, 1/2\}$ for $i \notin J$, and put

$$B^{\mathbf{e}} = B - \sum_{i \in I} (1 - e_i) S_i.$$

Ordinary global $\mathbb{Q}$ -factoriality makes this an actual rational-line operation. If $\widehat{G}$ is the crepant boundary of B , the new one is

$$\widehat{G}^{\mathbf{e}} = \widehat{G} - \pi^* \sum_{i \in I} (1 - e_i) S_i.$$

The subtracted pullback is effective. Exceptional coefficients remain below one, and the coefficient-one primes are exactly the retained strict floor primes. Its support stays inside the resolved SNC support.

SNC adjunction along the indices in J defines a subboundary $\widehat{B}_Z^{\mathbf{e}}$ on $\widehat{Z}$ and an actual residue identity

$$(K_{\widehat{V}} + \widehat{G}^{\mathbf{e}})|_{\widehat{Z}} = K_{\widehat{Z}} + \widehat{B}_Z^{\mathbf{e}}. \quad (6.1)$$

Every coefficient is at most one. Its coefficient-one divisors are precisely the one-index intersections for unused indices with $e_i = 1$. The equality means equality of the meromorphic maps from the same restricted pluriadjoint line in a sufficiently divisible even degree.

Proposition 6.3 (Adjunction and full-floor descent). *Every stratum Z is normal and compact Kähler. For every allowed choice $\mathbf{e}$ there is an effective rational boundary $B_Z^{\mathbf{e}}$ such that*

$$\pi_Z^*(K_Z + B_Z^{\mathbf{e}}) = K_{\widehat{Z}} + \widehat{B}_Z^{\mathbf{e}}, \quad (6.2)$$

$$\mathcal{O}_Z(m(K_Z + B_Z^{\mathbf{e}})) \simeq \mathcal{O}_V(m(K_V + B^{\mathbf{e}}))|_Z \quad (6.3)$$

for all sufficiently divisible even m . Both identities are the actual meromorphic residue identifications. For the full weights write $B_Z = B_Z^{\mathbf{1}}$ and $C_Z = \lfloor B_Z \rfloor$. The pair (Z, B_Z) is dlt; its floor components are exactly the next incident strata. In a common degree, sections on those components that agree by residue on every subordinate stratum descend uniquely to a section on the entire reduced C_Z . The same assertion holds for the floor $S \subset V$.

Proof. We induct on the number of indices. The assertions for the empty index set are the given normality, effective boundary, and crepant identity on V . Assume them for a stratum Z and all allowed weights there. Lower all unused floor weights. The resulting crepant SNC boundary on $\widehat{Z}$ has every coefficient below one. The effective pair $(Z, B_Z^{\mathbf{e}})$ is therefore klt. Lemma 2.1 shows that Z has rational singularities and is Cohen–Macaulay.

To add an index j , retain only that unused weight at one and lower the others. The coefficient-one locus on $\widehat{Z}$ is $R = \widehat{Z} \cap \widehat{S}_j$, a disjoint union of smooth components. Apply Lemma 2.1 to π_Z . Its connected fibers prevent the images of two such components from meeting. On each image D , the equality $\pi_{Z,*} \mathcal{O}_R = \mathcal{O}_{\pi_Z(R), \text{red}}$ identifies the proper bimeromorphic pushforward of the structure sheaf of its smooth resolving component $\widehat{D}$ with $\mathcal{O}_D$. Factoring through the finite normalization

shows that D is normal. It is a compact analytic subspace of V , so it inherits a Kähler form by restricting local strictly plurisubharmonic potentials.

For arbitrary allowed weights retaining j , define $B_D^e = \pi_{D,*} \hat{B}_{\hat{D}}^e$. At this point it is a rational divisor whose effectivity remains to be shown. In a divisible even degree, a local frame of the ambient adjoint line pulls back and takes SNC residue to a meromorphic pluriform on $\hat{D}$. The proper bimeromorphic map $\hat{D} \rightarrow D$ is an isomorphism at the generic points of divisors of the normal D . There the same map identifies the divisorial adjoint for B_D^e with the restriction of the existing invertible line. Normal reflexive extension gives (6.3) on all of D . Pulling the line back to $\hat{D}$ recovers the original meromorphic map, since the maps agree on a dense open. This proves (6.2), including its meromorphic interpretation.

We verify that B_D^e is effective. It is enough to test the coefficient at a general point of each prime in D , a codimension-two locus in Z . Apply Lemma 6.2, using the Cohen–Macaulay property already proved for Z . It reduces exactly that coefficient, with the actual residue order preserved, to the smooth cut curve in a normal surface germ with effective boundary. On a minimal resolution of this surface germ, write its crepant boundary as the effective strict transform plus an exceptional divisor E . For every exceptional curve C ,

$$E \cdot C = -(K_{\tilde{T}} + B^{\text{str}}) \cdot C \leq 0.$$

Indeed $B^{\text{str}} \cdot C \geq 0$, while surface adjunction gives $K_{\tilde{T}} \cdot C = 2p_a(C) - 2 - C^2 \geq 0$: exceptional self-intersections are negative and a smooth rational exceptional (-1) -curve is excluded by minimality. The negative-definite exceptional intersection matrix, with nonnegative off-diagonal entries, then implies $E \geq 0$. Explicitly, if $E = P - N$ has disjoint nonnegative parts and $N \neq 0$, then $E \cdot N = P \cdot N - N^2 > 0$, contradicting $E \cdot C \leq 0$ on each component of N . The strict transform of the smooth cut curve is finite birational over it and hence isomorphic. Adjunction to that smooth curve in a smooth surface with effective remaining boundary has a nonnegative coefficient. This is the coefficient originally tested. There is no coefficient to test when the new stratum is a point. Effectivity follows and completes this part of the induction.

For clarity, the full-weight pair on each stratum is dlt and has exactly the claimed floor. The SNC subboundary has coefficients at most one, so it is lc, and crepancy makes the pair on Z lc. On the smooth $\hat{Z}$, let a divisorial valuation have center of codimension c , and choose local coordinates $x_1, \dots, x_c, \dots$ at a general point of that center, with the SNC boundary components among the coordinate divisors. Give an unmarked coordinate weight zero and write the boundary weights as $b_i \leq 1$. The reduced coordinate arrangement is lc, so

$$a(v; \hat{Z}, \hat{B}_{\hat{Z}}) \geq \sum_{i=1}^c (1 - b_i) v(x_i).$$

All $v(x_i)$ here are positive. A zero discrepancy therefore forces every $b_i = 1$; its center is an intersection of coefficient-one components. Each such intersection has image meeting the isomorphism locus of the original special resolution. In particular each coefficient-one divisor is nonexceptional for π_Z ; the others have coefficients below one. Let $F_Z \subset Z$ be the complement of the largest open over which π_Z is an isomorphism. It has codimension at least two on the normal target Z , and contains no entire image of a zero-discrepancy center. Its inverse image on $\hat{Z}$ may already be divisorial. If necessary, principalize the pulled reduced ideal $\mathcal{I}(F_Z)\mathcal{O}_{\hat{Z}}$, and then apply the ordered-boundary resolution as in Lemma 5.6, marking each distinct prime in the prior SNC support, the newly principalized support, and every divisorial exceptional prime. All new exceptional centers lie over F_Z . The displayed inequality makes their discrepancies positive.

This is a dlt resolution in the standard sense. It also proves that the coefficient-one primes on Z are exactly the images of the one-index intersections.

It remains to check the asserted descent of sections. With full weights, let $R = \widehat{B}_Z^{-1}$. The preceding floor identification gives $(\pi_Z(R))_{\text{red}} = C_Z$; write $\pi_R : R \rightarrow C_Z$ for the induced restriction of π_Z . Sections on the incident strata pull back to sections on its smooth components. Residue compatibility means equality on the entire scheme-theoretic pairwise intersections, which are reduced smooth SNC strata. The local equalizer for an SNC union therefore glues them to a section on R ; it is the elementary equalizer for the coordinate ideals of its components. Even-degree iterated residues make all paths to deeper intersections identical. By Lemma 2.1 and projection formula,

$$(\pi_R)_* \pi_R^*(L|_{C_Z}) = (L|_{C_Z})$$

for any ambient Cartier line L under consideration. The section descends uniquely to all of C_Z . This reasoning applies also to $Z = V$ and $C_Z = S$. In particular it retains the conductor coefficient at a general double crossing on each normalized component; it does not posit an additional conductor boundary on the nonnormal union itself. $\square$

The proposition supplies a finite collection of normal compact Kähler dlt strata, all carrying restrictions of the same actual ambient line. It also specifies exactly what compatibility is needed to descend to a whole floor. The next section proves that, after imposing a small list of birational residue comparisons on the lower strata, these compatible boundary sections extend to their parent stratum.

7 Extension from a stratum and its residue link

Proposition 6.3 gives normal dlt pairs on the strata and a single actual adjoint line along every incidence. We now show that a section on the floor of a stratum extends after it satisfies at most one birational residue comparison. The comparison comes from two markings of a projective-line fiber on a Mori model. In the next section we will control the actions generated by these comparisons.

7.1 Semiample lines on the normal strata

We use a small model both to place lower-dimensional abundance inside its explicit sheaf conventions and to run the later perturbed program. Let Z be a positive-dimensional stratum of dimension at most three. Lower its remaining floor coefficients as in the preceding section, obtaining an effective klt boundary B_Z^- with actual adjoint. There is a projective small ordinary $\mathbb{Q}$ -factorialization $q_Z : Z_0 \rightarrow Z$ of this pair. Here is a construction that also checks the actual line used on it.

On a projective log resolution $r : W \rightarrow Z$, keep the effective strict boundary and assign to each exceptional prime a rational weight just below one but strictly above its crepant coefficient. This produces an effective SNC klt boundary C_W with

$$K_W + C_W = r^*(K_Z + B_Z^-) + F,$$

where F is effective, exceptional, and positive on every divisorial exceptional component. The equality is the meromorphic actual-line identity. The adjoint is relatively pseudo-effective. Apply the relative MMP for a projective morphism of compact analytic spaces in dimension at most three [11, Proposition 2.26]. Its source is the ordinary $\mathbb{Q}$ -factorial smooth W , its pair is effective SNC klt, and its morphism is a projective surjection of normal compact analytic spaces. On the relative

ordinary $\mathbb{Q}$ -factorial minimal model over Z , the transform of F is exceptional and relatively nef, since the pulled-back adjoint is relatively trivial. Negativity makes that transform zero. No step extracts a divisor, so the resulting morphism $q_Z : Z_0 \rightarrow Z$ is small. It is projective, hence Z_0 is compact Kähler. Codimension-one comparison and reflexive extension identify its perturbed adjoint with the actual pullback from Z .

The same smallness gives the crepant actual pullback for the full boundary. Moreover q_Z is an isomorphism over a smooth germ of Z . To see this, take a relatively very ample line over a small Stein neighborhood of the germ. It has a meromorphic section there: after large relative ample twists, relative Serre generation and Cartan generation supply sections whose quotient is such a section. Its divisor pushes to a Cartier divisor on the smooth factorial target. Smallness leaves no exceptional prime, so the original divisor and line are its pullback. A pulled-back line cannot be relatively ample on a positive-dimensional fiber. Any nonisomorphism fiber here would have positive dimension: otherwise properness and the fiber-dimension theorem make the morphism finite near that fiber, and a finite bimeromorphic morphism to the normal target is an isomorphism. Thus the small model is unchanged over this smooth germ. The lc centers of the full pair meet the smooth crossing locus established in Proposition 6.3; the small model is generically unchanged there. The full transformed pair (Z_0, B_0) is therefore dlt, and its floor C_0 is the strict transform of C_Z .

For a three-dimensional stratum, apply Das–Ou’s lc threefold abundance theorem [14, Corollary 1.3] to this full pair. Its adjoint is the analytically nef actual pullback of the ambient restriction: the metric lower bound restricts to Z and then pulls back. The detailed conventions in [14, Section 2.1] use the reflexive canonical sheaf, the invertible reflexive pluriadjoint, actual restriction isomorphisms, and smooth-potential nefness. Thus this application does not choose a global canonical Weil divisor. Generation descends to Z by normality and projection formula. Lower strata inherit generation by restriction from a three-dimensional stratum containing them.

Choose a common sufficiently divisible even integer q for the finite list of strata and set

$$L = \mathcal{O}_V(q(K_V + B)), \quad L_Z = L|_Z = \mathcal{O}_Z(q(K_Z + B_Z)).$$

Enlarge q so all L_Z in dimension at most three are generated. Their complete systems followed by Stein factorization give

$$f_Z : Z \longrightarrow Y_Z, \quad L_Z \simeq f_Z^* N_Z, \quad (7.1)$$

where f_Z has connected fibers, Y_Z is normal projective, and N_Z is ample. Indeed the image in projective space is projective by Chow, and the finite analytic Stein space is projective by algebraizing its finite coherent algebra with GAGA. Its line is the pullback of the tautological ample line. In particular, for every $k \geq 0$, all sections of L_Z^k come from N_Z^k .

7.2 A torsion-free obstruction to restriction

The following lemma is what lets general-fiber matching control all parameters, including those where the floor has a singular or nonreduced scheme fiber. Its hypothesis that the pair is klt off the floor is essential to the canonical character calculation.

Lemma 7.1. *Let (T, G) be a normal irreducible compact Kähler lc pair with effective rational boundary, and put $C = \lfloor G \rfloor$ with its reduced structure. Assume that the pair is klt on $T \setminus C$. Let $f : T \rightarrow P$ be a surjective holomorphic map with connected fibers to a normal irreducible compact complex space. Suppose an actual positive adjoint multiple is pulled back from a line on P . Then*

$$R^1 f_* \mathcal{O}_T(-C) \quad \text{is torsion-free on } P.$$

Here $\mathcal{O}_T(-C)$ is the ideal of the reduced floor.

Proof. The equality with the ideal holds because on the normal T both sheaves consist of holomorphic functions vanishing at every height-one prime in C . Work on a small connected base open trivializing the line whose pullback is $m(K_T + G)$. Its nonvanishing pulled-back frame is a local meromorphic m -pluricanonical form θ .

Adjoin the full set of m th roots of this form. This construction is local even without a canonical Cartier frame. At a normal local domain A , choose a meromorphic canonical generator η and write $\theta/\eta^m = a/b$. The monic algebra

$$A[w]/(w^m - ab^{m-1}) \subset \text{Frac}(A)[t]/(t^m - a/b), \quad w = bt,$$

is finite and reduced; its analytic normalization is finite [34, Part B, Section 4, Corollaries 2–3]. Changes of meromorphic canonical generator multiply the root coefficient by an m th power, and identify the total algebras and their integral closures. They therefore glue to the normalized cover. Keep every component and the full μ_m -action. The tautological form $\tau = t\eta$ is a meromorphic top form on a projective resolution with smooth source; take this resolution functorially and equivariantly, after shrinking near the compact fibers. Write $l : H \rightarrow T$ for the composite.

At a general prime of T with boundary coefficient b , the order of τ upstairs is

$$e(1 - b) - 1, \tag{7.2}$$

where e is the ramification index. This follows from the pole order $-eb$ of the root coefficient and the differential ramification order $e - 1$. It is -1 when $b = 1$ and is an integer between zero and $e - 1$ when $0 \leq b < 1$. A meromorphic form in the τ -character has the form $h\tau$, where h is an invariant meromorphic function and hence descends to T ; this uses the entire root algebra, whose invariants are the normal base. Regularity in codimension one forces h to be holomorphic and to vanish on C .

Conversely, suppose $h \in \mathcal{O}_T(-C)$. On a log resolution of the pair, every crepant coefficient is at most one. A coefficient-one exceptional prime has center in C , because the pair is klt away from C ; the pullback of h has positive order there. Thus in SNC coordinates the density $|h|^2|\theta|^{2/m}$ has every exponent strictly above the integrability threshold. It is locally integrable. Change of variables makes $l^*h\tau$ locally square integrable on the smooth resolution, and a square-integrable meromorphic top form is holomorphic. This proves the exact character equality

$$(l_*\omega_H)_\tau = \mathcal{O}_T(-C)\tau. \tag{7.3}$$

Each component of the full normalized cover dominates the normal base, and each resolved component does so as well. The inverse image in T of the connected base open is connected, because f is proper with connected fibers; normality makes it irreducible. Finite maps followed by projective resolutions have Kähler sources near the whole inverse image of a sufficiently small base neighborhood: patch relative ample metrics and add a large pulled-back Kähler form. Canonical torsion-freeness [21, Theorem 2.9 and Proposition 2.11] applies componentwise. For the generically finite map to T , its higher canonical images vanish generically and hence vanish. For the map to P , its first canonical image is torsion-free. Leray and the character projector in (7.3) make $R^1f_*\mathcal{O}_T(-C)$ a direct summand of that torsion-free sheaf. This proves the assertion locally, and hence on P . $\square$

Here is its restriction consequence. Suppose $L_T = f^*N$ in the lemma, where P is projective and N ample. The ideal sequence gives

$$\mathcal{Q} := \text{coker}(\mathcal{O}_P \longrightarrow f_*\mathcal{O}_C) \hookrightarrow R^1f_*\mathcal{O}_T(-C). \tag{7.4}$$

If C is vertical, meaning that it does not dominate P , then $\mathcal{Q}$ is supported on a proper analytic subset and is zero by torsion-freeness. Serre vanishing for the kernel of $\mathcal{O}_P \rightarrow f_*\mathcal{O}_C$, after tensoring by N^k , shows that every section of $L_T^k|_C$ extends to T for all sufficiently large k . The bound is uniform over the sections.

If C dominates P , the left arrow into $f_*\mathcal{O}_C$ is injective, and $f_*\mathcal{O}_C$ is torsion-free by (7.4). The finite Stein space of the reduced $C \rightarrow P$ is reduced and every irreducible component dominates P : a vertical minimal prime of a finite reduced algebra over a domain would supply a nonzero torsion element. Off a proper analytic subset this finite map is étale. A section of $L_T^k|_C$ whose values agree throughout the reduced floor fiber over every point of some dense open of P has zero image in $\mathcal{Q} \otimes N^k$ on that open, and hence everywhere by torsion-freeness. Its local lifts from N^k are unique and glue. Thus it extends. This argument neither asserts that special scheme fibers are reduced nor uses cohomology base change at a special point.

7.3 A Mori contraction over the semiample target

The torsion-free argument has settled extension when the floor is vertical over the semiample target. For a horizontal floor, it remains to make a section take the same value on the connected components of a general floor fiber. We first construct a Mori model on which those components can be counted.

Fix a stratum Z of dimension one, two, or three and its small model $q_Z : (Z_0, B_0) \rightarrow (Z, B_Z)$. Put $C_0 = \lfloor B_0 \rfloor$ and $f_0 : Z_0 \rightarrow Y = Y_Z$. The full adjoint is the actual pullback of the semiample line on Y .

Lemma 7.2 (A Mori model for a horizontal floor). *Assume that C_0 dominates Y . There is a finite sequence of projective divisorial contractions and flips from Z_0 to an ordinary $\mathbb{Q}$ -factorial compact Kähler model T , followed by a projective Mori contraction*

$$u : T \longrightarrow W, \quad g : W \longrightarrow Y,$$

where W is normal compact Kähler and both maps have connected fibers. Write G for the transform of B_0 and $C^+ = \lfloor G \rfloor$. The pair (T, G) is effective lc and klt off C^+ ; the divisor C^+ is relatively ample for u and still dominates Y . All steps factor over Y and extract no divisor. On a common projective resolution of the sequence, the full adjoints are equal as meromorphic pullbacks of the same actual line from Y .

Proof. Choose a small rational $\varepsilon > 0$ for which $B_0 - \varepsilon C_0$ is effective klt. This uses dlt and the fact that all its lc centers are contained in the floor. Let H be an integral very ample divisor on Y and choose a rational effective $H' \sim_{\mathbb{Q}} dH$ with $d > 2 \dim Z$ such that

$$B_0 - \varepsilon C_0 + f_0^* H'$$

is still klt. A sum of sufficiently many general members of $|H|$ with small rational weights does this by Bertini on a log resolution. If Y is a point take $H' = 0$.

The adjoint of this perturbed pair is not pseudo-effective. On a smooth general fiber of a resolution of f_0 , its class is the negative of the nonzero effective divisor εC_0 restricted to that fiber: the full adjoint and H' are pulled back from Y , and the floor is horizontal. Its integral against a Kähler power is strictly negative. If the ambient class were pseudo-effective, the potential of a positive current representing its pullback would restrict to a positive current on almost every such smooth fiber, contradicting this integral. The smooth horizontal-fiber conditions hold on a nonempty open, so the almost-everywhere restriction is sufficient.

This same integral also produces a negative class in the cone used by the Kähler cone theorem. If the smooth fiber has dimension $v > 0$, push its positive closed bidimension- $(1, 1)$ current ω_F^{v-1} through the fiber inclusion and the resolution to Z_0 . It represents a class in $\overline{NA}(Z_0)$, and its pairing with the perturbed adjoint is exactly that strictly negative integral. The cone decomposition therefore supplies a negative extremal ray.

Run the klt Kähler program for this perturbed adjoint. In dimension three, the negative extremal-ray and contraction theorems [12, Corollary 5.3 and Theorem 5.5] are used through Subsection 4.4, which incorporates Proposition 4.3 and the specified external results. They give connected projective contractions to normal compact Kähler targets. The supporting class differs from the klt adjoint by a Kähler class, which is the projectivity condition in that theorem. The supporting class need not be big, and the perturbed adjoint here is not pseudo-effective. Flips and termination of flip sequences are provided by [11, Theorems 2.24–2.25]. In dimensions at most two the non-pseudo-effective pair is projective and the ordinary projective program applies. For a surface, otherwise a non-Moishezon Kähler resolution has a nonzero holomorphic two-form: if $H^{2,0} = 0$, rational approximation of a Kähler class would make it projective. That form descends reflexively and, with the effective boundary, gives a section of a positive adjoint multiple, contradicting non-pseudo-effectivity. A Moishezon Kähler klt surface is projective by Namikawa’s criterion [41, Corollary 6’]. Curves are projective.

We specify the ordinary $\mathbb{Q}$ -factorial and finiteness details used by this program. For a projective negative extremal contraction, a global rational line of degree zero on all contracted curves descends in a multiple by [12, Theorem 2.44(3c)]. Here we apply its projective relative theorem to the effective klt pair, with the compact set equal to the entire compact target. For this projective surjection, property Q of [12, Definition 2.42] requires a normal source and compact source and target, as here. All fiber curves span the one relative ray, so the theorem’s contraction is the given contraction. Indeed its map is constant on each connected projective fiber of the given contraction, and rigidity factors it through that contraction. Its structure map back to the relative base supplies the reverse factorization; surjectivity makes the two factorizations inverse. This use of Theorem 2.44 does not assert a factoriality hypothesis for that theorem. At a divisorial step, adjust the transform of a destination Weil divisor by a $\mathbb{Q}$ -Cartier exceptional prime of nonzero degree on the ray, and descend the resulting degree-zero line. Comparison in codimension one makes the destination divisor $\mathbb{Q}$ -Cartier. At a flip, adjust on the negative side by the negative adjoint, descend, and pull to the positive side; the positive adjoint is $\mathbb{Q}$ -Cartier and the same comparison applies. Boundary components and the adjoint then also give the canonical reflexive power. This preserves ordinary $\mathbb{Q}$ -factoriality.

The rank of global rational divisor classes modulo curve numerical equivalence is finite. Each running model is a connected compact complex analytic space. Its reduced underlying real analytic space has bounded Zariski tangent dimension by a finite analytic chart cover. Acquistapace–Broglia–Tognoli [1, Theorem 1] embed it closedly in Euclidean space, and Łojasiewicz [38, Theorem 1] gives a compatible locally finite triangulation; the subcomplex corresponding to the compact model is finite. Hence its rational H^2 is finite-dimensional. The first Chern class sends global rational Cartier divisors to that group. Every divisor in its kernel has degree zero on each compact curve, by restriction to the curve’s normalization. The rational divisor space modulo curve numerical equivalence is consequently a quotient of the Chern-class image, and has finite rank. The rank drops at a divisorial contraction, by the nonzero exceptional ray degree, and does not increase at a flip. For the latter assertion a numerically trivial line on the negative side descends as above; the descended line is curve-numerically trivial because every curve downstairs is covered by a curve upstairs, using projectivity over that curve. Its pullback is trivial on curves on the positive side,

and every divisor there is a transform by smallness. There can therefore be only finitely many divisorial steps. Together with termination of flip sequences this gives a finite program to a nef model or a Mori contraction. All steps extract no divisors.

Every step factors over Y . Inductively write the running perturbed adjoint as

$$A_i^\varepsilon = K_{X_i} + B_i - \varepsilon C_i + f_i^* H'.$$

For a negative contraction apply the relative cone and length theorem [12, Theorem 2.45] to $K_{X_i} + B_i - \varepsilon C_i$ over its entire compact contraction target. The source is ordinary $\mathbb{Q}$ -factorial and the pair is klt. Property Q holds as above; because the compact set is the whole target, Theorem 2.45's factoriality over that set is precisely the global factoriality of the source, including its canonical reflexive power. A negative generator Γ has

$$-(K_{X_i} + B_i - \varepsilon C_i) \cdot \Gamma \leq 2 \dim Z.$$

If $f_i(\Gamma)$ were a curve, then $f_i^* H' \cdot \Gamma \geq d$, contradicting the negativity of A_i^ε . On the nonnegative part of that relative cone, both $K_{X_i} + B_i - \varepsilon C_i$ and the nef $f_i^* H'$ are nonnegative, while their sum is nonpositive on the contracted cone; its H' -degree is zero there as well. All contracted curves are vertical over Y . A holomorphic map from a connected projective fiber that is nonconstant would map a curve nontrivially, so f_i is constant on each fiber. Rigidity and normality of the target give the factorization, and the positive side of a flip factors through the same base.

We check the boundary properties needed to repeat the step. No prime is extracted, so the transformed full boundary is effective and its floor is exactly the transform C_{i+1} of C_i . The running MMP preserves the klt pair $B_i - \varepsilon C_i + f_i^* H'$. Removing the effective divisor $f_{i+1}^* H'$ shows that $B_{i+1} - \varepsilon C_{i+1}$ is klt. In particular the full pair is klt away from its floor.

The full adjoints remain actually crepant pullbacks from Y . Indeed the chosen meromorphic pluriadjoint identity extends to the new model in codimension one, since no prime is extracted, and then reflexively. On a common graph both meromorphic maps start from the same line pulled back from Y and agree on the dense isomorphism open. They agree everywhere as meromorphic maps. Equality of discrepancies therefore preserves log canonicity of the full pair. If r, s are the graph projections, negativity for the perturbed step gives

$$0 \leq r^* A_i^\varepsilon - s^* A_{i+1}^\varepsilon = \varepsilon(s^* C_{i+1} - r^* C_i). \quad (7.5)$$

A horizontal component of C_i cannot disappear into an entirely vertical C_{i+1} , since the coefficient of the right side along its strict valuation would then be negative. Thus the floor remains horizontal. On every subsequent model the full adjoint and H' still come from Y . Repeating the resolved smooth-fiber integral and pushing its Kähler-power current as above gives a negative class in its $\overline{\text{NA}}$ cone and hence a negative extremal ray. In particular a nef endpoint is impossible.

We have obtained a Mori contraction

$$u : T \longrightarrow W, \quad g : W \longrightarrow Y,$$

with T normal compact Kähler, u projective of fiber type, and both maps having connected fibers. For g , this follows also from $(gu)_* \mathcal{O}_T = \mathcal{O}_Y$ and $u_* \mathcal{O}_T = \mathcal{O}_W$. The full boundary G is effective lc, its adjoint is an actual pullback from Y , and it is klt away from $C^+ = [G]$: subtracting εC^+ is klt. The running perturbed adjoint satisfies

$$A^\varepsilon \equiv_u -\varepsilon C^+, \quad -A^\varepsilon \equiv_u +\varepsilon C^+.$$

The left side of the second equivalence is relatively ample for the Mori contraction. Thus C^+ is relatively ample for u . The graphs of the divisorial steps and flips are projective over both sides. A main component of their iterated fiber product, followed by normalization and projective resolution, gives the common projective resolution in the statement. Its smooth source is compact Kähler. $\square$

7.4 The restriction criterion

We now use the Mori model only to compare the values of a boundary section on general fibers. The torsion-free obstruction then extends that comparison over every parameter.

Proposition 7.3 (Restriction criterion). *For every stratum Z of dimension one, two, or three, there is either no comparison or one proper bimeromorphic comparison between two normal components of C_Z , allowing a component to be compared with itself, with the following properties.*

1. *On a resolution of its graph with smooth source, the two pulled-back actual boundary adjoint lines are equal as invertible subsheaves of meromorphic pluricanonical forms.*
2. *For all sufficiently large divisible k , uniformly over sections, a section of $L_Z^k|_{C_Z}$ extends to Z if its two pullbacks agree under this comparison. With no comparison there is no matching condition.*
3. *When present, the comparison pairs two coefficient-one markings on general projective-line fibers of a projective Mori contraction on a bimeromorphic compact Kähler model of Z .*

Proof. If C_Z is vertical over Y_Z , the consequence of (7.4) gives extension in a uniform large-degree tail without a comparison. Suppose therefore that it is horizontal, and use the small model q_Z and the Mori model $u : T \rightarrow W$, $g : W \rightarrow Y$ of Lemma 7.2. Extending the pulled-back section on C_0 suffices: a section on Z_0 descends to Z , and equality of its restriction after pullback implies equality on the reduced C_Z , since every component is covered birationally.

The floor on general Mori fibers

We now compare the connected components of a general floor fiber with the markings of one Mori fiber. On a common projective log resolution of the program, the full crepant boundary is the same on both sides. Its coefficient-one union maps to both C_0 and C^+ with connected fibers by Lemma 2.1. After restriction over any $y \in Y$, proper closed maps with connected fibers preserve connected components. Thus the connected components of the underlying spaces of $C_{0,y}$ and C_y^+ correspond. This is a topological statement, not a reducedness claim for their scheme fibers.

The finite Stein space of $C^+ \rightarrow W$ has every component dominating W . Indeed relative ampleness makes $C^+ \rightarrow W$ surjective, and Lemma 7.1 and (7.4) give the torsion-free property used in the preceding discussion. For general y , every irreducible component of the finite Stein fiber has dimension $\dim W - \dim Y$, by the dimension theorem. The fiber W_y is irreducible of that dimension: a resolution of W has connected fibers over Y , and a general one is smooth and connected, hence irreducible, and surjects onto W_y . Each component of the finite Stein fiber therefore dominates W_y . It follows that every connected component of C_y^+ meets $u^{-1}(w)$ for a common general $w \in W_y$.

The underlying space of a general fiber F of u is irreducible by the same resolution argument. If $\dim F \geq 2$, a Cartier multiple of C^+ restricts to a nonzero effective ample Cartier divisor on F_{red} . Its support is connected. To recall the reason, if an ample effective divisor split into two disjoint

nonzero parts, general hyperplane sections would reduce to an irreducible projective surface F_2 . Write the two induced nonzero effective Cartier parts on it as D_1, D_2 , and let $v : \tilde{F} \rightarrow F_2$ be a projective resolution with smooth source. Their pullbacks are effective and orthogonal. Although $v^*(D_1 + D_2)$ is only nef and big, the projection formula gives

$$v^*(D_1 + D_2) \cdot v^*D_i = (D_1 + D_2) \cdot v_*[v^*D_i] > 0 \quad (i = 1, 2) :$$

the pushforward is a nonzero effective curve cycle and $D_1 + D_2$ is ample on the integral surface. Orthogonality then gives $(v^*D_i)^2 > 0$ for both i . The surface Hodge index theorem forbids two such orthogonal positive classes. This argument does not require normality of F_{red} .

If $\dim F = 1$, choose F also in the generic-smoothness open. Generality avoids the singular locus of T , the singularities and intersections of the boundary, and the ramification of its horizontal primes. The fiber is a smooth connected curve, and the full pullback identity gives

$$0 = \deg(K_F + G|_F) = 2g(F) - 2 + \deg(G|_F).$$

There is at least one coefficient-one point. Hence $F \simeq \mathbb{P}^1$, and its floor has one or two points. In the two-point case these points exhaust the horizontal boundary on F . In all connected cases, every connected component of C_y^+ meets the same connected subset $C^+ \cap F$, so C_y^+ , and therefore $C_{0,y}$, is connected. Then no comparison is needed. In the two-point case each connected component of $C_{0,y}$ is represented by at least one of these two markings, after their transfer through the common graph.

The two-marking comparison

Normalize each surviving horizontal prime $D \subset C^+$ and take the Stein factorization

$$D^\nu \longrightarrow E_D \longrightarrow W.$$

The first map is projective bimeromorphic and E_D is normal; the second is finite. If there are two horizontal primes, each finite map has degree one and is an isomorphism over the normal W . The main component of the fiber product of their normalizations over W gives a proper bimeromorphic comparison. If there is one horizontal prime, $E_D \rightarrow W$ has degree two. On a dense open it is étale and has an exchange. The reduced horizontal non-diagonal component of $E_D \times_W E_D$ has finite generically one-to-one projections to the normal E_D ; both are isomorphisms. It defines the exchange involution globally, including across the branch. The main component of $D^\nu \times_{E_D, \text{exchange}} D^\nu$ lifts it to a proper bimeromorphic graph. It is the Stein space of the normalized prime that is used here, not the possibly nonnormal Stein space of the entire floor.

Compose this graph with the common graphs of the program and with the small model. It gives the asserted comparison between normal primes of C_Z , possibly a self-comparison. Each surviving prime is generically finite over W ; the image in W of the proper subset where its transfer is not an isomorphism is proper. Thus both markings on a general Mori fiber lie in the common transfer isomorphism loci.

These graph projections are projective. The maps $D^\nu \rightarrow E_D$ are projective and the maps $E_D \rightarrow W$ are finite; their fiber products and closed main components are therefore projective over the branches. The small-model and MMP graphs have the same property. Iterated main components and finite normalization preserve it, so the comparison admits a projective resolution with smooth compact Kähler source.

We verify the actual meromorphic identity required by property (1). The two residues of a logarithmic fiber form give the same comparison as the even Poincaré-residue diagrams of

[37, Section 3, Definition 13 and Proposition 14]; we include the calculation for the normalized two-branch construction above. On a smooth ruled open of W , order the two markings after an étale local cover when necessary, choose a base volume form ξ , and a fiber coordinate with markings at zero and infinity. The full q -pluriadjoint frame coming from Y has the form

$$a(w)(dz/z)^{\otimes q} \otimes \xi^{\otimes q}.$$

There is no other horizontal divisor on the general fiber. Its two residues are $a(w)\xi^{\otimes q}$ and $(-1)^q a(w)\xi^{\otimes q}$; they agree because q is even. The calculation is invariant under exchanging the two local markings. Full crepancy transfers it to the original primes. On a resolution of their comparison graph, both actual adjoint lines are the same pullback of N_Z from Y . Their meromorphic embeddings agree on this dense ruled open, hence agree everywhere. Thus the invertible subsheaves of meromorphic pluriforms are equal even at exceptional primes. An abstract $\mathbb{Q}$ -linear equivalence without this residue calculation would not suffice.

A section satisfying this comparison has equal values at the two representatives over a general w . On every connected component of a compact reduced floor fiber it is constant in the trivialized pulled-back line: holomorphic functions on a compact connected reduced complex space are constant. The representatives meet every component, so its values agree on $C_{0,y}$ for every y in a dense base open. The horizontal consequence of (7.4) extends it to Z_0 , and then it descends to Z . The only asymptotic bound on k came from Serre vanishing in the vertical case. Finitely many strata admit a common sufficiently divisible tail. This proves all three properties. $\square$

For the remainder of the boundary proof, call each comparison supplied by Proposition 7.3 a *link*. This word refers to its proper bimeromorphic graph together with the proved equality of meromorphic adjoint subsheaves. It does not mean only an abstract linear equivalence. We will use all curve comparisons, but on surfaces only the groupoid generated by links of three-dimensional strata. That restriction is what allows the common ambient Kähler class to control their scalar action.

8 Finite residue actions from one Kähler class

The restriction criterion asks for equality under links on lower strata. To build enough sections with all these equalities, we need finiteness of the induced actions on each fixed section space. On projective strata this is the usual log pluricanonical representation theorem. On nonprojective surfaces it holds here because the links all come from the boundary of one compact Kähler fourfold and therefore compare restrictions of one ambient Kähler class.

Fix the common even degree q and lines L_Z from (7.1). For $d = 0, 1, 2$, form a groupoid $\mathcal{G}_d$ whose objects are the finitely many d -dimensional strata. In dimension zero use all identifications of points, with the canonical zero-form generator 1. In dimension one use all crepant residue comparisons between the indicated curve pairs. In dimension two use only the groupoid generated by the links of three-dimensional strata from Proposition 7.3, together with their inverses. Here a comparison includes equality of the pulled-back invertible meromorphic adjoint subsheaves, as specified there.

Such a comparison induces an isomorphism

$$H^0(Z_1, L_{Z_1}^k) \longrightarrow H^0(Z_2, L_{Z_2}^k)$$

for every $k > 0$. Pull a section to a resolution of the graph, use the equality of invertible meromorphic subsheaves, and descend to the other normal stratum by projection formula. This

transport is compatible with composition, products of sections, and residue restriction along boundary divisors whose centers are birational on both strata. Lemma 9.1 proves the compatibility also when a floor curve is contracted, before that case enters the construction. The same argument shows that evaluation after a graph pullback is evaluation in the identified actual line fibers, a fact needed for generation later.

Proposition 8.1 (Finite isotropy images). *For every $d \in \{0, 1, 2\}$, every d -stratum Z , and every fixed integer $k > 0$, the image of the self-comparisons in $\mathcal{G}_d$ on $H^0(Z, L_Z^k)$ is a finite group.*

The restriction on $\mathcal{G}_2$ is essential. A nonprojective K3 surface with trivial canonical line can have an automorphism acting by a non-root-of-unity scalar on its holomorphic two-form [39, Theorems 3.5 and 4.1]. Thus the proposition would be false for arbitrary bimeromorphic self-comparisons of nonprojective surfaces.

8.1 Points, curves, and projective surfaces

On a point the zero-form generator is 1 and every comparison acts as the identity. A normal compact curve is smooth, and a bimeromorphic map is an isomorphism preserving the boundary coefficients. For genus at least two the automorphism group is finite. For genus one, the stabilizer of a nonempty finite boundary support is finite; with empty boundary translations act trivially on holomorphic forms and the linear automorphism group is finite. For genus zero, nefness of the adjoint and coefficients at most one require at least two marked points. If there are exactly two, both coefficients are one and the log line is trivial with generator $(dt/t)^{\otimes qk}$. Scaling fixes it, and exchange has sign one in the even degree. With at least three marked points the stabilizer is finite. This proves Proposition 8.1 in dimensions zero and one.

Suppose a surface stratum Z is Moishezon. Its klt perturbation in Proposition 6.3 gives rational singularities. Namikawa’s projectivity criterion [41, Corollary 6’] makes the compact Kähler Z projective. Its effective full boundary is dlt and its adjoint is semiample. The analytic comparison graphs are algebraic by Chow, and their meromorphic equality is the usual crepant B-birational equality for compatible canonical identifications. Fujino–Gongyo’s log pluricanonical representation theorem [24, Theorem 1.1] gives finite image in every fixed Cartier degree. It applies exactly in this projective case.

8.2 The class attached to a nonprojective surface

Let Z now be a non-Moishezon surface stratum, and let P be the minimal smooth surface of a compact Kähler resolution of Z . Point blowdowns preserve Kählerness, for example by the even- b_1 criterion [5, Theorem 11]. Algebraic dimension is bimeromorphically invariant, so P is nonprojective and has algebraic dimension zero or one. It has a nonzero holomorphic two-form: otherwise $H^2(P, \mathbb{R}) = H^{1,1}(P, \mathbb{R})$, and rational approximation of a Kähler class followed by Kodaira embedding would make P projective. Thus $\kappa(P) \geq 0$. The minimal model is neither rational nor ruled; its bimeromorphic maps are automorphisms, and bimeromorphic maps between such minimal surfaces are isomorphisms [44, Proposition 3.5]. Under a common resolution, $H^0(Z, L_Z^k)$ embeds as a finite-dimensional space of meromorphic qk -pluricanonical forms on P , compatibly with all transports.

Fix once and for all a Kähler form ω_V on the ambient fourfold. For a common resolution $h : R \rightarrow P$, $\rho : R \rightarrow Z$, define

$$c_Z = h_*[\rho^*(\omega_V|_Z)] \in H^{1,1}(P, \mathbb{R}). \quad (8.1)$$

The brackets denote the de Rham class of the pulled-back local-potential form on the smooth resolution. All cohomology in the comparison below is taken on smooth manifolds. This is independent of further resolutions, since $v_*v^* = 1$ for a modification v . Put $\beta = [\rho^*(\omega_V|_Z)]$. Its smooth representative is semipositive and is strictly positive on a nonempty open: the original stratum is generically immersed in the smooth isomorphism locus of the special resolution. Hence $\beta^2 > 0$. The blowup orthogonal decomposition has $\beta = h^*c_Z + e$, with e in the negative-definite exceptional subspace. Consequently

$$c_Z^2 = \beta^2 - e^2 > 0. \quad (8.2)$$

Lemma 8.2 (The same ambient class across a link). *Let a link of a three-dimensional stratum compare non-Moishezon surface strata Z_1, Z_2 , allowing $Z_1 = Z_2$. Let P_i be their smooth minimal models and $\tau : P_1 \simeq P_2$ the induced isomorphism. For the classes in (8.1),*

$$c_{Z_1} - \tau^*c_{Z_2} \in \text{NS}(P_1)_{\mathbb{R}}. \quad (8.3)$$

Here $\text{NS}(P)_{\mathbb{R}}$ is the real span of the first Chern classes of holomorphic line bundles in $H^{1,1}(P, \mathbb{R})$.

Proof. Let $i : Z \hookrightarrow V$ be the parent stratum, and choose a common projective resolution with smooth compact Kähler source U , with maps $a : U \rightarrow Z$ and $b : U \rightarrow T$ to its original and Mori models. Use the single class

$$\alpha = [(i \circ a)^*\omega_V] \in H^{1,1}(U, \mathbb{R}).$$

Resolve the marked-branch comparison graph and its maps to U and the minimal surfaces. We obtain a smooth compact Kähler surface D with maps $r_i : D \rightarrow U$, bimeromorphic maps $d_i : D \rightarrow P_i$, and bimeromorphic maps to the original Z_i , such that ar_i is the inclusion of that original branch after its resolution and $d_2 = \tau d_1$. These identities hold on the common marked open and hence everywhere. This construction includes a self-link: its two maps to the same original prime may differ by the normalized exchange. Resolution independence and projection formula give

$$c_{Z_1} - \tau^*c_{Z_2} = d_{1,*}(r_1^* - r_2^*)\alpha. \quad (8.4)$$

The exceptional image of the modification $b : U \rightarrow T$ has codimension at least two in the normal threefold, and hence dimension at most one. The Mori base in this two-marking case has dimension two. After removing its image, the discriminant, and the singular base locus, U is therefore a smooth proper $\mathbb{P}^1$ -fibration over a dense open of the base. For such a fibration $\pi : U^\circ \rightarrow W^\circ$, the differential sequence and $H^0(\mathbb{P}^1, \Omega^1) = 0$ give

$$\pi_*\Omega_{U^\circ}^2 = \Omega_{W^\circ}^2.$$

Indeed the quotient of $\Omega_{U^\circ}^2$ by $\pi^*\Omega_{W^\circ}^2$ is $\pi^*\Omega_{W^\circ}^1 \otimes \Omega_{U^\circ/W^\circ}^1$, whose direct image is zero, while $\pi_*\mathcal{O}_{U^\circ} = \mathcal{O}_{W^\circ}$. Thus the two restrictions of a holomorphic two-form on U agree on the dense paired-branch open of D , even for the double branch after an étale local ordering. They agree on all of D .

Holomorphic pullback and proper pushforward on compact Kähler manifolds preserve Hodge type. The map

$$d_{1,*}(r_1^* - r_2^*) : H^2(U, \mathbb{Q}) \longrightarrow H^2(P_1, \mathbb{Q})$$

is therefore a rational Hodge morphism; pushforward here is between equal-dimensional surfaces. The preceding equality kills its $(2, 0)$ part and, by conjugation, its $(0, 2)$ part. Its rational image is of type $(1, 1)$, hence belongs to $\text{NS}(P_1)_{\mathbb{Q}}$ by Lefschetz $(1, 1)$. Extend scalars to $\mathbb{R}$ in (8.4) to obtain (8.3). $\square$

Swapnajit Das's positive-class and ruled-branch arguments [15, Lemmas 7.2–7.3 and Corollary 7.4] are close predecessors of this mechanism for two disjoint degree-one branches. The proof above uses the same ambient class on both restrictions and includes the single normalized degree-two branch. It requires no rational polarization.

8.3 A uniform scalar bound

We spell out why the class congruence gives finite image for a whole group, not just a finite-order scalar for each individually chosen comparison.

Lemma 8.3. *Let P be a smooth connected nonprojective compact Kähler surface with a nowhere-vanishing holomorphic two-form η . Put $N = \text{NS}(P)_{\mathbb{R}}$. Let $G \subset \text{Aut}(P)$ be a subgroup. Assume either that N is degenerate, or that there exists $c \in H^{1,1}(P, \mathbb{R})$ with $c^2 > 0$ and $\sigma^*c - c \in N$ for every $\sigma \in G$. Then the image of G on $\mathbb{C}\eta^{\otimes m}$ is finite for every $m > 0$. More precisely, every volume scalar has order in the finite set of integers n with $\varphi(n) \leq b_2(P)$.*

Proof. The intersection form on $H^{1,1}(P, \mathbb{R})$ has Lorentz signature $(1, h^{1,1} - 1)$. Its restriction to N is nonpositive. To see this, a rational positive-square class in N , after scaling and choosing its sign, is $c_1(L)$ with positive Kähler degree. Riemann–Roch and Serre duality give $h^0(P, L^v) \gg v^2$: the Euler characteristic has positive quadratic leading term and $H^0(P, K_P \otimes L^{-v}) = 0$ for large v by its negative Kähler degree. This would make P Moishezon and hence projective. Rational approximation in N rules out a positive real class as well.

If N is negative definite, orthogonally project c to $N^{\perp}$. The projection $c_{\perp}$ has positive square and is fixed by every σ^* : the congruence kills its projected difference, and each automorphism preserves N and the intersection form. The perpendicular of $c_{\perp}$ is negative definite, so all eigenvalues on $H^{1,1}(P, \mathbb{R})$ have modulus one.

If N is degenerate, its radical is a rational isotropic line ℓ ; nonpositivity in a Lorentz space permits no larger radical. An integral automorphism preserves a primitive integral generator of ℓ up to sign. The invariant flag

$$\ell \subset \ell^{\perp} \subset H^{1,1}(P, \mathbb{R})$$

has eigenvalues ± 1 on the first and last quotients, which pair dually, and a negative-definite middle quotient $\ell^{\perp}/\ell$. All $(1, 1)$ eigenvalues again have modulus one. This argument permits unipotent action and makes no finiteness assertion for all cohomology.

Write $\sigma^*\eta = \delta_{\sigma}\eta$. Integration of $\eta \wedge \bar{\eta}$ gives $|\delta_{\sigma}| = 1$, and the conjugate eigenvalue has the same modulus. Since a nowhere-vanishing canonical form spans $H^{2,0}(P)$, all eigenvalues on $H^2(P, \mathbb{C})$ now have modulus one. The action on $H^2(P, \mathbb{Z})/\text{torsion}$ is integral. Kronecker's theorem makes each eigenvalue a root of unity; if its order is n , its cyclotomic polynomial has degree $\varphi(n) \leq b_2(P)$. There are only finitely many such n . The scalars of every element of G therefore lie in one finite set of roots of unity, which proves finite image on $\mathbb{C}\eta^{\otimes m}$. $\square$

If P has algebraic dimension zero, its minimality and [36, Theorem 4] imply that its canonical line is trivial, so it has a nowhere-vanishing volume η . Every meromorphic pluriform is a constant multiple of a power of η , since the quotient is a meromorphic function and $a(P) = 0$. In particular the section space under consideration has dimension at most one. Composing (8.3) along a returning sequence of links gives $\sigma^*c_Z - c_Z \in \text{NS}(P)_{\mathbb{R}}$; each intervening isomorphism preserves Néron–Severi. Lemma 8.3 applies, with the fixed cohomology rank of this P . It gives a finite scalar image for all returning compositions at once.

8.4 Algebraic dimension one

Suppose $a(P) = 1$. Its holomorphic algebraic reduction is an elliptic fibration $v : P \rightarrow J$ with connected fibers over a smooth compact curve, and every curve on P is vertical [36, Theorem 2]. Every automorphism preserves this reduction, which is determined by the meromorphic function field. Let G be the group of returning link automorphisms and let H be the corresponding finite-dimensional invariant space of meromorphic qk -pluriforms on P .

On a smooth base open avoiding the finitely many polar fibers of a basis of H , division by a base differential makes these forms holomorphic powers of the elliptic differential on each fiber. An automorphism over the base multiplies that differential by a root of unity of order 1, 2, 3, 4, or 6; its multiplier is holomorphic with values in a finite set, hence locally constant. It acts on all of H by the corresponding common scalar. Thus the kernel of $G \rightarrow \text{Aut}(J)$ has finite image on H . If the image on J is finite, its finitely many cosets make the full image finite as well.

It remains to consider an infinite base image. A curve of genus at least two has finite automorphism group. If $J = \mathbb{P}^1$, the finite set of nonsmooth fibers, including multiple fibers, is invariant, so it would have at most two points. This contradicts the theorem that a nonalgebraic compact Kähler elliptic surface over $\mathbb{P}^1$ has at least three singular fibers [7, Proposition A.1]. This theorem is a short form of the elliptic canonical-degree exclusion needed here and does not assume a section of the fibration.

If J has genus one, the infinite base group contains infinitely many translations. Its invariant finite special-fiber set must be empty. Choose a nonzero holomorphic two-form η on P , whose existence was proved above, and write its effective integral zero divisor as D_P . Every component is vertical. Each fiber is now smooth and connected, hence an irreducible reduced elliptic curve; a vertical prime is that entire fiber, and smoothness gives it multiplicity one in the pullback of its base point. Thus $D_P = v^*D_J$ for an effective integral divisor D_J on J , and the actual section gives

$$\omega_P \simeq \mathcal{O}_P(D_P) \simeq v^*\mathcal{O}_J(D_J).$$

Pullback $v^* : \text{Pic}(J) \rightarrow \text{Pic}(P)$ is injective. Indeed, if $v^*M \simeq \mathcal{O}_P$, connected fibers and projection formula give

$$M \simeq M \otimes v_*\mathcal{O}_P \simeq v_*v^*M \simeq v_*\mathcal{O}_P \simeq \mathcal{O}_J.$$

For an automorphism $\sigma \in G$ covering $h \in \text{Aut}(J)$, the identities $v\sigma = hv$ and $\sigma^*\omega_P \simeq \omega_P$ therefore imply $h^*\mathcal{O}_J(D_J) \simeq \mathcal{O}_J(D_J)$.

Put $d = \deg D_J$ and write $J = \mathbb{C}/\Lambda$. If translation t_x stabilizes $\mathcal{O}_J(D_J)$, then $t_x^*D_J - D_J$ is principal. Its point sum in J is $-dx$, whereas a principal divisor has point sum zero. To recall the latter fact, lift its meromorphic function to an elliptic function F , and choose a fundamental parallelogram $\mathcal{P}$ with boundary avoiding its zeros and poles. For a lattice basis ω_1, ω_2 , the residue theorem and pairing opposite edges give

$$\sum_{z \in \mathcal{P}} \text{ord}_z(F)z = \frac{1}{2\pi i} \int_{\partial \mathcal{P}} z \frac{F'(z)}{F(z)} dz = \omega_1 n_2 - \omega_2 n_1 \in \Lambda,$$

where $n_i = (2\pi i)^{-1} \int_{z_0}^{z_0 + \omega_i} F'(z)/F(z) dz \in \mathbb{Z}$, since the endpoint values of F agree. It follows that $dx = 0$ in J . If $d > 0$, all stabilizing translations lie in the finite group $J[d] = (\frac{1}{d}\Lambda)/\Lambda$. The infinitely many translations above therefore force $d = 0$. Effectivity gives $D_J = 0$, so η is nowhere vanishing.

The ratio of any holomorphic two-form to η is holomorphic on the compact connected P , hence constant. Thus η spans $H^0(P, \omega_P)$, and every $\sigma \in G$ satisfies $\sigma^*\eta = c_\sigma \eta$ for a nonzero constant c_σ .

Every element of H is $\eta^{\otimes qk}$ times a meromorphic function from J . The corresponding function space is therefore preserved by the base action. The union of the pole sets of a basis is finite and intrinsic to this vector space, hence invariant under the base group. Infinitely many translations preserve no nonempty finite subset. Thus these functions have no poles and are constant.

The fiber class belongs to $\text{NS}(P)_{\mathbb{R}}$, has square zero, and is nonzero by its positive Kähler area. Nonpositivity of $\text{NS}(P)_{\mathbb{R}}$ makes it a radical vector. The degenerate case of Lemma 8.3 gives the same finite scalar image on H . This completes the algebraic-dimension-one case and the proof of Proposition 8.1.

For later use, finiteness of isotropy also controls all transports in a fixed degree between two objects of the same orbit. Choose one comparison from an orbit representative to each member. Every other transport to that member is a self-transport of the representative, whose image is finite, followed by this chosen transport. Thus only finitely many linear transport actions occur in that degree, even if the groupoid has infinitely many bimeromorphic arrows.

9 Compatible sections on the whole floor

We finish Theorem 6.1 by constructing sections on all strata at one common degree. The construction follows the pre-admissible and admissible section induction of Fujino [19, Section 4], using the restriction criterion and finite images proved in the preceding sections. The point to retain is that compatibility holds on every intersection before the sections descend to the existing line on the reduced floor.

We proceed from lower strata to higher ones. At each stage the restriction criterion lets us choose all extensions of an already compatible lower collection. Those extensions need not be invariant under comparisons. Finite products of their transports will impose invariance while raising every prescribed lower section to the same power. To make this possible, we first show that transport preserves the prescribed lower restrictions, including when a surface comparison contracts a floor curve.

For a positive degree k and $d \in \{0, 1, 2, 3\}$, a *system of tuples through dimension d* is a vector subspace

$$\mathcal{V}_d(k) \subset \bigoplus_{\dim Z \leq d} H^0(Z, L_Z^k)$$

whose tuples satisfy the residue restriction equality along every incidence among these strata. We call the system invariant if, for each $e \in \{0, \dots, \min(d, 2)\}$, each tuple is compatible with every arrow of $\mathcal{G}_e$: the arrow carries its component at the source to its component at the target. It *generates* if for every such stratum Z and every $z \in Z$, some tuple has its Z -component nonzero at z . These are linear conditions except for generation. The vector space is finite-dimensional because there are finitely many strata and each is compact.

If a system generates, the span of the componentwise v th powers of its tuples generates in degree kv . Restriction and transport commute with products, so compatibility and invariance persist. We may therefore enlarge a successful degree to any sufficiently divisible later degree. Empty collections of strata impose no condition.

Lemma 9.1 (Transport preserves lower restrictions). *Fix $k > 0$ and $d \in \{1, 2\}$. Let a tuple through dimension d be compatible along all incidences, and suppose that its part through dimension $d - 1$ is invariant. For any arrow of $\mathcal{G}_d$, transport of the tuple's source component has the assigned lower restriction on every floor stratum of its target.*

Proof. For curves, the comparison is an isomorphism preserving the marked points and their residue generators. Invariance makes the lower point values equal, so the assertion follows.

For surfaces, take a target floor curve and view its valuation on the source of the comparison. If its center is a floor curve there, adjunction of the equality of meromorphic adjoint subsheaves on the graph gives a crepant residue comparison of the two curve pairs. This is an arrow of $\mathcal{G}_1$, and the assigned curve tuple is invariant.

If the center is a point, it is a zero-dimensional lc center of the source dlt surface, by crepancy. It is one of the point strata and is an actual smooth crossing of two coefficient-one curves: the resolution used in Proposition 6.3 is an isomorphism at such a point. In coordinates (x, z) for the crossing, a divisorial lc place above it arises by successive blowups of crossings of two coefficient-one branches. To verify this, factor a surface resolution into point blowups. In an SNC surface boundary the new crepant coefficient at the blowup of a crossing is the sum of the two coefficients minus one; at a point on just one branch it is that coefficient minus one. Starting with coefficients at most one, a new coefficient one can arise only at a crossing of two coefficient-one branches. This remains true at every subsequent step.

The corresponding exceptional curve is a $\mathbb{P}^1$ whose different has exactly the two adjacent coefficient-one points. Residue of $(dx/x \wedge dz/z)^{\otimes qk}$ along it is $(dt/t)^{\otimes qk}$, up to a sign removed in the even degree: at each crossing blowup the logarithmic change-of-coordinate determinant is ± 1 . Further point blowups do not change this meromorphic residue on its strict transform. The normal target floor curve is bimeromorphic, hence isomorphic, to this $\mathbb{P}^1$, and its actual log line is trivial with that generator. Its section is determined by the common residue at the two point strata. Pulling a local surface section through the crossing restricts on the exceptional curve to its value at the point times that generator. Compatibility of the original tuple identifies this value with its assigned point component, and invariance makes it the assigned value at both target point strata. Thus the entire target curve receives exactly the prescribed lower section.

The two alternatives apply to every composite surface comparison, so the assertion holds for the whole groupoid, not only for a generating link. $\square$

Proposition 9.2. *There is a degree $k > 0$ and a compatible generating system of tuples through dimension three which is invariant in dimensions zero, one, and two.*

Proof. For point strata take the same scalar in the canonical zero-form generator 1 on every point. The even iterated residue convention identifies these generators along all paths. This gives an invariant generating system in dimension zero.

Suppose a system has been constructed through dimension $d - 1$, for $1 \leq d \leq 3$. Replace it by powers and their span in a sufficiently large common degree, still denoted k . For a d -stratum Z , the components of any lower tuple glue to a section on its entire floor C_Z by Proposition 6.3. They satisfy its link comparison: for $d = 1$ the link is a point comparison, for $d = 2$ a curve comparison, and for $d = 3$ one of the generating surface links. Lower invariance supplies each equality. Proposition 7.3 therefore extends every such boundary section to Z , once the degree is sufficiently large. There are only finitely many strata, so the degree can be chosen uniformly.

Let $\mathcal{P}_d(k)$ be the vector space of all tuples through dimension d whose lower part belongs to the chosen lower system and whose d -components restrict to those glued floor sections. It maps surjectively to the lower system, since the finitely many extensions can be chosen independently. This is a linear space of lifts; no choice of a nonlinear extension operator is involved.

The pre-system $\mathcal{P}_d(k)$ generates. For $z \in Z$ with $y = f_Z(z) \in f_Z(C_Z)$, choose a floor point above y and a lower tuple nonzero there. Any extension to Z is the pullback of a section of N_Z^k , by (7.1). Its value at y is nonzero, and hence it is nonzero at z . If $y \notin f_Z(C_Z)$, the sheaf

$\mathcal{I}_{f_Z(C_Z)} \otimes N_Z^k$ is generated for large k by Serre's theorem. A section nonzero at y pulls back to a section vanishing on C_Z ; with zero lower tuple and all other new components zero, it belongs to $\mathcal{P}_d(k)$. This also covers an empty floor. Lower points remain generated because the projection to the lower system is surjective.

For $d = 3$ this pre-system is the required final system. For $d = 1, 2$ we impose invariance by norm products. Lemma 9.1 applies to every pre-tuple: all transported top components have the same assigned lower restrictions. A product of h such factors will therefore have the original lower tuple raised componentwise to h .

For each orbit of d -strata choose a representative Z , and let G_Z be the finite isotropy image on $H^0(Z, L_Z^k)$ from Proposition 8.1. Given a pre-tuple with component s_Z , form its norm

$$P_Z(s_Z) = \prod_{g \in G_Z} g(s_Z) \in H^0(Z, L_Z^{k|G_Z|}).$$

Choose a common integer h divisible by every $|G_Z|$, and use $P_Z(s_Z)^{h/|G_Z|}$. Transport this section to every member of its orbit. This is well defined: a change of the chosen transport by an isotropy arrow permutes the factors in degree k , and transport commutes with multiplication. No finiteness statement about the possibly larger representation in degree kh is needed. Each factor has the assigned lower restriction proved above, so the resulting lower tuple is the original lower tuple raised componentwise to h . Thus all these sections are compatible across different orbits and all deeper intersections, and they are invariant in dimension d .

They still generate. Fix a point z of an orbit member. For each of the finitely many transported factors, choose a point over z on a resolution of its comparison graph. Equality of the pulled-back invertible adjoint subsheaves identifies the line fibers. Nonvanishing of that factor at z is therefore the nonvanishing of the representative pre-section at a definite point of Z . Each such condition is a nonzero linear evaluation functional on the generating vector space $\mathcal{P}_d(k)$. A finite union of the proper kernels of these functionals cannot cover a complex vector space. One pre-tuple satisfies all of them, and its norm is nonzero at z . At a lower point choose a pre-tuple whose assigned lower value is nonzero; its h th power remains nonzero. Finally take the linear span of all the constructed norm tuples. Compatibility and invariance are linear conditions, so this span is an invariant generating system in degree kh .

This completes the induction through dimensions one and two; the pre-system at dimension three then has all the required properties. $\square$

Proof of Theorem 6.1. Apply Proposition 9.2. Its three-dimensional components agree on every subordinate stratum, so Proposition 6.3 descends each tuple uniquely to a section of $L^k|_S$. For any $x \in S$, choose a component through x . The generating system has a tuple whose value there is nonzero. It is the pullback of the descended value in the actual line fiber at x , so that descended value is nonzero. Nakayama's lemma makes the evaluation map onto $L^k|_S$ surjective at x . The finite-dimensional span of these global sections therefore generates at every point, proving semiampleness on the whole reduced floor. $\square$

For the supported model in Proposition 5.1, choose a common multiple of the actual adjoint index, the coefficients and actual equivalence of P , and the generated boundary degree just obtained. It gives a line $L = \mathcal{O}_V(qA)$, a section with divisor qP , and a generated restriction on S . Its boundary sections define a holomorphic map $S \rightarrow \mathbb{P}^b$ with the actual pullback identity for $L|_S$. The gluing argument has not extended these sections to V ; that is the distinct lifting task to which we now turn.

10 Root neighborhoods and a split residue map

We begin the proof of Theorem 1.2. A generated multiple of the actual adjoint restriction constructs a morphism from the reduced boundary to projective space. Near its compact fibers we will replace the supported divisor by a reduced Cartier covering divisor and study sections on its finite neighborhoods. The use of neighborhood covers and successive finite thickenings has its threefold antecedents in [40, Sections 1–2 and 4] and [35, Section 4]. Here no extension of the boundary map off the reduced boundary is assumed.

Fix the data of Theorem 1.2, and put $S = \lfloor B \rfloor$, so $\text{Supp } P = \text{Supp } S$. Choose a sufficiently divisible even integer $q > 0$ which clears the actual adjoint and equivalence indices, the boundary and P coefficients, and a generated degree of $A|_S$. There are then an actual line and section

$$L = \mathcal{O}_V(qA), \quad s \in H^0(V, L), \quad G = \text{div}_L(s) = qP = \sum_i d_i S_i, \quad (10.1)$$

where every d_i is a positive integer, and a morphism

$$f : S \longrightarrow T = \mathbb{P}^b, \quad L|_S \simeq f^* \mathcal{O}_T(1). \quad (10.2)$$

The morphism is constructed from a generating system on S ; it is used only on that reduced subspace and is not required to be surjective. Choose $r > 0$ divisible by q and every d_i , and put $a = r/q$. In particular a is a positive integer and $d_i \leq r$.

10.1 Root pairs near a compact analytic subspace

Lemma 10.1 (A root with a prescribed boundary comparison). *Let A be a compact analytic subspace of a Hausdorff complex analytic space X , and let L be a holomorphic line on a neighborhood of A . Suppose a line Q on A and an isomorphism $\beta : Q^{\otimes r} \simeq L|_A$ are given, with $r > 0$. There are a neighborhood U of A , a line P on U , and isomorphisms*

$$\alpha : P^{\otimes r} \simeq L|_U, \quad \gamma : P|_A \simeq Q, \quad \alpha|_A = \beta \circ \gamma^{\otimes r}.$$

If two root pairs are already defined near A , every prescribed power-compatible isomorphism between their restrictions to A extends to an isomorphism on a neighborhood, uniquely as a germ about A . The root pair itself is not asserted unique.

Proof. The analytic Kummer sequence

$$1 \longrightarrow \mu_r \longrightarrow \mathcal{O}_X^\times \xrightarrow{(\cdot)^r} \mathcal{O}_X^\times \longrightarrow 1$$

is exact also for singular or nonreduced analytic spaces. A unit germ has a local holomorphic root; the kernel is the locally constant sheaf of r th roots of unity, since $u^r - 1$ has distinct roots, including in a local ring with nilpotents. Line bundles are classified by $H^1(\mathcal{O}^\times)$, and abelian torsors by H^1 of the corresponding sheaf [54, Tags 09NU and 02FQ].

For an abelian sheaf F and a compact subset whose distinct points have disjoint neighborhoods, SGA 4 continuity [51, Exposé Vbis, Lemma 4.1.3] gives

$$\varinjlim_{U \supset A} H^i(U, F) = H^i(A, F|_A) \quad \text{for every } i. \quad (10.3)$$

The separation hypothesis holds in the Hausdorff analytic space. Although the intrinsic structure sheaf of A need not be the inverse image of $\mathcal{O}_X$, the inverse image of μ_r is exactly μ_r on A .

Naturality of Kummer therefore makes the obstruction to a root of L restrict to zero in $H^2(A, \mu_r)$, because Q is such a root there. Continuity in degree two kills that obstruction on a smaller neighborhood, giving a root pair (P, α) .

The difference between $(P|_A, \alpha|_A)$ and (Q, β) is a μ_r -torsor. By continuity in degree one it extends to a torsor on a smaller neighborhood. The associated line with its trivialized r th power twists P to give the prescribed pair on A . Finally, the sheaf of compatible isomorphisms of two root pairs is itself a locally trivial finite μ_r -torsor. A prescribed section on A trivializes its restriction there. Continuity in degree one therefore trivializes this torsor on a smaller neighborhood; choose a section there. The ratio of its boundary restriction to the prescribed section is a section of μ_r on A . Continuity in degree zero extends that ratio uniquely as a germ, and correcting the chosen section gives the prescribed extension. The same degree-zero injectivity gives uniqueness as a germ. These sections are holomorphic, being locally solutions of $z^r = u$ for a holomorphic unit. $\square$

Apply the lemma near a parameter $t \in T$ to $A = (f^{-1}(t))_{\text{red}}$, using a local frame ℓ of $\mathcal{O}_T(1)$ and the trivial root whose r th power is $f^*\ell$. We obtain a root pair $P_1^r \simeq L$ on a neighborhood of that compact fiber. The prescribed compatible frame on the fiber extends as a germ inside S , by the last part of the lemma. If it is defined on $N_S \subset S$ containing the fiber, properness of f gives a smaller base neighborhood $U \ni t$ with $S_U = f^{-1}(U) \subset N_S$: remove the closed set $f(S \setminus N_S)$. Shrink once more so this whole S_U lies in the ambient root neighborhood W_0 , and replace W_0 by $W = W_0 \setminus (S \setminus S_U)$. Thus

$$W \cap S = S_U, \quad P_1^r \simeq L|_W, \quad p_1 \text{ a frame of } P_1|_{S_U}, \quad p_1^r = f^*\ell. \quad (10.4)$$

Comparisons between two choices, with prescribed power-compatible boundary frames, extend uniquely as ambient germs near a compact fiber. Only this torsion comparison is extended; no arbitrary boundary trivialization is extended off S .

10.2 The first normalized cover and the lifting target

On a chart (10.4), take the full normalized cyclic cover $\pi : Z \rightarrow W$ defined by

$$y^r = \pi^*s \quad \text{in } \pi^*P_1^r,$$

retaining all components and the μ_r -action. Finite analytic normalization is available by [34, Part B, Section 4, Corollaries 2–3]. At a general point of S_i , a root of a local unit reduces the equation to $y^r = x^{d_i}$. Because $d_i \mid r$, each normalized branch is

$$x = t^{r/d_i}, \quad y = \zeta t.$$

The ramification index is r/d_i and y has order one. Outside S the equation roots a unit and is étale. A finite map preserves the dimension of a prime analytic subset, so no further divisor above a codimension-two set is missed. The zero divisor

$$E = (y = 0)$$

is Cartier: y is a nonzerodivisor on each normal component. A Cartier divisor on a normal space is S_1 , and the calculation makes it generically reduced; hence it is reduced. Its canonical line identity is

$$\mathcal{O}_Z(E) \simeq \pi^*P_1, \quad (10.5)$$

sending the canonical section of $\mathcal{O}_Z(E)$ to y .

Put

$$I = \mathcal{O}_Z(-E), \quad g = f \circ \pi|_E : E \rightarrow U, \quad \mathcal{A}_j = I^j/I^{j+1} = \mathcal{O}_E(-jE) \quad (j \in \mathbb{Z}). \quad (10.6)$$

The map $E \rightarrow S_U$ is finite and g is proper. Each $\mathcal{A}_j$ is invertible on the possibly singular reduced E . In the Laurent graded algebra of these layers, $u = y/p_1$ denotes the degree-one frame. This is intrinsic on the associated graded: locally divide the equation of E by the boundary value of a frame of P_1 . It does not assert that p_1 is an ambient frame.

For $j \in \mathbb{Z}$ and $k \geq 1$, put

$$\mathcal{T}_{j,k} = I^j/I^{j+k}. \quad (10.7)$$

We view this as a sheaf of complex vector spaces on the underlying topological space of E . A sheaf on Z supported on the closed subset E is canonically such a sheaf. Thus g_* and its derived functors are defined for $\mathcal{T}_{j,k}$, although there is no holomorphic map from its thickening to U . For a coherent layer $\mathcal{A}_j$, this is the usual coherent analytic direct image. The finite-neighborhood assertion we will prove is the following.

Proposition 10.2 (All finite lifting orders). *For every local root chart (10.4), every integer j , and every $k \geq 1$, the map*

$$g_*\mathcal{T}_{j,k+1} \longrightarrow g_*\mathcal{T}_{j,k} \quad (10.8)$$

is an epimorphism of sheaves of complex vector spaces on U . The neighborhoods used to lift a particular germ may depend on j and k .

The proof is in Section 12. To see what geometry it requires, consider the exact sequence

$$0 \longrightarrow \mathcal{A}_{j+k} \longrightarrow \mathcal{T}_{j,k+1} \longrightarrow \mathcal{T}_{j,k} \longrightarrow 0. \quad (10.9)$$

Its connecting map takes a section of $g_*\mathcal{T}_{j,k}$ to an obstruction in $R^1g_*\mathcal{A}_{j+k}$. In order k , choosing $j = -a - k$ puts that obstruction in the fixed sheaf $R^1g_*\mathcal{A}_{-a}$. We next construct a split insertion of this sheaf into the first cohomology of an SNC dualizing sheaf. The Hodge argument will apply there, and the induction will return from these negative degrees to all integer degrees.

10.3 A canonical root and its resolved support

Take a second full normalized cover, this time adjoining a q th root of π^*s as a meromorphic q -pluricanonical form. The local construction in Lemma 7.1 also proves its existence on the normal analytic Z : if its coefficient in a meromorphic canonical generator is a_0/b_0 , normalize the finite reduced monic algebra

$$A[w]/(w^q - a_0b_0^{q-1}) \subset \text{Frac}(A)[t]/(t^q - a_0/b_0).$$

The total meromorphic algebra is separable and its components all dominate. A change of generator identifies the total meromorphic algebras by multiplying the root by a meromorphic unit. The displayed finite monic orders need not coincide under this identification, but their integral closures do: each is the integral closure of the base ring in that same total algebra. These closures therefore glue. The forms $t\eta$ glue to the tautological meromorphic top form τ . For the total meromorphic field K of a normal base component, the full μ_q -action has invariant algebra K and root-character space Kt ; the corresponding meromorphic form space is $K\tau$. The invariant subalgebra of the normalized holomorphic algebra is the base ring A , by normality. These statements are read componentwise on a disconnected base. The earlier μ_r -action lifts functorially, preserving the

pulled-back pluriform, and commutes with μ_q . Choosing one component instead would in general destroy these assertions.

Resolve this second cover and principalize the pulled ideal of E , equivariantly for these finite actions. We use the analytic smooth-functorial resolution and principalization of [53, Theorems 1.1.11 and 1.1.13], together with its nonembedded analytic construction [52, Sections 5.2.2 and 5.3.2]. We retain the final indexed complete boundary in the construction of F_{princ} : after the blowup of the ideal, boundary desingularization makes the final strictly monomial support a union of components of this SNC boundary. Its closed indexed intersections are regular by [53, Lemma 2.1.10]. Splitting the disjoint connected components of each boundary component near the compact fiber gives globally smooth component labels; their intersections remain smooth, and the finite group may permute the labels. On a neighborhood of a compact parameter fiber only finitely many stages of the locally finite blowup hypersequence occur; a finite-group invariant shrink preserves equivariance. Write the composite as $\rho : \hat{Z} \rightarrow Z$, with $\hat{Z}$ smooth, and put

$$D_E = \rho^* E, \quad H = (D_E)_{\text{red}}, \quad \rho_H : H \rightarrow E, \quad h = g\rho_H. \quad (10.10)$$

Here D_E is the scheme inverse Cartier divisor and H is its globally simple SNC reduction. The tautological form satisfies $\tau^{\otimes q} = (\pi\rho)^* s$ with differential pullback understood.

These constructions commute with restrictions to opens and ordinary products by a complex manifold. For normalization, the product of the normal normalization with a manifold is finite, normal, and bimeromorphic to the reduced product, hence is its normalization by uniqueness; normality of these analytic products is part of [34, Part C, Theorem 4(b)]. The resolution is functorial for smooth morphisms, including products. We make no assertion about normalization or resolution under a ramified base change.

We will also need the Kähler property near the resolved support. A finite analytic map is projective locally: generators of its finite algebra embed it in a relative projective space, and the affine coordinate 1 makes the trivial line relatively ample. The finite maps and the finitely many blowups above are therefore projective near the preimage of a compact set. Patch local positive metrics of a relatively ample line by partitions pulled from the base. Their curvature remains positive on vertical tangent directions. A sufficiently large multiple of the pulled-back Kähler form of W makes it positive in all directions near the compact preimage; compactness bounds the mixed terms uniformly. This gives a closed Kähler form on an ambient neighborhood $\mathcal{N}$ of the compact resolved fiber under consideration. Properness of $h : H \rightarrow U$ lets us shrink U so that all of H above the smaller U lies in $\mathcal{N}$: remove the closed image of $H \setminus \mathcal{N}$. We make this shrink in each root chart. Every closed intersection of components of H is then smooth and proper over U , and inherits one global relative Kähler form there. A connected component can split after a further restriction of the base, so we make the indexing choice only after these neighborhood restrictions.

10.4 Finite indexing near a parameter fiber

The filtered direct-image construction will use the closed intersections of globally indexed smooth components. The following lemma makes that indexing finite after a base restriction, even when some strata map only to special parameter loci.

Lemma 10.3 (Components near a compact fiber). *Let $h : H \rightarrow T$ be a proper holomorphic map to a complex manifold, and fix $t_0 \in T$. Suppose that near $h^{-1}(t_0)$ the reduced support H has a locally finite family of closed smooth labels D_λ which locally are its distinct SNC branches, and that every closed intersection of these labels is smooth. After restricting to a suitable open neighborhood U of t_0 , the support has finitely many globally smooth indexed irreducible components. Every closed*

intersection of the new components has finitely many connected components, all smooth and proper over U . A global relative Kähler form already given on the old strata restricts to such a form on the new ones. The open U can be chosen inside any previously prescribed neighborhood of t_0 .

Proof. Local finiteness and compactness give an open neighborhood $\mathcal{N}_1$ of the central fiber which meets only finitely many labels. Remove from the base the closed image of $H \setminus \mathcal{N}_1$; properness then puts the entire restricted support in $\mathcal{N}_1$. There are now finitely many closed indexed intersections to consider. The connected components of each form a locally finite family in H : the intersection is closed and smooth, so near one of its points a small smooth neighborhood meets only its own component, and near a point outside it a neighborhood avoids it. Compactness, followed by the same properness argument, gives a second neighborhood $\mathcal{N}_2$ meeting only finitely many such components and a base restriction on which the whole support lies in $\mathcal{N}_2$.

Let $C_1, \dots, C_N$ be the whole connected components of the pre-restriction closed intersections which meet $\mathcal{N}_2$. We retain the whole components, not just their intersections with $\mathcal{N}_2$. Each C_a is closed in a closed stratum and is a connected complex manifold. Thus $f_a = h|_{C_a}$ is proper over the preceding base and C_a is irreducible. Their restrictions cover every closed intersection after the current restriction, including the singleton intersections.

Put $S_a = f_a(C_a)$ with its reduced structure. Remmert's proper mapping theorem makes S_a closed analytic [25, Section 7, no. 1, Satz 1, p. 60], and it is irreducible because C_a is. Inside any prescribed open coordinate neighborhood $V \ni t_0$ contained in the preceding restrictions, take the locally finite irreducible decompositions of all $S_a \cap V$. Let B be the union of those components which do not contain t_0 . A union of a subfamily of the locally finite irreducible components of an analytic set is closed analytic. Since the list of a 's is finite, B is closed analytic in V and misses t_0 . Set $U = V \setminus B$.

This removal does not split any retained irreducible component A of an $S_a \cap V$. Its regular locus is connected, and its intersection with B is a proper analytic subset: otherwise closedness and density would imply $A \subset B$, contrary to $t_0 \in A \setminus B$. The complement of a proper analytic subset in a connected complex manifold is connected. One may see this by perturbing a path, in a finite chain of coordinate balls, off a subset of positive complex codimension. Hence $\text{Reg}(A) \setminus B$ is connected and $A \cap U$ is irreducible. Only finitely many components of $S_a \cap V$ contain t_0 , by local finiteness. Their restrictions give a finite cover of $S_a \cap U$ by irreducible closed analytic subsets, all containing t_0 . Therefore every irreducible component of $S_a \cap U$ contains t_0 ; if S_a misses t_0 , its restriction is empty.

Let W be a connected component of $f_a^{-1}(U)$. It is open in the connected smooth C_a , is closed in $f_a^{-1}(U)$, and is smooth and irreducible. Its map to U is proper. Put $r_a = \dim S_a$. The maximal-rank locus of f_a is dense in C_a : the generic rank theorem gives maximal rank r_a , and the lower-rank locus is a proper analytic subset defined by the maximal minors. The nonempty open W meets the maximal-rank locus. Its proper image is consequently an irreducible closed analytic subset of $S_a \cap U$ of dimension r_a . Every component of $S_a \cap U$ has this dimension because S_a is irreducible and hence pure-dimensional. The image is therefore an irreducible component, since a proper analytic subset of such a component has smaller dimension. It therefore contains t_0 , and W meets $f_a^{-1}(t_0)$.

The compact analytic fiber has finitely many connected components: on its reduction, irreducible components are locally finite and compactness makes their number finite. Each $W \cap f_a^{-1}(t_0)$ is a nonempty open-and-closed subset of this fiber. Different W 's give disjoint such subsets, so

$$\#\pi_0(f_a^{-1}(U)) \leq \#\pi_0((f_a^{-1}(t_0))_{\text{red}}) < \infty. \quad (10.11)$$

This includes zero-dimensional and other vertical images and permits nonreduced scheme fibers.

Reindex the singleton strata by these finitely many components W . An intersection of the new labels is an open-and-closed part of the restriction of the corresponding old closed intersection, hence is a union of some of its finitely many connected components. It remains smooth and proper, and the relative form restricts. Distinct new pieces from one old label are disjoint, so the local SNC branches and their ordering are unchanged. This proves the assertion. $\square$

Apply Lemma 10.3 after the preceding Kähler neighborhood and properness restrictions, to all regular labels and their closed intersections. The resulting local support has the finite global component indexing used in Section 11. The final open need not be Stein or a polydisc: the roots and covers have already been constructed, and their remaining local uses only restrict the existing coordinate charts.

10.5 A residue insertion that retains all cohomology

Proposition 10.4 (Split residue insertion). *For the data (10.6)–(10.10), multiplication by τ gives a morphism*

$$\rho^* \mathcal{O}_Z(aE) \longrightarrow \omega_{\widehat{Z}}(H),$$

and residue gives $\rho_H^ \mathcal{A}_{-a} \rightarrow \omega_H$. The resulting map in the derived category of E has a retraction in the τ -character. Consequently, for every j , and in particular for $j = 1$, it gives a split injection*

$$\tau_* : R^j g_* \mathcal{A}_{-a} \hookrightarrow R^j h_* \omega_H. \quad (10.12)$$

The construction is natural under the power-compatible root comparisons of Lemma 10.1. Under an ordinary product with a complex manifold B_0 , it is the pullback construction for canonical sheaves relative to B_0 . For absolute canonical sheaves the insertion and its retraction are tensored with ω_{B_0} ; in particular the product insertion has source $\mathcal{A}_{-a} \boxtimes \omega_{B_0}$ and target $R(\rho_H \times \text{id})_ \omega_{H \times B_0}$.*

Proof. Let v be a local meromorphic frame of $\mathcal{O}_Z(aE)$. By (10.5) and $aq = r$, the q -pluriform $v^q \pi^* s$ is, up to a holomorphic unit, the pullback of a local frame ℓ_V of L . On a log resolution of (V, B, G) , write b_F for a crepant boundary coefficient and $m_F = \text{ord}_F(s/\ell_V)$. Then $m_F \geq 0$, $b_F \leq 1$, and $b_F = 1$ implies $m_F > 0$: the dlt pair is klt off S , and the section vanishes exactly on S . Thus the local density

$$|\ell_V|^{2/q} |s/\ell_V|^{2\epsilon}$$

has exponent $-b_F + \epsilon m_F > -1$ at each prime for every $\epsilon > 0$, and is locally integrable. Change of variables under the proper generically finite map $\pi\rho$ preserves this integrability. For $\alpha = (\rho^* v)\tau$, let k_F be its integral order at a prime upstairs and n_F the order of the pulled section ratio. Integrability says $k_F + \epsilon n_F > -1$ for all $\epsilon > 0$. The order n_F is positive exactly on H . Hence $k_F \geq -1$ on H and $k_F \geq 0$ elsewhere. This proves the logarithmic map. Cartier adjunction on the reduced SNC divisor gives its residue map on H , with $\rho_H^* \mathcal{A}_{-a} = \rho^* \mathcal{O}_Z(aE)|_H$.

Let χ be the τ -character of the second cover. At a general point of E over S_i , differential pullback gives

$$\text{ord}_E(\pi^* s) = \frac{r}{d_i}(d_i - q) + q \left(\frac{r}{d_i} - 1 \right) = r - q = q(a - 1).$$

The second root is unramified there and $\text{ord}_E \tau = a - 1$. At any other codimension-one prime the first cover is étale and the original boundary coefficient is $\beta \in [0, 1)$. For $e = q/\text{gcd}(q, q\beta)$, the second-cover order is $e(1 - \beta) - 1 \in \{0, \dots, e - 1\}$, by (7.2). A meromorphic form in character χ is $f\tau$ for an invariant meromorphic f descended to Z . The orders force $\text{ord}_E f \geq -(a - 1)$ and nonnegative orders at all other primes. Normality extends it as a section of the indicated

invertible line. Conversely, the logarithmic map multiplied by the equation of E has no pole because $D_E \geq H$. We obtain the exact sheaf identities

$$(\rho_* \omega_{\widehat{Z}})_\chi = \mathcal{O}_Z((a-1)E)\tau, \quad (10.13)$$

$$(\rho_* \omega_{\widehat{Z}}(D_E))_\chi = \mathcal{O}_Z(aE)\tau. \quad (10.14)$$

The second is projection formula for the invertible line $\mathcal{O}_Z(E)$.

Canonical torsion-freeness [21, Theorem 2.9 and Proposition 2.11] applies locally near E to each dominating resolved component. More precisely, around a compact fiber choose the Kähler neighborhood just constructed. Properness of ρ supplies a smaller normal target neighborhood whose entire inverse image lies in it, by removing the closed image of its complement. After a base shrink this neighborhood contains all of E in the root chart. The restricted morphism is proper with Kähler smooth source, so the cited theorem applies there. These neighborhoods cover E . On each, the higher direct images for the generically finite resolution vanish because they vanish generically. Finite pushforward is exact. Thus on their union $Z^\circ \subset Z$, an open neighborhood of E ,

$$(R^i \rho_* \omega_{\widehat{Z}})|_{Z^\circ} = 0 \quad (i > 0).$$

Projection formula gives the same restricted vanishing for $\omega_{\widehat{Z}}(D_E)$.

Put $A_E = \mathcal{A}_{-a} = \mathcal{O}_E(aE)$. Cartier adjunction for the possibly nonreduced divisor D_E is the exact sequence

$$0 \longrightarrow \omega_{\widehat{Z}} \longrightarrow \omega_{\widehat{Z}}(D_E) \longrightarrow \omega_{D_E} \longrightarrow 0.$$

Since D_E is the scheme inverse image of E , it has a morphism $\rho_D : D_E \rightarrow E$. The preceding vanishings and character identities identify

$$(R(\rho_D)_* \omega_{D_E})_\chi \simeq A_E \tau[0] \quad \text{in } D(\mathcal{O}_E). \quad (10.15)$$

One can see this first after exact closed pushforward from E to Z° , where it is the quotient of (10.14) by (10.13). Closed pushforward detects its cohomology sheaves, and the canonical truncation of a complex concentrated in degree zero gives (10.15) on E . No derived full-faithfulness assertion is needed.

Because A_E is invertible, its pullback followed by the residue map defines $A_E[0] \rightarrow R(\rho_H)_* \omega_H$. The inclusion $\omega_H \hookrightarrow \omega_{D_E}$, followed by the character projection and (10.15), gives a map back to $A_E \tau[0]$. On degree zero their composition sends f to $f\tau$ in the quotient of the two character lines. Under the chosen identification it is the identity. An endomorphism of the sheaf $A_E[0]$ in degree zero is determined by its sheaf map, so this is a derived retraction. Applying Rg_* proves the split injections (10.12). All maps were defined by the tautological form, the actual divisor ideal, and residue, so they respect root comparisons. Under an ordinary product the tautological form is relative to the new factor; wedging with a local frame of its canonical line gives the absolute map and the stated ω_{B_0} factor. $\square$

The derived retraction is stronger than an injection only on functions or at general fibers. It is this strength that allows the later argument to retain obstruction classes supported on special parameters.

10.6 One compact graph on a parameter cover

We next prepare the geometric setting for a global symbol test. For any chosen parameter t_* , the construction gives a projective parameter cover unbranched over t_* and one compact SNC graph

over the entire cover. Global vanishing can then be tested back at t_* without discarding a class supported there. The cover may change with t_* . We form and resolve the covers before imposing the graph equations; this avoids assuming that normalization commutes with ramified base change.

Proposition 10.5 (Compact transverse graph). *Assume $b > 0$, and fix any $t_* \in T = \mathbb{P}^b$. There is a coordinate-power map $\psi : T' = \mathbb{P}^b \rightarrow T$ of degree r in each coordinate, with $R = \mathcal{O}_{T'}(1)$ and an actual identity $R^r \simeq \psi^* \mathcal{O}_T(1)$, which is unbranched over t_* , and the following data. On a neighborhood of the compact graph $\Xi = S \times_T T' \subset V \times T'$ there are the two full root covers and a projective resolution with smooth ambient source $\mathcal{W}$. The graph cut H' inside the reduced inverse section divisor is compact and reduced SNC of pure dimension $\dim V - 1$, with finitely many globally smooth components. The projection $p' : \mathcal{W} \rightarrow T'$ is a submersion near H' , and every nonempty closed component intersection is smooth and proper over T' , carrying the restriction of one Kähler form on a neighborhood of H' .*

Locally near every compact graph slice over $w \in T'$, the resolved ambient data, including their projection and full tautological divisor ideal, are isomorphic to an open restriction of the ordinary product $\hat{Z} \times U'$ for a local root chart. There the cut is $H^b = H \times_U U'$, and its actual dualizing line is

$$\omega_{H^b} \simeq \text{pr}_H^* \omega_H \otimes \text{pr}_{U'}^* (\omega_{U'} \otimes \psi^* \omega_U^{-1}). \quad (10.16)$$

Proof. Choose finitely many pairs of parameter opens $U'_i \in U_i$, with the U'_i covering $f(S)$, such that a local root chart and its proper map $h_i : H_i \rightarrow U_i$ are defined on U_i . Each $h_i^{-1}(\overline{U'_i})$ is compact. Local finiteness therefore leaves only finitely many smooth component intersections meeting these compact sets. For each such stratum C and each subset of variable hyperplanes in T , consider the incidence

$$\{(z, H_1, \dots, H_v) : h(z) \in H_1 \cap \dots \cap H_v\}.$$

It is smooth, since varying each hyperplane independently supplies its normal direction. Sard's theorem for the projection to the hyperplane parameter space gives transversality for a general tuple; a countable atlas handles any noncompact stratum. Choose a tuple satisfying all the finitely many conditions and also the open conditions that it forms a coordinate basis and that none of its hyperplanes contains t_* . Use those coordinates for

$$\psi([w_0 : \dots : w_b]) = [w_0^r : \dots : w_b^r].$$

Its derivative at a point has image the tangent space to the intersection of exactly the coordinate hyperplanes corresponding to vanishing coordinates, or is invertible when none vanish. The chosen transversality therefore gives, at $h(z) = \psi(w)$,

$$dh(T_z C) + d\psi(T_w T') = T_{h(z)} T \quad (10.17)$$

for every smooth SNC stratum. The map is unbranched over t_* .

In local coordinates t_i on U , extend the coordinate functions of h holomorphically off H to functions G_i on local opens of $\hat{Z}$. In $\hat{Z} \times U'$, the equations $\psi_i(w) - G_i = 0$ have surjective differential on each H -stratum by (10.17). They cut a smooth submanifold of codimension b transverse to the SNC divisor. The cut $H^b = H \times_U U'$ is therefore reduced SNC of pure dimension $\dim V - 1$, and has codimension $b + 1$ in the product ambient. Complete-intersection adjunction gives (10.16). In coordinates its last factor has the wedge frame dw/dt ; this means a frame of $\omega_{U'} \otimes \psi^* \omega_U^{-1}$, not an inverse Jacobian.

The graph Ξ is a compact closed analytic subspace. Its actual line identity is

$$\mathrm{pr}_V^* L|_{\Xi} \simeq \mathrm{pr}_{T'}^* R^r|_{\Xi}.$$

Lemma 10.1 gives a root P_* of $\mathrm{pr}_V^* L$ on a neighborhood of Ξ , identified with $\mathrm{pr}_{T'}^* R$ along Ξ . Form the same two full normalized covers there. For the second cover use coefficients of s in meromorphic canonical generators from the first factor V . This defines its relative tautological form without requiring a Cartier relative canonical line on the singular cover; changing to a generator after differential pullback changes the coefficient by a q th power and identifies the total root algebra and its normalization. Resolve and principalize the full tautological ideal functorially on this neighborhood, before cutting by the graph.

Near w_0 , choose a frame $\bar{p}$ of R whose r th power is $\psi^* \ell$ for a downstairs frame ℓ ; a local root of a unit gives such a choice. Along the compact graph slice, compare the pullback of P_1 with P_* by sending the boundary frame p_1 to $\bar{p}$. Lemma 10.1 extends this to an ambient root isomorphism as a germ. On the graph, its ratio with the prescribed comparison is a μ_r -section equal to one on the compact slice, hence equal to one on an open graph neighborhood of that slice. Properness of $\Xi \rightarrow T'$ permits a base shrink on which the ambient comparison is defined and agrees with the prescribed one along the entire graph: remove the closed image of the complement of that neighborhood. This argument also applies to a nonreduced graph. The comparison identifies the first cover with the local ordinary product cover, and product normality identifies the second cover as well. Apply the same fixed smooth-functorial principalization to the same full ideal and ordered boundary data on both sides; functoriality identifies the results. These are isomorphisms of resolved ambient germs over the product base preserving the whole divisor ideal, not only isomorphisms of their supports.

Let H' be the graph cut inside the reduced inverse section divisor on this global resolution. Locally it is exactly H^b above, so it is reduced SNC with the asserted pure dimension. It is compact, being over the compact Ξ by proper maps. The ambient projection is a submersion near it by the local product comparison. The transverse cuts of the globally smooth inverse-divisor components are smooth; split them into their disjoint connected components. Compactness and local finiteness leave only finitely many. Their closed intersections are smooth and compact, hence proper over T' . Finally, the relative metric construction above, applied over the Kähler product neighborhood of Ξ , gives one Kähler form near H' . Its restrictions supply all the required relative Kähler forms. $\square$

The local and global graph supports in this proposition are locally a reduced SNC divisor in a smooth complete intersection, with submersive ambient projection. The next section develops the filtered direct-image statement for precisely that geometry.

11 Filtered direct images on simple normal-crossing supports

The residue insertion has placed the finite lifting obstruction in $R^1 h_* \omega_H$. We need two facts about this absolute dualizing direct image: it embeds as the lowest step of a filtered right $\mathcal{D}$ -module, and on a projective base the kernel of first-symbol multiplication admits no map from an ample line. In Section 12, differentiation of representatives of the finite obstruction will produce just such a symbol relation. We prove the two facts here at the geometric scope of the graph supports.

Let $p : A \rightarrow T$ be a holomorphic map of complex manifolds, where A has pure dimension n . Let $i : H \hookrightarrow A$ be a reduced closed analytic subspace of pure dimension d , and put $c = n - d > 0$. Assume the following.

1. The map p is a submersion near H , and $h = p|_H : H \rightarrow T$ is proper.
2. Locally in A , the space H is a reduced SNC divisor in a smooth submanifold of dimension $d + 1$. It has finitely many globally smooth indexed irreducible components $H_1, \dots, H_s$, and every closed intersection $H_I = \bigcap_{i \in I} H_i$ is smooth, possibly empty or disconnected.
3. Every connected component of every nonempty H_I is proper over T and has a global closed real $(1, 1)$ -form which, locally on T , becomes Kähler after adding the pullback of a Kähler form from T .

The last condition is the relative-form convention for a proper Kähler map. A Kähler form on a neighborhood of H , as constructed in the preceding section, implies it. We may replace A by the submersion neighborhood of H . All direct images below are proper on their support; the ambient map p itself need not be proper.

We use right $\mathcal{D}$ -modules with the increasing order filtration. Set

$$K = \mathcal{H}_{[H]}^c(\omega_A), \quad F_0 K = \text{im}(\mathcal{E}xt_A^c(\mathcal{O}_H, \omega_A) \longrightarrow K), \quad F_p K = (F_0 K) F_p \mathcal{D}_A \quad (p \geq 0), \quad (11.1)$$

and $F_p K = 0$ for $p < 0$. The brackets mean algebraic local cohomology with finite pole orders. We will show that the first map is injective and its image is the absolute dualizing sheaf ω_H .

For filtered right $\mathcal{D}$ -modules, p_+ denotes the direct image; in this submersion setting it is computed by the relative right de Rham (Spencer) complex followed by Rp_* .

Proposition 11.1 (The filtered SNC direct image). *For the preceding data, (K, F) is a good filtered regular holonomic right module. Put $M^j = \mathcal{H}^j p_+ K$ for any integer j , and give it the filtration induced by the filtered direct image. Every level cohomology map*

$$\mathcal{H}^j F_p p_+(K, F) \longrightarrow M^j$$

is injective, with image $F_p M^j$. The module (M^j, F) has a finite filtration by $\mathcal{D}_T$ -submodules, strict at every F_p , whose quotients are polarizable real pure Hodge modules. More precisely, the quotient with support-filtration index ℓ has weight $\ell + j$. There are canonical identities, with the first realized by that injection,

$$F_0 M^j = R^j h_* \omega_H \hookrightarrow M^j, \quad F_p M^j = 0 \quad (p < 0). \quad (11.2)$$

Here $R^j = 0$ for $j < 0$, and ω_H is the absolute dualizing sheaf.

For the later use with $j = 1$, define the first-symbol map for the right action by

$$\sigma : F_0 M^1 \otimes T_T \longrightarrow \text{gr}_1^F M^1, \quad m \otimes \xi \longmapsto [m\xi].$$

When T is smooth projective, Corollary 11.3 will prove $\text{Hom}_{\mathcal{O}_T}(N, \ker \sigma) = 0$ for every ample line N on T .

11.1 Finite poles and the two filtrations

At a point of H , choose analytic coordinates

$$(z_1, \dots, z_{c-1}, x_1, \dots, x_t, y), \quad \mathcal{I}_H = (z_1, \dots, z_{c-1}, f), \quad f = x_1 \cdots x_t.$$

The z -list is empty when $c = 1$. Write R for the local analytic ring and η for its coordinate volume form. These equations are a regular sequence. Their localization Čech complex, equivalently the direct limit of their Koszul complexes, has cohomology only in degree c . In that degree it is

$$\frac{R[z_1^{-1}, \dots, z_{c-1}^{-1}, f^{-1}]\eta}{\sum_{j=1}^{c-1} R[z_1^{-1}, \dots, \widehat{z_j^{-1}}, \dots, z_{c-1}^{-1}, f^{-1}]\eta + R[z_1^{-1}, \dots, z_{c-1}^{-1}]\eta}. \quad (11.3)$$

The same assertion holds for a subunion of the branches, with the product of its selected x_i 's. An intersection of k branches instead has $c + k - 1$ regular equations.

Successive finite Taylor divisions give a unique additive polar normal form in (11.3): a finite sum of terms

$$a_{\mathbf{u}, \mathbf{v}, J}(x^{J^c}, y) \prod_{j=1}^{c-1} z_j^{-u_j} \prod_{i \in J} x_i^{-v_i} \eta, \quad u_j, v_i \geq 1, \quad \emptyset \neq J \subset \{1, \dots, t\}. \quad (11.4)$$

The coefficient is a holomorphic germ in exactly the displayed variables. Each fraction has bounded negative orders, so only finitely many Taylor coefficients in its denominator variables enter this description. It is an additive normal form, not an assertion that the polar types form an R -linear direct sum.

The Koszul generator of $\mathcal{E}xt_A^c(\mathcal{O}_H, \omega_A)$ maps to the class

$$\frac{a\eta}{z_1 \cdots z_{c-1} f}. \quad (11.5)$$

If it is zero in (11.3), the expansion of $a|_{z=0}/f$ has no negative x_i -power. Every Taylor monomial of $a|_{z=0}$ is then divisible by every x_i , hence by f . Thus $a \in (z_1, \dots, z_{c-1}, f)$, proving injectivity. Complete-intersection adjunction, including its determinant of the conormal, identifies its image intrinsically with ω_H .

For a polar term define the excess order

$$e(\mathbf{u}, \mathbf{v}, J) = \sum_{j=1}^{c-1} (u_j - 1) + \sum_{i \in J} (v_i - 1).$$

The simple fractions (11.5) are exactly the terms of excess zero. On a right canonical module a coordinate derivative acts by minus differentiation of the coefficient of η . A derivative in a denominator variable raises its pole order by one with a nonzero scalar. Conversely, in (11.4) the coefficient is independent of those variables, so these derivatives produce each desired term from an excess-zero term without a tangential error. It follows that

$$F_p K = \{\text{finite sums of terms (11.4) of excess at most } p\}, \quad F_p K = 0 \ (p < 0). \quad (11.6)$$

This proves the generated formula in (11.1) locally, and proves that it is a good filtration. The generated definition makes it independent of the coordinates.

For a closed immersion of smooth manifolds $j : Y \hookrightarrow A$, the right transfer convention is

$$F_p j_+(N, F) = \sum_{\nu} F_{p-|\nu|} N \partial_{\perp}^{\nu}. \quad (11.7)$$

There is no codimension shift. For example, the codimension shift for left transfer cancels the two dimension shifts in changing sides. These right conventions are recorded in [23, Section 1.1, equations (1.1.2)–(1.1.4)].

We define an increasing support filtration on K by summing the images of the local-cohomology modules of subunions of at most k of the indexed components:

$$W_{-d+k-1}K = \sum_{|I| \leq k} \operatorname{im} \left(\mathcal{H}_{[\bigcup_{i \in I} H_i]}^c(\omega_A) \longrightarrow K \right) \quad (1 \leq k \leq s). \quad (11.8)$$

Set the lower steps to zero and the upper steps to K . In the polar normal form the summand for I consists of the types $J \subset I$. The support maps are locally injective, and hence

$$W_{-d+k-1}K = \{\text{terms with } |J| \leq k\}.$$

The intersection $F_p K \cap W_{-d+k-1}K$ imposes this bound and the excess bound simultaneously. Keeping exactly the polar type $J = I$, $|I| = k$, leaves $c + k - 1$ normal denominators. By (11.7) the resulting filtered quotient is

$$\operatorname{gr}_{-d+k-1}^W(K, F) \simeq \bigoplus_{|I|=k} (i_I)_+(\omega_{H_I}, F^{(0)}), \quad F_p^{(0)}\omega_{H_I} = \begin{cases} 0, & p < 0, \\ \omega_{H_I}, & p \geq 0. \end{cases} \quad (11.9)$$

Here $i_I : H_I \hookrightarrow A$, empty intersections contribute zero, and $d_I = \dim H_I = d - k + 1$. Distinct sets I give distinct polar types even at a point incident to further components. This proves the exact induced F in (11.9), not just the unfiltered quotient.

The identifications glue intrinsically. For an ordered I of size k , the iterated connecting maps for Mayer–Vietoris triangles of supports give

$$\frac{\mathcal{H}_{[\bigcup_{i \in I} H_i]}^c(\omega_A)}{\sum_{J \subsetneq I} \operatorname{im} \mathcal{H}_{[\bigcup_{j \in J} H_j]}^c(\omega_A)} \longrightarrow \mathcal{H}_{[H_I]}^{c+k-1}(\omega_A). \quad (11.10)$$

Locally this map selects the term with every $i \in I$ in the polar type, up to the sign of the order, and is therefore an isomorphism. The one fixed ordering of the global smooth components fixes those signs on overlaps. Disconnected intersections are treated componentwise. This identifies (11.10) with (11.9) globally. The smooth-support modules on the right are regular holonomic; their finite extension K is regular holonomic as well.

11.2 The real structure and strict direct image

The filtration W has a real realization. With the absolute right de Rham complex ending in degree zero, $\operatorname{DR}_A \omega_A \simeq \mathbb{C}_A[n]$. The ordinary unshifted finite-pole de Rham complex along one coordinate has the degree-zero constant and the degree-one logarithmic class of a punctured disk. Tensoring these local comparisons and then taking the localization Čech complexes gives, naturally for all inclusions of subunions,

$$\operatorname{DR}_A K \simeq R\Gamma_H \mathbb{C}_A[n + c] \simeq i_* \mathbb{D}_H(\mathbb{C}_H[d]). \quad (11.11)$$

Here $R\Gamma_H$ is the sheaf-valued support functor on A . The shift in the second identity is fixed by the complex orientation: $n + c = 2n - d$. The same support complexes with $\mathbb{R}$ supply a real perverse sheaf, since complexification is exact and faithful and its complexification is the de Rham realization of the regular holonomic module. Taking real perverse images of the subunion maps defines a real filtration whose complexification is (11.8).

The real version of (11.10) is an isomorphism because its complexification is. Apply the support comparison anew to its graded target $\mathcal{H}_{[H_I]}^{c+k-1}(\omega_A)$, now with support dimension d_I . Its real realization is

$$(i_I)_* \mathbb{D}_{H_I}(\mathbb{R}_{H_I}[d_I]) \simeq (i_I)_* \mathbb{R}_{H_I}[d_I],$$

using smoothness and the complex orientation of H_I . This is the shifted rank-one constant real local system of the graded target. The residue normalization may multiply its complex realization by a nonzero constant on a connected component; it introduces neither a varying factor nor monodromy. Such a real rank-one form is polarizable for its single Hodge type. With the filtration in (11.9), the right module on H_I is the dual constant

$$\mathbb{R}_{H_I}^Hd_I.$$

The untwisted constant has weight d_I and first right step F_{-d_I} ; the twist by d_I moves that step to zero and changes the weight by $-2d_I$. Thus its weight is $-d_I = -d + k - 1$. These dual-constant conventions agree with [23, equations (1.2.1)–(1.2.6) and Theorem 2.3].

There is also a useful support check. If a holomorphic germ g vanishes on reduced H , then

$$gF_p K \subset F_{p-1} K. \quad (11.12)$$

Indeed $g \in (z_1, \dots, z_{c-1}, f)$. Multiplication by z_j lowers the excess or kills the polar class; multiplication by f does the same, since a surviving class has some remaining x_i -pole. This is the filtered support condition for treating the object on H through its local embeddings [46, Section 1.17]. In particular properness on H , rather than properness of A , is the relevant properness here.

We now prove Proposition 11.1. Apply the proper Kähler direct-image theorem [47, Theorem 1 and Remark 1.2] after restricting to each connected component of T , and then separately to the constant real Hodge module on each connected smooth component of H_I over it, using the stipulated global relative form. These componentwise sums are locally finite on T : near any parameter, apply Lemma 10.3 to the proper support and its smooth closed intersections. Thus the componentwise direct images and their polarizations give those for the full H_I ; in the cover and graph applications the final indexing is already finite. The theorem supplies strict filtered direct image, Lefschetz decomposition, and real primitive polarizations. The latter give a polarization on each full cohomology module. After twisting by d_I and using filtered closed-embedding transitivity, the degree- v direct image of a summand of $\mathrm{gr}_\ell^W K$ is polarizable real pure of weight

$$d_I + v - 2d_I = -d_I + v = \ell + v. \quad (11.13)$$

We use the convention fixed by the constant input in Section 1.1 and the direct image in Section 2.2 of that source. The opposite sign for the untwisted dimension in its displayed theorem statement is a typographical error, also ruled out by applying it to the identity map. This application is only to constant modules on smooth sources.

Write $C = p_+(K, F)$ for the filtered direct-image complex. Since p is a submersion near its support, before applying Rp_* its right relative de Rham complex has at level F_p and degree $-v$ the term

$$F_{p-v} K \otimes_{\mathcal{O}_A} \bigwedge^v T_{A/T}. \quad (11.14)$$

The tangent wedges are locally free. The simultaneous pole and branch bounds show that the quotient of this level complex by W is exactly the same level of the relative complex for $\mathrm{gr}^W K$.

Use the increasing W spectral sequence, written as

$$E_1^{-\ell, v+\ell} = \mathcal{H}^v p_+ \operatorname{gr}_\ell^W K \implies \mathcal{H}^v p_+ K. \quad (11.15)$$

Each term on the first page is a strict filtered direct image and a polarizable real pure module of weight $\ell + v$, by (11.13). Strictness says that the corresponding level- F_p first page injects into it with image F_p . The real support realization (11.11), its proper direct image on the support, and the de Rham comparison make the unfiltered differentials real. Comparison with the level spectral sequences makes them filtration preserving.

The differential d_r sends (ℓ, v) to $(\ell - r, v + 1)$. For $r = 1$ these weights are equal. The differential is therefore a morphism of real pure Hodge modules, hence is strict for F , and its kernels and cokernels remain polarizable pure modules [46, Propositions 1.10 and 1.14]. Consequently the level second page injects into the unfiltered second page with image F_p , and the latter consists of pure pieces of the indicated weights.

For $r \geq 2$ the target weight $\ell + v - r + 1$ is smaller than the source weight $\ell + v$. Here is why a real filtered differential between those pieces is zero. Decompose both into their locally finite strict-support summands. A perverse map between distinct strict supports has image supported properly in at least one of them, and hence is zero. On a common smooth dense support of dimension e , the map is a real map of local systems preserving their Hodge filtrations; this follows also directly from (11.7), which recovers the intrinsic filtered module as the sections annihilated by the normal ideal. The two variation weights are the module weights minus e . A real map preserving $F^\bullet$ preserves $\bar{F}^\bullet$ as well. The image of a vector of source type (a, b) lies in $F^a \cap \bar{F}^b$ of the target. Since $a + b$ is the larger source weight, that intersection in the pure target is zero. Thus the generic map is zero, and strict support makes the map zero everywhere.

Inductively, injection of a level page implies its differential is zero when the unfiltered differential is zero. Hence both sequences degenerate at the second page and the level infinity page injects with image F_p . The filtration W is finite. If the abutment map $\mathcal{H}^j(F_p C) \rightarrow \mathcal{H}^j(C)$ had a nonzero kernel element, choose the least ℓ containing it. Its nonzero leading class in gr_ℓ^W would contradict the injection on the infinity page. This proves the asserted injection of every level.

The same leading-class argument proves strictness of the induced W : if an element of $\mathcal{H}^j(F_p C)$ maps into $W_\ell M^j$, it already lies in $W_\ell \mathcal{H}^j(F_p C)$. The resulting finite filtration of (M^j, F) is therefore strict for every level and has exactly the pure second-page quotients, of weights $\ell + j$. It follows as well that M^j is regular holonomic and its filtration is good: these properties are preserved under this finite filtered extension. Taking gr^F , and then $\operatorname{gr}^F \operatorname{DR}_T$, preserves its short exact sequences. If indexed by their pure weights, the output weight filtration is the shifted filtration $W[j]$.

For $p < 0$, every term in (11.14) vanishes. For $p = 0$, its only term is $F_0 K = i_* \omega_H$ in degree zero. The just-proved level injection consequently gives exactly (11.2). This completes the proof of Proposition 11.1. A relative dualizing sheaf would appear only after tensoring by ω_T^{-1} ; the line in (11.2) is absolute.

The first residue cohomology $R^1 h_* \omega_H$ is therefore embedded as $F_0 M^1$. For lifting, it remains to prove the symbol-kernel vanishing when T is smooth projective.

11.3 Projective vanishing for the pure quotients

The projective vanishing we will use is stated for the filtered components of the real or complex theory of Sabbah–Schnell. The Kähler direct-image theorem above supplied real polarizations; a rational polarization was not part of its output. We therefore need vanishing in the real theory

and must identify the actual filtration, not merely the underlying regular holonomic module. The following comparison does this for each strict-support pure summand.

Lemma 11.2 (Pure-component comparison and vanishing). *Let T be smooth projective, and let (Q, F) be the right filtered module of one strict-support summand of a polarizable real pure quotient in Proposition 11.1. Write Z for its support and w for its weight. It is isomorphic, as a right filtered module, to the prime holomorphic component of a pure object in $\text{pHM}_Z(T, \mathbb{R}, w)$ of Sabbah–Schnell, attached to the same generic polarized real variation. In particular, for every ample line N on T , every integer p , and every $v < 0$,*

$$\mathbb{H}^v(T, N^{-1} \otimes \text{gr}_p^F \text{DR}_T Q) = 0. \quad (11.16)$$

Proof. We use Saito’s analytic category of pure real Hodge modules [46, Sections 1.4 and 1.8], which permits discrete real indices for the V -filtration. The pure properties supplied above are in that category. If the constant-source theorem is read with the stronger rationally indexed pure conditions, those imply the broad real specializability conditions, and induction on support dimension gives the pure clauses of Definition 1.8 as well. On a smooth dense analytic Zariski open Z° of Z , the piece Q gives a polarized real variation $\mathbb{H}$ of weight $w - \dim Z$, by Lemmas 1.9 and 1.13 of that source. Theorem 16.2.1 and Corollary 16.3.6 of [45] give a pure object $B \in \text{pHM}_Z(T, \mathbb{R}, w)$ extending this same variation with its real polarization. Use its prime holomorphic filtered component.

Both underlying complex regular holonomic modules are the intermediate extension $\text{IC}_Z(\mathbb{H}_{\mathbb{C}})$. For the Saito filtered module use strict support and regularity in Remark 1.11 of [46]; for the Sabbah–Schnell component use [45, Section 14.2.13 and Theorem 14.7.1]. Regular Riemann–Hilbert and uniqueness of intermediate extension [45, Theorem 13.2.11 and Section 13.2.12] give the unique unfiltered identity extending the chosen identity on Z° . It remains to prove that this identity preserves F at the boundary.

On a smooth support U of dimension e , write $\mathcal{H}$ for the flat holomorphic bundle of the variation, with decreasing Hodge filtration. Its right filtration is

$$F_p(\omega_U \otimes \mathcal{H}) = \omega_U \otimes F^{-p-e} \mathcal{H}. \quad (11.17)$$

Indeed the left filtration is $F_p^L = F^{-p}$, and the side change is $F_p^R = \omega_U \otimes F_{p+e}^L$. The same formula is recorded in [45, Section 16.3.12]. The right closed transfer (11.7) introduces no further shift. Thus the two right filtrations agree away from the boundary of Z° .

Work locally on T . Choose a holomorphic g whose zero set contains that boundary but no local component of Z , and split into the finitely many local strict-support factors if necessary. Such a g exists by prime avoidance: the boundary has smaller dimension than each local support branch. If the boundary is empty there is nothing to prove. Use the graph embedding in $T \times \mathbb{C}_t$, and let Q_g denote the transferred module. Put $b = \dim T$. Saito’s left convention uses increasing F^L and decreasing V_L^α with $t\partial_t - \alpha$ nilpotent on its α -grade. On the graph ambient the conversion is

$$F_p^R = \omega_{T \times \mathbb{C}} \otimes F_{p+b+1}^L, \quad V_L^\alpha \longleftrightarrow V_{-\alpha-1}^R. \quad (11.18)$$

The second rule follows because transposition sends $t\partial_t$ to $-t\partial_t - 1$. In particular $V_L^{>-1}$ becomes $V_{<0}^R$. The sign change of a normal ∂_t does not change a generated sum.

We verify the filtered surjectivity needed in Saito’s reconstruction. Put $\Psi = \psi_{g,1}Q$ and $\Phi = \phi_{g,1}Q$. Strict support gives a surjective $\text{can} : \Psi \rightarrow \Phi$ and an injective Var , by [46, Proposition 1.5].

Forgetting F , this is the nilpotent middle-extension monodromy quiver of holonomic modules. Its monodromy filtrations centered at zero obey

$$\text{can}(M_k\Psi) = M_{k-1}\Phi \quad (11.19)$$

by [45, Lemma 3.3.13]. The centers of the weight filtrations in Saito's pure definition are $w - 1$ for Ψ and w for Φ . Thus (11.19) says $\text{can}(W_i\Psi) = W_i\Phi$. Condition (1.8.2) of Definition 1.8 and the direct-factor Lemma 1.15 of [46] put both $\text{gr}_i^W\Psi$ and $\text{gr}_i^W\Phi$ in Saito's pure real category of weight i . The induced surjection on each grade is a filtered pure morphism, hence is F -strict by its Proposition 1.10.

At a stalk, take $y \in F_p\Phi$ and choose i with $y \in W_i\Phi$. Strictness on the i th grade lifts its class by an element of $F_p\Psi \cap W_i\Psi$. Subtract its image and repeat on the next lower weight. The finite monodromy filtration terminates this procedure. It proves

$$\text{can}(F_p\Psi) = F_p\Phi \quad \text{for all } p.$$

The finite lifting argument obtains filtered surjectivity from pure strictness on each weight grade.

Saito's filtered middle-extension formula [46, equation (1.7.2)] now applies: the local pure factor satisfies its specializability conditions, the zero set of g does not contain its support, and can is filtered surjective. After (11.18), that formula is

$$F_p Q_g^R = \sum_{v \geq 0} \left(V_{<0}^R Q_g^R \cap (j_{\text{op}})_* F_{p-v}(Q_g^R|_{t \neq 0}) \right) \partial_t^v, \quad (11.20)$$

where $j_{\text{op}} : T \times \mathbb{C}^* \hookrightarrow T \times \mathbb{C}$ and the intersection is inside $(j_{\text{op}})_* j_{\text{op}}^{-1} Q_g^R$. The strict-support module and its V -pieces embed there by restriction.

For the Sabbah–Schnell component, strict real specializability, pure support, and strictness of can follow respectively from Definition 14.2.2, Theorem 14.2.19, and Corollary 14.2.23 of [45]. Its unfiltered intermediate-extension property makes can surjective, so it is a filtered middle extension. Definition 10.6.1, Remark 10.6.2, and Proposition 10.6.5 of the same source give precisely (11.20) for its right component: for $\alpha < 0$, the filtered V_α is the intersection with the filtration off the divisor, and the full module is generated from $V_{<0}$ by normal derivatives.

The unfiltered intermediate-extension identity preserves the canonical V -filtration and, by (11.17), preserves F off the divisor. The two right sides of (11.20) are therefore identical. This proves equality of F on the graph. Formula (11.7) recovers the module before graph transfer, and its filtration, as the sections annihilated by the normal ideal. Hence the identity is filtered on T . The local identities glue by uniqueness of the unfiltered identity. The comparison uses the actual generic variation and its filtration; the two theories' internal twist symbols need not have the same weight convention.

A pure object of Sabbah–Schnell is an object of $\text{WHM}(T)$ with its one-step weight filtration, by Section 14.2.14 of [45]. Its Theorem 16.3.10, on the smooth projective T , gives negative-ample graded de Rham vanishing for either filtered component. Sections 8.4.1–8.4.9 of that source identify the perverse right de Rham convention with the Spencer complex ending in the module in degree zero, with its filtration grading preserved. The theorem therefore gives (11.16) for every homogeneous degree p in our convention. This proves the lemma. $\square$

Corollary 11.3 (No ample line in the first symbol kernel). *In Proposition 11.1, assume in addition that T is smooth projective, and fix j . Put $M = M^j$, and let*

$$\sigma : F_0 M \otimes T_T \longrightarrow \text{gr}_1^F M, \quad m \otimes \xi \longmapsto [m\xi]$$

be principal-symbol multiplication for the right action. For every ample line N on T ,

$$\mathrm{Hom}_{\mathcal{O}_T}(N, \ker \sigma) = 0. \quad (11.21)$$

This assertion includes a coherent torsion subsheaf of the kernel.

Proof. On compact T the locally finite strict-support decomposition of each pure quotient of Proposition 11.1 is finite. Apply Lemma 11.2 to each factor. The finite filtration of M is F -strict, so its short exact sequences remain exact after $\mathrm{gr}^F \mathrm{DR}_T$. Their hypercohomology long exact sequences give (11.16) for M itself. This deduction uses the comparison for the pure quotients and the strict filtration of M .

By (11.2), the $p = 1$ right graded de Rham complex has exactly two terms:

$$\mathrm{gr}_1^F \mathrm{DR}_T M = \left[F_0 M \otimes T_T \xrightarrow{\sigma} \mathrm{gr}_1^F M \right] \quad \text{in degrees } -1, 0. \quad (11.22)$$

Since N is invertible, its tensor is exact, and negative hypercohomology gives

$$0 = \mathbb{H}^{-1}(T, N^{-1} \otimes \mathrm{gr}_1^F \mathrm{DR}_T M) = H^0(T, N^{-1} \otimes \ker \sigma) = \mathrm{Hom}_{\mathcal{O}_T}(N, \ker \sigma).$$

The two-term calculation uses no local freeness of $\ker \sigma$, so it applies to torsion as well. $\square$

12 Lifting through every boundary neighborhood

We retain the data L, s, G, q, r, a , the local root charts, and the maps $g : E \rightarrow U$, $h : H \rightarrow U$ of Section 10. In particular E is reduced Cartier, $I = \mathcal{O}_Z(-E)$ is invertible, and $\mathcal{A}_j = I^j/I^{j+1}$ has the Laurent graded frame u^j for every integer j . The frame uses the prescribed boundary root comparison; it is a frame on the associated graded, not an extension of that boundary frame to Z .

The finite quotients $\mathcal{T}_{j,k} = I^j/I^{j+k}$ have the lifting target in Proposition 10.2. The intervening constructions supplied the split residue insertion and the projective symbol-kernel vanishing. We now use them to kill the connecting maps of (10.9) at every finite order. The quotients remain sheaves on the underlying space of E ; no map from an infinitesimal thickening to the parameter space is required.

12.1 The obstruction is a graded derivation

We prove the proposition by induction on k , simultaneously in all integer degrees and all root charts. The following description uses only the orders strictly smaller than the current one.

Lemma 12.1. *Fix $k \geq 1$, and assume (10.8) for every smaller order, every integer degree, and every root chart. The connecting maps for the current order factor uniquely as maps of complex vector space sheaves*

$$\delta_k : g_* \mathcal{A}_j \longrightarrow R^1 g_* \mathcal{A}_{j+k} \quad (j \in \mathbb{Z}). \quad (12.1)$$

They form a degree- k derivation from the Laurent graded algebra $\bigoplus_j g_ \mathcal{A}_j$ to its graded cohomology module $\bigoplus_j R^1 g_* \mathcal{A}_j$. If $t_1, \dots, t_b$ are local coordinates on U and F is holomorphic in those coordinates, then*

$$\delta_k(F(t)) = \sum_{i=1}^b F_{t_i}(t) \delta_k(t_i). \quad (12.2)$$

These maps and formulas commute with restriction and with the ambient power-compatible root comparisons. Vanishing of δ_k in all degrees proves the current order of (10.8).

Proof. Use (10.9) together with

$$0 \longrightarrow \mathcal{T}_{j+1,k-1} \longrightarrow \mathcal{T}_{j,k} \longrightarrow \mathcal{A}_j \longrightarrow 0 \quad (k > 1). \quad (12.3)$$

All are valid for negative j because I is invertible. Write $\partial_{j,k} : g_*\mathcal{T}_{j,k} \rightarrow R^1g_*\mathcal{A}_{j+k}$ for the connecting map of the first sequence. The shorter orders make $g_*\mathcal{T}_{j,k} \rightarrow g_*\mathcal{A}_j$ surjective. For $k > 1$, its kernel is the image of $g_*\mathcal{T}_{j+1,k-1}$, by left exactness in the second sequence. The shorter order $k-1$ in degree $j+1$ lifts that image through $g_*\mathcal{T}_{j+1,k}$, which maps into $g_*\mathcal{T}_{j,k+1}$. The connecting map vanishes on the kernel. For $k=1$, the leading map is the identity and its kernel is zero. This proves the unique factorization (12.1). If it is zero, exactness of the first sequence gives the desired epimorphism.

Here is the derivation calculation on germs. A leading section in degree j lifts to a section of $\mathcal{T}_{j,k}$ after a base shrink. Choose local representatives $x_\alpha \in I^j$ on an ambient open cover near E . Their differences belong to I^{j+k} , and their classes modulo I^{j+k+1} represent $\delta_k(x)$. For a second section z of degree l , take simultaneous representatives. On an overlap,

$$x_\beta z_\beta - x_\alpha z_\alpha = x_\alpha(z_\beta - z_\alpha) + z_\alpha(x_\beta - x_\alpha) + (x_\beta - x_\alpha)(z_\beta - z_\alpha).$$

The last term lies in $I^{j+l+2k} \subset I^{j+l+k+1}$. The first two terms reduce to multiplication by the leading coefficients. Their Čech classes therefore give

$$\delta_k(xz) = x\delta_k(z) + z\delta_k(x).$$

This uses the ordinary multiplication action of layer sections on their first cohomology, and gives the asserted graded derivation.

For the chain rule, choose simultaneous shorter representatives $G_{i,\alpha} \in \mathcal{O}_Z$ of the degree-zero sections t_i . Their differences lie in I^k . Shrink the local opens so that $F(G_{1,\alpha}, \dots, G_{b,\alpha})$ is defined. Its holomorphic Taylor difference, modulo the square of the differences, is $\sum_i F_{t_i}(G_\alpha)(G_{i,\beta} - G_{i,\alpha})$. The square lies in $I^{2k} \subset I^{k+1}$, and the coefficient restricts to $F_{t_i}(t)$ on E . This gives (12.2) before any restriction to a generic parameter. It is valid when the target contains torsion. The connecting map and all representative calculations are natural for restrictions and the prescribed ambient root isomorphisms. $\square$

12.2 The local differential-operator identity

Fix the current order k , and put $m = a + k > 0$. For a local leading section $x \in g_*\mathcal{A}_{-m}$, define, in coordinates $t = (t_1, \dots, t_b)$ on U ,

$$c(x) = \tau_*\delta_k(x), \quad e_i(x) = \tau_*(x\delta_k(t_i)) \quad \text{in } R^1h_*\omega_H. \quad (12.4)$$

Both arguments of τ_* have layer degree $-a$, since $-m + k = -a$. The map is the residue insertion of Proposition 10.4; it is injective on every parameter germ.

Consider a holomorphic $\psi : U' \rightarrow U$ between manifolds of the same dimension b , in coordinates $w = (w_1, \dots, w_b)$, satisfying the transversality (10.17) for every smooth component intersection of H . The identity map is one such map. As in Proposition 10.5, after a shrink about a compact fiber the support

$$H^b = H \times_U U' \subset \widehat{Z} \times U'$$

is reduced SNC, locally a divisor in a smooth complete intersection, of codimension $b+1$ in the product. Its ambient projection p^b is a submersion and its closed strata are proper with global

relative Kähler forms. After the restrictions needed for the root comparisons and the Kähler neighborhood, apply Lemma 10.3 anew to the smooth closed strata of the graph cut over U' , and use its final open neighborhood. Properness survives this base change, and the lemma gives finitely many globally smooth irreducible component labels and finitely many smooth proper connected pieces of each closed intersection. The existing coordinates and relative forms restrict to that open. Set

$$K^b = \mathcal{H}_{[H^b]}^{b+1}(\omega_{\widehat{Z} \times U'}), \quad M^b = \mathcal{H}^1 p_+^b K^b.$$

Proposition 11.1 applies with no additional codimension shift and gives $F_0 M^b = R^1 h_*^b \omega_{H^b} \hookrightarrow M^b$.

The adjunction identity (10.16) supplies the factor $\omega_{U'} \otimes \psi^* \omega_U^{-1}$. We denote its wedge frame by dw/dt ; this notation is a ratio of canonical frames, not an inverse Jacobian. Pull the Čech classes representing $c(x), e_i(x)$ to H^b and tensor with this frame. Write their images in $F_0 M^b$ as $\tilde{c}, \tilde{e}_i$. These are the natural pullbacks of these classes; no base-change isomorphism is asserted for a ramified ψ .

Lemma 12.2 (The adjugate identity). *Let $J = (\partial \psi_i / \partial w_j)$ and $J^\# = \text{adj}(J)$. Under the preceding assumptions, the following identity holds in the right module M^b :*

$$\tilde{c} \det J + \sum_{i,j} (\tilde{e}_i \partial_{w_j}) J_{ji}^\# = 0. \quad (12.5)$$

In particular,

$$\sigma \left(\sum_{j,i} J_{ji}^\# \tilde{e}_i \otimes \partial_{w_j} \right) = 0 \quad \text{in } \text{gr}_1^F M^b. \quad (12.6)$$

For $\psi = \text{id}$, after the natural graph identification, the exact identity is

$$c(x) + \sum_i e_i(x) \partial_{t_i} = 0 \quad \text{in } M^b. \quad (12.7)$$

When $b = 0$, the same construction without graph equations gives $c(x) = 0$ in its degree-one direct-image module.

Proof. All statements are on germs, so choose a common parameter shrink and simultaneous shorter lifts of the finitely many sections x, t_i . Choose local ambient representatives near E , indexed by α , with

$$x_\alpha \in I^{-a-k}, \quad x_\beta - x_\alpha \in I^{-a}, \quad G_{i,\alpha} \in \mathcal{O}_Z, \quad G_{i,\beta} - G_{i,\alpha} \in I^k. \quad (12.8)$$

They exist by the smaller orders in Lemma 12.1; for $k = 1$ the shorter truncation is already the layer. Pull them to $\widehat{Z}$, suppressing pullback symbols, and put

$$\eta_\alpha = x_\alpha \tau, \quad Q_{i,\alpha} = \psi_i(w) - G_{i,\alpha}, \quad P_\alpha = \prod_i Q_{i,\alpha}.$$

On H the $G_{i,\alpha}$ restrict to the coordinate functions of h . The Q_i 's and a reduced equation of H are therefore the regular graph-support equations. The logarithmic insertion and $D_E = \rho^* E \geq H$ give the precise pole bounds

$$\begin{aligned} \eta_\alpha &\in \omega_{\widehat{Z}}(H + kD_E), \\ \eta_\beta - \eta_\alpha, \quad \eta_\alpha(G_{i,\beta} - G_{i,\alpha}) &\in \omega_{\widehat{Z}}(H), \\ \eta_\alpha(G_{i,\beta} - G_{i,\alpha})(G_{l,\beta} - G_{l,\alpha}), \quad (\eta_\beta - \eta_\alpha)(G_{i,\beta} - G_{i,\alpha}) &\in \omega_{\widehat{Z}}(H - kD_E) \subset \omega_{\widehat{Z}}. \end{aligned} \quad (12.9)$$

The last inclusion uses $k \geq 1$. Thus the quadratic and cross difference terms have no pole on the first-factor support H .

In algebraic local cohomology consider the local sections

$$S_\alpha = \left[\frac{\eta_\alpha \wedge dw}{P_\alpha} \right] \quad \text{of } K^b.$$

The H -pole is already in η_α . These are finite-pole generalized fractions with a fixed ordering of the graph equations. To expand their differences, first quotient the localization along H by forms regular there, and then localize for the Q_i 's. Each polar class is killed by a power of a reduced equation of H , and every $G_{i,\beta} - G_{i,\alpha}$ is a multiple of that equation. The change from $Q_{i,\alpha}$ to $Q_{i,\beta} = Q_{i,\alpha} - (G_{i,\beta} - G_{i,\alpha})$ therefore has a finite geometric expansion on each polar class. By (12.9) only its linear terms survive. This proves the exact generalized-fraction equality on an overlap

$$S_\beta - S_\alpha = \left[\frac{(\eta_\beta - \eta_\alpha) \wedge dw}{P_\alpha} \right] + \sum_i \left[\frac{\eta_\alpha (G_{i,\beta} - G_{i,\alpha}) \wedge dw}{Q_{i,\alpha} P_\alpha} \right]. \quad (12.10)$$

There is no infinite series or division by a Jacobian in this identity.

Let C be the first Čech cochain in (12.10). Let E_i have the numerator of its i th summand but only the denominator P_α , and let $E_i^{(l)}$ denote that cochain with the additional denominator $Q_{l,\alpha}$. The simple fractions C, E_i are cocycles: their reductions are respectively the cocycles for $\delta_k(x)$ and $x\delta_k(t_i)$; on a triple overlap the possible change of the representative of x is a cross term with no H -pole by (12.9). The description of the residue insertion on representatives and complete-intersection adjunction identify their classes with $\tilde{c}, \tilde{e}_i$ in $F_0 M^b$. More explicitly, the residue of a log form η on H , pulled with dw/dt , is represented by $[\eta \wedge dw/P_\alpha]$; the determinant of the graph conormal gives exactly that frame. Thus this identification holds also at ramification. The cochains $E_i^{(l)}$ need not individually be cocycles.

The numerator of E_i is independent of w . The right canonical action is minus differentiation of the coefficient of the volume form, so at the cochain level

$$E_i \partial_{w_j} = \sum_l J_{lj} E_i^{(l)}. \quad (12.11)$$

Write $\check{d}$ for the Čech differential. Equation (12.10) is $\check{d}S = C + \sum_i E_i^{(i)}$. Multiply it by $\det J$ and use $JJ^\# = (\det J)\text{id}$ together with (12.11). The result, still at the cochain level, is

$$\check{d}(S \det J) = C \det J + \sum_{i,j} (E_i \partial_{w_j}) J_{ji}^\#. \quad (12.12)$$

The placement of $J_{ji}^\#$ after the right derivative is part of this exact formula.

These cochains lie in the end term of the relative right de Rham complex for the product projection, with first-factor coordinates held fixed. That term has no outgoing relative differential, so a Čech coboundary there is a total coboundary in direct image. The action in (12.11) induces the base right action there. Passing (12.12) to M^b gives (12.5). This uses the natural map from these Čech classes to direct image, and does not require that all direct-image classes be computed by this cover.

The term $\tilde{c} \det J$ is in F_0 . Moving a holomorphic function past a right vector field changes an operator by order zero. Taking gr_1^F of (12.5) gives (12.6). With unchanged coordinates $J = \text{id}$, the exact formula itself gives (12.7). If $b = 0$, there are no Q_i ; the undivided difference of the η_α in the support module of H is the cochain for $c(x)$ and is a total coboundary. This proves the last assertion as well. $\square$

12.3 One global symbol kills the current obstruction

Proof of Proposition 10.2. Assume the shorter orders and use Lemma 12.1. First suppose $b > 0$, and fix an arbitrary point $t_* \in T = \mathbb{P}^b$. Proposition 10.5 gives the coordinate-power map $\psi : T' = \mathbb{P}^b \rightarrow T$, the line $R = \mathcal{O}_{T'}(1)$ with $R^r \simeq \psi^* \mathcal{O}_T(1)$, and a single compact SNC graph support $H' \subset \mathcal{W}$ with submersive projection $p' : \mathcal{W} \rightarrow T'$. It is unbranched over t_* . Put

$$K' = \mathcal{H}_{[H']}^{b+1}(\omega_{\mathcal{W}}), \quad M' = \mathcal{H}^1 p'_+ K'.$$

The global graph proposition verifies every hypothesis of Proposition 11.1: the support has pure dimension $\dim V - 1$, globally smooth finitely many components, and proper smooth closed intersections with one ambient Kähler form. Thus $F_0 M' = R^1 h'_* \omega_{H'} \hookrightarrow M'$, in the absolute canonical convention, and Corollary 11.3 applies.

Take the local leading section $x = u^{-m}$. On a local graph chart choose a frame $\bar{p}$ of R with $\bar{p}^r = \psi^* \ell$ for a downstairs frame ℓ . The local formula

$$\bar{p}^m \mapsto \sum_{j,i} J_{ji}^\# \tilde{e}_i(u^{-m}) \otimes \partial_{w_j} \quad (12.13)$$

defines a global morphism

$$R^m \longrightarrow F_0 M' \otimes T_{T'}. \quad (12.14)$$

We verify the transition, including on the branch locus.

Let $t' = t'(t)$ and $w' = w'(w)$ be changes of coordinates, and put $A_t = \partial t' / \partial t$, $B_w = \partial w' / \partial w$. These matrices are invertible. If $\bar{p}' = \lambda \bar{p}$, then $\lambda^r = \psi^* \mu$ for the downstairs unit μ changing the frame of $\mathcal{O}_T(1)$. Locally even at a branch point, λ descends as a holomorphic unit: take an r th root of μ on a small downstairs neighborhood, and observe that the remaining ratio has r th power one and hence is locally constant. Denote the resulting downstairs unit also by λ .

Compare the local root pairs by sending their new boundary frame to λp_1 . Lemma 10.1 extends this power-compatible comparison to the ambient germs. Its pullback agrees with the transition of the global root near the graph, by the uniqueness and properness argument in Proposition 10.5. It therefore identifies both full covers, their tautological forms, and the fixed functorial principalizations. Under this identification $u'^{-m} = \lambda^m u^{-m}$. In the definition of $e_i(x)$, the factor x multiplies outside the derivation. Hence no derivative of λ occurs. Naturality and the chain rule (12.2) transform the downstairs column of e_i 's by $\lambda^m A_t$.

The absolute graph factor changes by $dw' / dt' = (\det B_w / \det A_t) dw / dt$. Consequently, for the pulled columns and Jacobians,

$$\tilde{e}' = \frac{\det B_w}{\det A_t} \lambda^m A_t \tilde{e}, \quad J' = A_t J B_w^{-1}, \quad (12.15)$$

$$(J')^\# = \frac{\det A_t}{\det B_w} B_w J^\# A_t^{-1}, \quad (J')^\# \tilde{e}' = \lambda^m B_w J^\# \tilde{e}. \quad (12.16)$$

The adjugate identity holds for every J : it follows for invertible J from the inverse formula and then for all J as a polynomial identity. Only the coordinate changes A_t, B_w are inverted. Since the vector basis changes by B_w^{-1} , the last equality is exactly the transition for the image of the frame $\bar{p}^m$ in (12.13). This proves (12.14) on all of T' . At ramification the classes are the natural Čech pullbacks under the resolved ambient germ comparisons; a flat base-change isomorphism for $R^1 h_* \omega_H$ is not used.

By Lemma 12.2, the image of (12.14) lies in $\ker \sigma$. The line $R^m = \mathcal{O}_{\mathbb{P}^b}(a+k)$ is ample because $a+k > 0$. Corollary 11.3 makes the global map zero. At a point above t_* , the map ψ is locally

biholomorphic and $J^\#$ is invertible. The ambient product comparison then identifies the graph family and its Čech classes with the downstairs family. Thus the zero map implies $e_i(u^{-m}) = 0$ as a downstairs germ near t_* , including a class supported at that parameter. The point t_* was arbitrary, and the root comparisons allow any chart at it. It follows that these classes vanish locally everywhere.

The injection τ_* and the invertible layer frame u^{-m} now imply $\delta_k(t_i) = 0$ on every chart: multiplication by that frame is an isomorphism from the layer $\mathcal{A}_k$ to $\mathcal{A}_{-a}$, including on first direct images. For any local x of degree $-m$, all $e_i(x)$ vanish. The exact unchanged coordinate identity (12.7) gives $c(x) = 0$ in M^b . The injection (11.2) brings this back to $R^1 h_* \omega_H$, and the residue injection gives $\delta_k(x) = 0$. When $b = 0$, Lemma 12.2 gives $c(x) = 0$ directly, and the same two injections give this conclusion; there are no parameter coordinates to treat.

In particular $\delta_k(u^{-m}) = 0$. The derivation rule for the invertible Laurent frame gives

$$0 = \delta_k(u^{-m}) = -mu^{-m-1}\delta_k(u),$$

so $\delta_k(u) = 0$. For any local $F \in g_* \mathcal{O}_E$, the section Fu^{-m} also has degree $-m$, and

$$0 = \delta_k(Fu^{-m}) = u^{-m}\delta_k(F).$$

Thus $\delta_k(F) = 0$. Every local graded section is Fu^j , so δ_k is zero in every integer degree. Lemma 12.1 closes the current order. Induction from $k = 1$ proves Proposition 10.2. $\square$

12.4 Invariant layers and the growth of sections

Finite lifting is now available in every degree. To obtain ambient sections, we pass to cyclic invariants and count the resulting finite quotients downstairs. Translation by the Cartier divisor G will make the same finitely many coherent layers recur with increasing positive twists on T . Those twists supply the section growth and bound the higher-cohomology loss.

Completion of Theorem 1.2. Define global divisorial subsheaves of meromorphic functions on the normal space V by

$$J_j = \mathcal{O}_V(-[jG/r]) \quad (j \in \mathbb{Z}). \quad (12.17)$$

They agree in each root chart with $(\pi_* I^j)^{\mu_r}$. Indeed, at a prime over S_i , an invariant meromorphic function of downstairs order v_i has order $(r/d_i)v_i$. Membership in I^j is equivalent to

$$v_i \geq [jd_i/r].$$

At all other primes it is regular. The full invariant meromorphic algebra is the downstairs algebra, and normality extends the codimension-one test. Finite coherent pushforward and averaging by μ_r are exact. This also proves coherence of the divisorial sheaves in (12.17) and of their quotients locally, and their valuation descriptions glue globally.

Since $0 < d_i \leq r$, the sheaf $J_1 = \mathcal{O}_V(-S)$ is the ideal of the reduced divisor S . The same coefficient inequality gives $J_1 J_j \subset J_{j+1}$ for every j . Hence

$$\mathcal{B}_j = J_j / J_{j+1}$$

is a coherent $\mathcal{O}_S$ -module, and $\mathcal{B}_0 = \mathcal{O}_S$. Taking invariants of Proposition 10.2 preserves its epimorphisms: lift an invariant local section upstairs and average the lift. For $l < n$, define the sheaf of complex vector spaces

$$\mathcal{F}_{l,n} = f_*(J_l / J_n) \quad \text{on } T, \quad F_j = f_* \mathcal{B}_j.$$

Set also $\mathcal{F}_{n,n} = 0$. The quotients J_l/J_n are considered on the underlying space of S , just as before. The invariant lifting surjections give exact sequences of complex vector space sheaves

$$0 \longrightarrow \mathcal{F}_{j+1,n} \longrightarrow \mathcal{F}_{j,n} \longrightarrow F_j \longrightarrow 0 \quad (j < n). \quad (12.18)$$

In particular $\mathcal{F}_{l,n}$ has a finite filtration with the full quotients F_j , $l \leq j < n$. Each F_j is a coherent $\mathcal{O}_T$ -module by properness of f ; coherence as an $\mathcal{O}_T$ -module is not asserted for $\mathcal{F}_{l,n}$.

Cartier translation by G in (12.17), together with the actual line identity $\mathcal{O}_V(G) = L$ defined by s , gives

$$J_{j-r} = J_j \otimes \mathcal{O}_V(G), \quad \mathcal{B}_{j-r} = \mathcal{B}_j \otimes L|_S.$$

Projection formula and (10.2) consequently give

$$F_{j-r} = F_j \otimes \mathcal{O}_T(1), \quad F_0 = f_* \mathcal{O}_S \neq 0. \quad (12.19)$$

This is an identity of the actual holomorphic lines and sheaves.

For $N > 0$, each $-Nr \leq j < 0$ is uniquely $j = l - tr$ with $0 \leq l < r$ and $1 \leq t \leq N$. By (12.19), $F_j = F_l(t)$. Analytic GAGA and Serre's coherent finiteness and vanishing theorems [50, Section 3, no. 12, Theorems 1 and 3], [49, Section 66, Theorems 1 and 2] imply, for every $i > 0$,

$$\sum_{j=-Nr}^{-1} h^i(T, F_j) \leq C_i := \sum_{l=0}^{r-1} \sum_{t \geq 1} h^i(T, F_l(t)) < \infty. \quad (12.20)$$

These constants are independent of N , and they are zero for $i > \dim T$. The long exact sequences of the finite filtration (12.18) give the same bounds for $h^i(T, \mathcal{F}_{-Nr,0})$, as well as finiteness and vanishing above that fixed dimension. Serre vanishing is being applied only to the coherent quotients $F_l(t)$.

There is at least one section of $F_0(t)$ for every $t \geq 1$. Choose a point of the nonempty S and a section of $\mathcal{O}_T(t)$ nonzero at its image; its pullback is nonzero there. Thus $\sum_{j=-Nr}^{-1} h^0(T, F_j) \geq N$. Additivity of the finite Euler characteristics in (12.18), and the uniform higher bounds for both the quotients and $\mathcal{F}$, give a constant C independent of N with

$$h^0(T, \mathcal{F}_{-Nr,0}) \geq N - C. \quad (12.21)$$

For example one can take $C = 2 \sum_{i=1}^{\dim T} C_i$.

Finally, $J_{-Nr} = \mathcal{O}_V(NG)$ and $J_0 = \mathcal{O}_V$. Degree-zero direct image on the underlying support gives

$$H^0(T, \mathcal{F}_{-Nr,0}) = H^0(V, \mathcal{O}_V(NG)/\mathcal{O}_V).$$

The exact sequence on V loses at most the fixed finite dimension $h^1(V, \mathcal{O}_V)$ when lifting these quotient sections to $H^0(V, \mathcal{O}_V(NG))$; finiteness follows from proper coherent direct image to a point. Equation (12.21) therefore makes these section spaces unbounded. For some N the line $\mathcal{O}_V(NG)$ has two independent sections. Their quotient is a nonconstant meromorphic function on the irreducible normal V , so its Iitaka dimension is at least one. Since $\mathcal{O}_V(G) = L = \mathcal{O}_V(qA)$ is the actual adjoint identity, this proves $\kappa(V, A) \geq 1$, as claimed. $\square$

13 Proof of abundance after nonvanishing

Proof of Theorem 1.1. If $\kappa(X, D) \geq 1$, Proposition 5.2 gives semiampleness on the original X , by descent of the generated actual Cartier multiple. It remains to treat $\kappa(X, D) = 0$.

Choose a positive integer m_0 for which the actual line $\mathcal{L} = \mathcal{O}_X(m_0 D)$ is invertible and has a nonzero section s_0 . A connected normal space is irreducible, so this section is locally a nonzerodivisor. Let

$$M = \frac{1}{m_0} \operatorname{div}_{\mathcal{L}}(s_0).$$

It is the normalized effective rational $\mathbb{Q}$ -Cartier divisor of the plurisecion, and the chosen line and section give the actual equivalence $M \sim_{\mathbb{Q}} D$.

Suppose $M \neq 0$. Proposition 5.1 produces a normal ordinary $\mathbb{Q}$ -factorial compact Kähler dlt fourfold (V, B) with effective rational boundary, analytically nef actual adjoint $A = K_V + B$, and a nonzero effective rational $\mathbb{Q}$ -Cartier divisor P such that

$$A \sim_{\mathbb{Q}} P, \quad \operatorname{Supp} P = \operatorname{Supp} \lfloor B \rfloor, \quad \kappa(V, A) = 0.$$

The space V is irreducible, being bimeromorphic to X , and has the special projective resolution in Lemma 5.6. Theorem 6.1 therefore makes the actual restricted adjoint semiample on the entire reduced floor $S = \lfloor B \rfloor$. This floor is nonempty because $P \neq 0$. Every hypothesis of Theorem 1.2 is now satisfied, so it gives $\kappa(V, A) \geq 1$, a contradiction.

Thus $M = 0$. The section s_0 is nowhere zero. Indeed, if a local representative were a nonunit at a point of the normal space, a minimal prime over its nonzero principal ideal would have height one, and would give a component of its zero divisor. Consequently s_0 supplies an isomorphism of holomorphic lines

$$\mathcal{O}_X \xrightarrow{\cong} \mathcal{O}_X(m_0 D).$$

This proves the asserted actual $\mathbb{Q}$ -linear triviality in Iitaka dimension zero, and in particular semiamplessness there. Together with the positive-Iitaka-dimensional case it proves the theorem. $\square$

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